Nonlinear Wave and Dispersive PDE

***EYNTKA* Series**

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Summary

It as assumed that the reader is comfortable with the concept introduced in a first course in analysis and linear algebra, importantly that smooth functions and vector-spaces are comfortable concepts.

Introduction Differential Equations

To avoid super-script counting, the book shall work with smooth functions (or else the order of differentiability must be kept track). This shall not change the theory for ODE's and PDEs, especially since later it shall be shown that even working with non-smooth differentiable functions shall end up having the same theory (due to the advent of Distribution Theory [1])

Let $f : \mathbb{R} \to \mathbb{R}$. When working smooth functions, we may take the derivative of the function. This gives the following function taking smooth functions and returning smooth functions:

$$f \mapsto f'$$

Due to the fundamental theorem of calculus, the inverse of this procedure is called *integration*, and gives a unique inverse up to a constant:

$$f \mapsto F + c \qquad c \in \mathbb{R}$$

where F may be defined as $F(x) = \int_t^x f(x)dx$ for appropriate t. Using this result, it is possible to find the smooth function that satisfies an arbitrary degree of differentiation, for example:

$$f^{(n)} = 0$$

whose solutions are the polynomials of degree at most $n - 1$, or more generally

$$f^{(n)} = g \qquad \text{equiv.} \qquad f^{(n)} - g = 0$$

for some smooth functions g[2].

Given this simple scenario, the question may arise if we may solve more complicated algebraic expression. For example, does there exist a solution to any of these:

$$f = f' \qquad f' + f = 0 \qquad f'' + f' + f = 0$$

[1] see chapter 3

[2] Note that the solution may not always come in an elementary form, that is expressible as a polynomial-combination and composition of x, log, e^x, or a trigonometric identity

More precisely, we ask if any arbitrary algebraic expression on a smooth function f combined by using the 4 usual arithmetic operators $(+, -, \times, \text{div})$ along with differentiation yields a solution:

$$p(f) = 0$$

more commonly written as:

$$F(x, f(x), f'(x), ..., f^{(n)}(x)) = 0$$

The study of the solution to such equations is called the study of *Ordinary Differential Equations* (ODEs). If we have a smooth function of the form $f : \mathbb{R} \to \mathbb{R}^m$, this is still the study of ODEs, where it is solving a *system of ODEs*. When working with multi-variable functions and partial derivatives, the study of the solution to such equations is called the study of *Partial Differential Equations* (PDEs). The study of ODEs can be thought of as a special case of the study of PDEs.

This book shall start with the study of ODEs in the first chapter, and proceed with the rest of the chapter with the study of PDEs. Only one chapter is dedicated to ODEs as they are a solved problem where most ODEs have a general solution that can be found algorithmically, while this is provably not the case for most PDEs.

From a more physical perspective, differential equations will give you how certain parameters relate to each, for PDEs it is usually over time. As a simple example, take:

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial x} \qquad \text{or} \qquad \frac{\partial}{\partial t} - \frac{\partial}{\partial x} = 0 \tag{1}$$

What this equation is giving us is a restriction we require a function $f(t, x)$ to obey, that is:

$$\frac{\partial f}{\partial t} - \frac{\partial f}{\partial x} = 0$$

For example $f(x, t) = x + t$ would satisfy this equation. One key skill to learn when solving problems in PDEs is to be able to "read" such an equation. For example, eq. (1) shows that the speed at which our "system" is evolving over time is identically propositional to the speed at which our "system" is moving. Sometimes, it is quite subtle what an equation may represent. For example:

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = 0$$

is very similar to eq. (1), however it represents how a *wave* evolves over time, with behavior that is very different from eq. (1).

The reason why finding solutions to PDEs is interesting to me is because the derivative can give lot's of local information about a system, and sometimes that's all the information we have. For example, it is much easier to describe the force of gravity as

$$F = ma$$

where $a = dv/dt$, and $v = d(\text{displacement})/dt = dD/dt$. We can use this equation to construct a gravitational field to how any object at any point will have a certain force on them represented by a vector pointing downwards times the mass of the object, but this "global solution" has a lot more information than is needed. More generally, it is often easier to come up with the "localizations", since they are easier to define or control most of the time.

Also, after seeing how heat equations and diffusion equations are described I have a new motivation: I wonder if at some point these techniques of relating "local information" in describing how it's dynamics evolve (in physics, it would be how it changes in time, in more general scenarios it can be a recursive function or a sequence) will at some point lend itself useful in solving more general problems. For example, let's say we have a particularly complicated mathematical object. Let's say we can solve it in some nice special cases, perhaps with some restricting assumptions. Can we then step-by-step evolve the problem (or model an evolution of the problem) to eliminate the restrictions? Can we localize in some way to simplify it, and glue that information together? These feel like modelling something like a PDE.

As a final motivator, every differential form has a "canonical representative" in that there always exists representative ω such that ω vanishes under the Laplace an: $\Delta\omega = 0$ (functions, and more generally forms, that vanishes under the Laplace an are called *harmonic*). This is the beginning of *Hodge Theory*, which is central in studying the cohomology groups of smooth manifolds. In the same vein as geometry but from another perspective, PDEs are also used to model *Ricci flow*, which is about the flow of a manifold given certain parameters as it changes across time.

PDEs

Partial differential equations, often abbreviated as PDEs, let's us *relate* quantities. Another way of seeing how PDEs are different is that ODE's usually only model one variable (ex. angle of a pendulum), while a PDE can angle a whole set of points at once (ex. all the temperatures of a rod, as they change over time). For example, we may hope that the position of a particle (locally) relates to the force put on that particle (locally), and we want to do this for every particle in a system (more generally a continuum). For a more mathematical motivation, they show up more and more often in differential geometry (think of lee bracket's, lie derivative, and geometric analysis), they let us study D-modules (modules over the ring of differential operators, which tries to find aglebraic properties of differential operators), and geometry topology (PDEs were important in defining Ricci flow to show that any loop on a locally $\mathbb{R}^3$ space which can contract to a point is in fact a 3-sphere).

There is no unified study of PDEs, instead there is an amalgamation of "models" or "problems" that were of interest that are represented as a PDE. In the most abstract form, PDEs are smooth functions $f : U \to \mathbb{R}$ for a set $U \subseteq \mathbb{R}^n$ (usually with some extra conditions like being compact or open) solving a function:

$$F\left(\vec{x}, f(\vec{x}), \frac{\partial f}{\partial x_1}, ...\right) = 0 \tag{2}$$

where F takes takes all possible partials of f. Restrictions on F are usually that it is a polynomials combination or a trigonometric combination, however this is not necessarily the case. It is often the case that the solution function is denoted by u. Example of PDEs include:

- transport equation: Models how something move's across time:

$$u_t + cu_x = 0$$

- wave equation: Models how a "wave" moves across time. Exapmles of wave are "literal" water waves, gravitaional waves, sound waves, physical waves (ex. a string or membrane vibrating), and so forth. What characterizes a wave is that the more "curved" any part of the wave is (i.e. the more concave it is), the more it accelerrates. To capture the curve-ness, the term $\frac{\partial^2}{\partial x^2}$ is

used (which you can think of as representing the average), while the acceleration is captured by $\frac{\partial^2}{\partial t^2}$. Hence put together, we get:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} \quad \Leftrightarrow \quad \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$$

if we add more dimensions, we would write:

$$\frac{\partial^2 u}{\partial t^2} - \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = \frac{\partial^2 u}{\partial t^2} - \triangle u = 0$$

where $\triangle = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$ is the Laplace operator. A square symbol is often used to denote the wave operator:

$$\square = \frac{\partial^2}{\partial t^2} - \triangle$$

Hence, a function $u(t, \mathbf{x})$ would represent a [homogeneous] wave equation if it satisfies:

$$\square u = 0$$

- heat equation: The heat equation, usually written (in 1 dimension) as $\frac{\partial}{\partial t} = \frac{\partial^2}{\partial x^2}$ represents how heat naturally averages over time. In higher dimensions, this is often written as:

  $$u_t = k \triangle u$$

  for some constant k and the *Laplace operator*: $\triangle u = \sum_i \frac{\partial^2 u}{\partial x_i^2}$

- Maxwell's equation: These are a more complicated version of the wave equations to take into account some interaction of the electromagnetic force:

  $$\begin{aligned} \nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

  where E and B represent the *electric* and *magnetic* fields.

- Other famous equations note remembering are: Schrödinger's equation, Navier-Stokes equation, Dirac Equations, Yang-Mills equations, Einsteins General Relativity equations, and the Black-Scholes equation.

As it stands, the set of solutions for such equations can be more general than we want. For example, when solving a heat equation or a wave equation, a moment in time and a finite area is usually considered. Thus, when solving a PDE, it usually comes with a set of questions and restrictions. This set along with the equation is usually called a *Problem*. The two most types of problems considered in this book are:

1. *Initial Value Problem (IVP)*: One of the variables is considered a variable of time, and we usually have an *initial state* at a certain specified time t_0:

$$u|_{t_0} = u_0$$

for example, if u is modelling the temperature in a rod, then at time t_0, half the rod may be at 100 degrees, while the other half at 0 degrees. This information would be captured in u_0.

2. *Boundary Value Problem*: If U is the domain of f, then a known function is imposed on this domain:

$$u|_{\partial U} = \varphi$$

where φ is a known (smooth) function defined on ∂U.

These two can also be combined, the resulting problem is sometimes called a *mixed problem* due to the prevalence of combining these two types of problems. When finding a solution to a problem, we often would want the function (or functions in the case of a system of PDEs) to satisfy the following:

1. *Existence*: Given the restrictions (ex. choices of boundary conditions or choices of time), there exists a solution.
2. *Uniqueness*: Given a specific choice of restrictions, there exists a unique solution
3. *Continuity*: Given small variations of the choices of conditions, there still exists a solution, and it varies continuously with the data (the solution map from data to solutions is continuous[3])

What "continuously" means here can depend: It may be that the desired solution varies continuously, smoothly, or any condition in between. A problem that satisfies all three of the above conditions is called a *well-posed* problem. Most problems that will be studied will be well-posed, however there will exist *ill-posed* problems when considering a type of problem called the *inverse problem* later in this book.

Classification of PDEs

As we saw, PDE's are very general and are not well suited for classifications. However, there are some broad categories that they may be put into that helps with understanding how to approach them. A *linear* PDE is one where F in eq. (2) is a linear transformation:

$$L\left(\vec{x}, f(\vec{x}), \frac{\partial f}{\partial x_1}, ...\right) = g(x) \tag{3}$$

if $g(x) \equiv 0$, then the above is called a *homogeneous* linear PDE. If $g(x) \neq 0$, then it is called an *inhomogeneous* PDE. Note that the coefficients of the linear combination may be *functions* (consider this a $C^\infty(U)$-module). Linear PDEs are the simplest type: sums and multiples of solutions of a homogeneous linear PDE are again solutions, so solutions can be built by superposition, for instance from Fourier modes when the coefficients are constant. Nonlinear PDEs are a more complicated story, and the following classes of nonlinear equations keep part of the linear structure.

[3] In the natural topologies imposed on these function spaces, see a course in functional analysis or an advanced analysis course for more details

Semi-linear equations are those in which the highest-order derivatives appear linearly, with coefficients that depend only on the independent variables. For example, a general 2nd order semi-linear PDE in two variables[4] would be:

$$a_1(x,y)u_{xx} + a_2(x,y)u_{xy} + a_3(x,y)u_{yx} + a_4(x,y)u_{yy} + f(u_x, u_y, x, y, u) = 0$$

Note that the lower-order terms have no linearity restrictions, and hence are put within the general function f. A *quasi-linear* PDE extends this concept: the highest-order derivatives still appear linearly, but their coefficients may also depend on u and its lower-order derivatives:

$$a_1(u_x, u_y, x, y, u)u_{xx} + a_2(u_x, u_y, x, y, u)u_{xy} + a_3(u_x, u_y, x, y, u)u_{yx} + a_4(u_x, u_y, x, y, u)u_{yy} + f(u_x, u_y, x, y, u) = 0$$

A semi-linear equation in which f is affine in (u, u_x, u_y) is a *linear equation*. The wave and heat equations above are linear, while the later chapters study nonlinear wave equations.

[4] Do not confuse the dimension of the problem with the order of the PDE. There *can* be more or fewer dimensions, hence variables.

1

1-dimensional PDEs

The goal of this chapter is to be able to solve basic homogeneous wave equations of the form $\Box u = 0$. This will build up bit by bit, form the simplest case of a one-dimensional standing wave in a "perfect" system, increasingly more complex and "realistic" situations.

1.1 1st order

Consider:

$$au_t + bu_x = 0$$

where a and b are smooth functions. Physically, the left hand side may be interpreted as the derivative of the vector-field $\ell = (a, b)$. The integral lines of this field (lines representing the "flow")would be of the form:

$$\frac{dt}{a} = \frac{dx}{b}$$

Constant variables

Starting with the toy-case of a, b, being constant, then we have the integral liens:

$$\frac{t}{a} - \frac{x}{b} = C$$

for some constant C. This makes solving for u particularity easy: given any smooth function φ:

$$u = \varphi\left(\frac{t}{a} - \frac{x}{b}\right)$$

This is considered the *general solution*. If we consider a IVP where $u|_{t=0} = f$. Then

$$\varphi(-x/b) = f(x) \quad \Rightarrow \varphi(x) = f(-bx)$$

giving us:

$$u = f(x - ct) \qquad c = b/a$$

which solves:

$$\begin{cases} au_t + bu_x = 0 \\ u(x,0) = f(x) \end{cases}$$

such a solution is called a *running wave*, due to the physical interpretation of this equation.

Let us now develop for more general cases. Some care must now be taken to insure our problem is *well-posed*

Example 1.1.1: Ill-posed Problem

Consider $u_t + tu_x = 0$. This equation has integral curves

$$tdt - dx = 0$$

which, solving by using the method of separable variables, we get:

$$x - \frac{1}{2}t^2 = C$$

Hence, we see that $u = \varphi(x - 1/2t^2)$ is a solution to this PDE. Notice that the input must be symmetrical across the t axis. This shall give some important constraints for possible IVP.

Consider this equation with IVP $\varphi(0 - 1/2t^2) = u(0,t) = g(t)$, that is we want the wave to follow the path $g(t)$ at point $x = 0$. Namely, we have $\varphi(-1/2t^2) = g(t)$, meaning g *must* be an even function of t, or this equation is not solvable. Notice that with a badly picked g (ex. g is odd with respect to t), this is not possible. Furthermore, even if we limit to even function or functions where $t > 0$, then there is no restriction on all points $x > 1/2t^2$, making us lack uniqueness for the solution.

Hence, more care must now be put with choosing our IVP. That being said, let us also consider the non-homogeneous case:

$$au_t + bu_x + f(x,t,u) = 0 \qquad \Leftrightarrow \qquad au_t + bu_x = f(x,t,u)$$

where the negative is incorporated into f on the right hand side. Then the integral curves follow:

$$du = u_t dt + u_x dx = (au_t + bu_x)\frac{dt}{a} = f\frac{dt}{a}$$

Which reduces our problem to:

$$\frac{dt}{a} = \frac{dx}{b} = \frac{du}{f}$$

Example 1.1.2: Non-homogeneous Solutions

1. Take $u_t + u_x = x$. This gives
$$dt = dx = \frac{du}{x}$$
The first equality gives $x - t = C$, and the second gives $du = x\,dx$, so $u - \frac{1}{2}x^2 = D$ along each integral curve. As D may vary with the curve, that is with $x - t$, the general solution is
$$u = \frac{1}{2}x^2 + \varphi(x - t)$$
for a smooth function φ. If the initial condition is $u|_{t=0} = 0$, then $\frac{1}{2}x^2 + \varphi(x) = 0$, so $\varphi(y) = -\frac{1}{2}y^2$ and
$$u = \frac{1}{2}x^2 - \frac{1}{2}(x - t)^2 = xt - \frac{1}{2}t^2$$
Indeed $u_t = x - t$ and $u_x = t$, so $u_t + u_x = x$ and $u|_{t=0} = 0$.

2. Take $u_t + xu_x = xt$. This gives:
$$\frac{dt}{1} = \frac{dx}{x} = \frac{du}{xt}$$
The first equality gives $x = Ce^t$ with $C = xe^{-t}$ constant along each integral curve. Along such a curve the second equality reads $\frac{du}{dt} = xt = Cte^t$, so $u = C(t-1)e^t + D = x(t-1) + D$ with D constant on the curve. The general solution is therefore
$$u = x(t - 1) + \varphi(xe^{-t})$$
Indeed $u_t = x - xe^{-t}\varphi'(xe^{-t})$ and $xu_x = x(t - 1) + xe^{-t}\varphi'(xe^{-t})$, which add up to xt.

Semi-Linear

Let us now consider the case where $a = a(x, t)$ and $b = (x, t)$ so that we are working with semi-linear 1st order equations. We it is also the case that $f = c(x, t)u + g(x, y)$ (that is, f is a linear combination of u with a constant[1]), then the equation is called linear.

Example 1.1.3: Semi-Linear and Linear Equations

1. Take $u_t + xu_x = u$ so that the integral curves are:
$$\frac{dt}{1} = \frac{dx}{x} = \frac{du}{u}$$
Taking the first two equations in this equality, we can solve it using the method of variable separation to get:
$$x = Ce^t$$
Taking the first and third, we get:
$$\frac{du}{u} = dt \Rightarrow u = De^t$$

[1] Recall our constants are usually smooth functions

as $C = xe^{-t}$ is a constant, we get:

$$u = \varphi(xe^{-t})e^{t}$$

form which we may impose an IVP problem, for example if $u|_{t=0} = x^2$ we would get $u = x^2e^{-t}$.

2. Consider

Quasi-Linear

In the 1-dimensional first order case, this means that the coefficients of u_t and u_x now also permit u as a free-variable. The general equation for integral curves becomes:

$$\frac{dt}{a} = \frac{dx}{b} = \frac{du}{f}$$

These may present more problems when looking for solutions:

Example 1.1.4: Quasi-linear Trouble
Consider:

$$u_t + uu_x = 0$$

Giving

$$\frac{dt}{1} = \frac{dx}{u} = \frac{du}{0}$$

The last equation is a shorthand for saying that u is constant: $u = C$. From the other equation, we get:

$$ut - x = D \tag{1.1}$$

We now have two equations with constants, which is enough to piece together the general solution, namely since C, D are the constants we get that $u = C = \varphi(D) = \varphi(ut - x)$:

$$u = \varphi(ut - x)$$

However, note that this is defined *implicitly* (notice the repetition of u). By the implicit function theorem, we may solve for u where

$$\frac{\partial}{\partial u}(u - \varphi(ut - x)) = 1 - t\varphi'(ut - x) \neq 0$$

If $\varphi' \leq 0$, this derivative is at least 1 for every $t > 0$, while if $\varphi'(s) > 0$ for some s, it vanishes at $t = 1/\varphi'(s)$. For $t < 0$ the derivative $1 - t\varphi'$ is at least 1 when $\varphi' \geq 0$, and it vanishes at $t = 1/\varphi'(s)$ when $\varphi'(s) < 0$. In terms of the profile $\psi(s) = \varphi(-s)$, for which $u = \psi(x - ut)$ and $u(x, 0) = \psi(x)$, solvability for all $t > 0$ requires ψ to be nondecreasing.

Initial Boundary Value Problem (IBVP)

Let us motivate this by going back to the transport equation but change up the restrictions:

$$\begin{cases} u_t + cu_x = 0 & x > 0,\ t > 0 \\ u|_{t=0} = f(x) & x > 0 \end{cases}$$

The general solution was found to be $u = \varphi(x - ct)$, and when plugging the initial data we get:

$$\varphi(x) = f(x)$$

as $x > 0$. Now, as f is only defined when $x > 0$, we have that u is defined when $x - ct > 0$. If c is negative (the wave propagates to the left) then we have no issue. However, if the wave propagates to the right, the wave shall run into a problem when $x < ct$. In this case, there is an extra condition we put to the *border* of our problem:

$$\begin{cases} u_t + cu_x = 0 & x > 0,\ t > 0 \\ u|_{t=0} = f(x) & x > 0 \\ u|_{x=0} = g(t) & t > 0 \end{cases}$$

This gives us that $u(0,t) = \varphi(-ct) = g(t)$ as $t > 0$, setting $x = -ct$ we get $\varphi(x) = g(\frac{-1}{c}x)$ when $x < 0$. Thus, we get that $u(x,t) = g(\frac{-1}{c}(x - ct)) = g(t - \frac{1}{c}x)$ as $x < ct$, giving a general solution of:

$$u = \begin{cases} f(x - ct) & x > ct \\ g(t - \frac{1}{c}x) & x < ct \end{cases}$$

Note that if $f(0) \neq g(0)$, we get a discontinuity (or no solution) along the line $x = ct$. There is still a way in which there is a valid solution, but it will have to wait for the notion of *Distribution*.

1.2 Multi-Dimensional

Let us now consider the same 1st order equations, but in the multidimensional case. The strategy is the same as in one dimension.

Example 1.2.1: Simple Example

1. Consider
$$u_t + 2tu_x + xu_y = 0$$
This gives:
$$\frac{dt}{1} = \frac{dx}{2t} = \frac{dy}{x} = \frac{du}{0}$$
From the first equation, we get $x - t^2 = C$. Taking $dt/1 = dy/x$ and substituting $x = C + t^2$, we get:
$$y - Ct - \frac{t^3}{3} = D$$

Eliminating the constant C, we get:

$$D = y - (x - t^2)t - \frac{t^3}{3}$$

Having found the integral lines, we may finish off:

$$u(x, y, t) = \varphi(x - t^2, y - xt + \frac{2t^3}{3})$$

2. Consider

$$u_t + 2tu_x + xu_y = 12xtu$$

This gives:

$$\frac{dt}{1} = \frac{dx}{2t} = \frac{dy}{x} = \frac{du}{12xtu}$$

The first two equations give, as before,

$$x = t^2 + C \qquad y = Ct + \frac{t^3}{3} + D$$

Then:

$$\begin{aligned} \frac{du}{u} &= 12xt\,dt = (12t^3 + 12Ct)\,dt \\ \Rightarrow \ln(u) &= \ln(E) + 3t^4 + 6Ct^2 \\ \Rightarrow u &= E\exp(3t^4 + 6Ct^2) \\ \Rightarrow u &= E\exp(6xt^2 - 3t^4) \end{aligned}$$

where the last line substitutes $C = x - t^2$, and E is constant along each characteristic, hence a function of the two invariants C and D. Hence, we get:

$$u = \exp(6xt^2 - 3t^4)\varphi(x - t^2, y - xt + \frac{2t^3}{3})$$

2

wave equations

The one-dimensional wave equation $\partial_t^2 u = \partial_x^2 u$ equates two second derivatives.[1] For u of class C^2, Taylor's theorem gives

$$\frac{1}{2}\big(u(t,x+h)+u(t,x-h)\big)-u(t,x)=\frac{h^2}{2}\partial_x^2 u(t,x)+o(h^2)\qquad h\to 0$$

so $\partial_x^2 u(t,x)$ measures how far the average of $u(t,\cdot)$ over the neighboring points $x\pm h$ exceeds $u(t,x)$, that is, the concavity of the graph of $u(t,\cdot)$ at x. The second time derivative $\partial_t^2 u(t,x)$ is the acceleration of the graph at x. The wave equation therefore says that each point of the graph accelerates towards the average of its neighbors, with acceleration equal to the concavity there. In higher dimensions the second derivative in x becomes the Laplacian, and we study the *wave operator*

$$\Box=\frac{\partial^2}{\partial t^2}-\sum_{i=1}^{n}\frac{\partial^2}{\partial x_i^2}=\frac{\partial^2}{\partial t^2}-\triangle$$

2.1 D'Alembert's Formula

We treat the one-dimensional case first:

$$\partial_t^2 u-\partial_x^2 u=0 \tag{2.1}$$

with the *initial value problem* (IVP), also known as the *Cauchy problem*,

$$u(0,x)=f(x)\qquad \partial_t u(0,x)=g(x)$$

[1] A visual explanation of why second derivatives appear is at https://youtu.be/ly4S0oi3Yz8?t=428.

for given data f and g. Suppose first that $u \in C^2(\mathbb{R}^2)$ solves this problem, and change variables to

$$r := x + t \qquad s := x - t$$

By the chain rule $\partial_t = \partial_r - \partial_s$ and $\partial_x = \partial_r + \partial_s$, so

$$\partial_t + \partial_x = 2\partial_r \qquad \partial_t - \partial_x = -2\partial_s$$

and eq. (2.1) becomes

$$(\partial_t + \partial_x)(\partial_t - \partial_x)u = -4\partial_r\partial_s u = 0$$

Hence $\partial_r u$ does not depend on s, that is, $\partial_r u = v(r)$ for a continuous function v. Integrating in r, with an integration constant $W(s)$ that may depend on s, gives

$$u = \int_0^r v(r')\,dr' + W(s) =: V(r) + W(s)$$

and in the original variables

$$u(t,x) = V(x+t) + W(x-t) \tag{2.2}$$

which decomposes u into a left-moving component $V(x+t)$ and a right-moving component $W(x-t)$. To express V and W through the data, evaluate eq. (2.2) and its time derivative at $t = 0$:

$$\begin{aligned} V(x) + W(x) &= f(x) \\ V'(x) - W'(x) &= g(x) \end{aligned}$$

Differentiating the first line in x and solving for V' and W' gives

$$V'(x) = \frac{1}{2}f'(x) + \frac{1}{2}g(x) \qquad W'(x) = \frac{1}{2}f'(x) - \frac{1}{2}g(x)$$

and integrating,

$$\begin{aligned} V(x) &= \frac{1}{2}f(x) + \frac{1}{2}\int_0^x g(x')\,dx' + c_1 \\ W(x) &= \frac{1}{2}f(x) - \frac{1}{2}\int_0^x g(x')\,dx' + c_2 \end{aligned}$$

The relation $V + W = f$ forces $c_1 + c_2 = 0$, and substituting into eq. (2.2) gives

$$u(t,x) = \frac{1}{2}\left(f(x+t) + f(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(x')\,dx' \tag{2.3}$$

This is *d'Alembert's formula* for solutions of the one-dimensional wave equation. Conversely, if $f \in C^2(\mathbb{R})$ and $g \in C^1(\mathbb{R})$, the right hand side of eq. (2.3) is a C^2 function solving the Cauchy problem: its second derivatives in t and in x both equal $\frac{1}{2}\big(f''(x+t) + f''(x-t) + g'(x+t) - g'(x-t)\big)$, and at $t = 0$ it equals f, with time derivative g.

The right hand side of eq. (2.3) makes sense for continuous f and g, and even for $f, g \in L^1_{\mathrm{loc}}(\mathbb{R})$, with u then defined almost everywhere. Such a u need not be C^2, so it is not a solution in the classical sense. It is a *weak solution*, a notion developed in chapter 3.

For a point (t_0, x_0), the set $\{(t,x) \mid |x - x_0| = |t - t_0|\}$ is the *light cone* with vertex (t_0, x_0), and its part with $t \le t_0$ is the *backwards light cone*. By d'Alembert's formula, the solution $u(t,x)$ depends

on the initial data (f, g) only through their values at points x' with $|x' - x| \le |t|$, that is, on the segment between the two points $x \pm t$ where the light cone of (t, x) meets the line $\{t = 0\}$. Figure 2.1 draws, for $t > 0$, the triangle that the backwards light cone of (t, x) encloses above this segment.

This property is the principle of *finite propagation speed*. In particular, if f, g have compact support within some interval $[-R, R]$, then

$$\operatorname{supp}(u(t, \cdot)) \subseteq [-R - |t|, R + |t|]$$

because for $|x| > R + |t|$ the points $x \pm t$ and the interval between them lie outside $[-R, R]$, so both terms of d'Alembert's formula vanish. The information carried by the initial data in $[-R, R]$ travels with speed at most 1, the slope of the light cone, and at time t it has not propagated beyond $[-(R + |t|), R + |t|]$.

2.2 Duhamel's Principle

We next solve the *inhomogeneous* one-dimensional wave equation

$$\partial_t^2 u - \partial_x^2 u = h(t, x) \tag{2.4}$$

with initial data

$$u(0, x) = f(x), \quad \partial_t u(0, x) = g(x)$$

where $f \in C^2(\mathbb{R})$, $g \in C^1(\mathbb{R})$, and h and $\partial_x h$ are continuous on $\mathbb{R}^2$. For $h = 0$, d'Alembert's formula gives a solution u_1. For $f = g = 0$, the construction below, *Duhamel's principle*, gives a solution u_2. Since eq. (2.4) is linear, $u = u_1 + u_2$ then solves the full problem.

For $\varphi \in C^1(\mathbb{R})$ write

$$W(t)\varphi(x) = \frac{1}{2} \int_{x-t}^{x+t} \varphi(y)\, dy$$

By eq. (2.3), $(t, x) \mapsto W(t)\varphi(x)$ is the solution of the homogeneous equation with data $u(0, \cdot) = 0$ and $\partial_t u(0, \cdot) = \varphi$, and we call $W(t)$ the *propagation operator* or *solution operator*. Its time derivative

$$\partial_t W(t)\varphi(x) = \frac{1}{2}\big(\varphi(x + t) + \varphi(x - t)\big)$$

satisfies $\partial_t W(0)\varphi = \varphi$, and for $\varphi \in C^2(\mathbb{R})$ it is the solution with data $(\varphi, 0)$. So d'Alembert's formula reads

$$u_1(t, x) = \partial_t W(t) f(x) + W(t) g(x)$$

For the inhomogeneous term, set

$$u_2(t, x) = \int_0^t W(t - s)\big(h(s, \cdot)\big)(x)\, ds = \frac{1}{2} \int_0^t \int_{x-(t-s)}^{x+(t-s)} h(s, y)\, dy\, ds$$

To check that u_2 solves eq. (2.4) with zero data, we differentiate under the integral sign with the rule

$$\frac{d}{dt} \int_0^t \psi(t, s)\, ds = \psi(t, t) + \int_0^t \partial_t \psi(t, s)\, ds \tag{2.5}$$

valid for continuous ψ on $\mathbb{R}^2$ with continuous $\partial_t \psi$. Indeed, for $\Phi(a, b) = \int_0^a \psi(b, s)\, ds$ the fundamental theorem of calculus gives $\partial_a \Phi(a, b) = \psi(b, a)$, and [8, Leibniz Integral Rule] gives $\partial_b \Phi(a, b) =$

$\int_0^a \partial_b \psi(b,s)\,ds$. Both are continuous, so Φ is C^1, and the chain rule applied to $t \mapsto \Phi(t,t)$ gives eq. (2.5). We use it together with $W(0) = 0$, $\partial_t W(0)\varphi = \varphi$ and $\partial_t^2 W(t)\varphi = \partial_x^2 W(t)\varphi$, where $\partial_t W(t-s)$ and $\partial_t^2 W(t-s)$ denote the time derivatives of W evaluated at time $t-s$:

$$\partial_t u_2(t,x) = W(0)\big(h(t,\cdot)\big)(x) + \int_0^t \partial_t W(t-s)\big(h(s,\cdot)\big)(x)\,ds = \int_0^t \partial_t W(t-s)\big(h(s,\cdot)\big)(x)\,ds$$
$$\partial_t^2 u_2(t,x) = \partial_t W(0)\big(h(t,\cdot)\big)(x) + \int_0^t \partial_t^2 W(t-s)\big(h(s,\cdot)\big)(x)\,ds = h(t,x) + \partial_x^2 u_2(t,x)$$

The last step differentiates twice in x under the integral, which is allowed because $\partial_x W(t-s)\big(h(s,\cdot)\big)(x) = \frac{1}{2}\big(h(s,x+t-s) - h(s,x-t+s)\big)$ and its derivative in x are continuous in (s,x). The same rule shows that $\partial_t \partial_x u_2$ is continuous, so $u_2 \in C^2(\mathbb{R}^2)$, and at $t=0$ both u_2 and $\partial_t u_2$ vanish. Adding u_1 and u_2 gives the solution of the full problem,

$$u(t,x) = \frac{1}{2}\big(f(x+t) + f(x-t)\big) + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy + \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} h(s,y)\,dy\,ds$$

For $t > 0$ the double integral runs over the triangle enclosed by the backwards light cone of (t,x) above the line $\{t=0\}$, drawn in fig. 2.1.

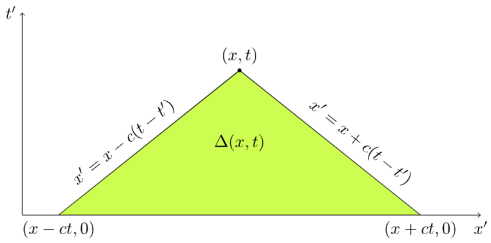


Figure 2.1: The backwards light cone of (t,x) and the triangle $\Delta(x,t)$ it encloses, drawn for propagation speed c, which is 1 here

2.2.1 Boundary Problems

A wave equation on a domain with boundary also needs *boundary conditions*, which turn the Cauchy problem into an *initial boundary value problem* (IBVP). On the half line $x \geq 0$ the two basic boundary conditions are the *Dirichlet condition*

$$u(t,0) = 0$$

and the *Neumann condition*

$$\partial_x u(t,0) = 0$$

For the Dirichlet problem with data $f \in C^2([0,\infty))$ and $g \in C^1([0,\infty))$, compatibility with the boundary condition at $t=0$ requires $f(0) = g(0) = 0$. Extend f and g to odd functions on $\mathbb{R}$ by $f(-x) = -f(x)$ and $g(-x) = -g(x)$. D'Alembert's formula for the extended data gives

$$u(t,x) = \frac{1}{2}\left(f(x+t) + f(x-t)\right) + \frac{1}{2}\int_{x-t}^{x+t} g(x')\,dx'$$

which is odd in x because f and g are, so $u(t,0)=0$ for all t. If moreover $f''(0)=0$, the extended f is C^2 and the extended g is C^1, so u is a C^2 solution of the IBVP. For $t\geq 0$ and $x\geq t$ the formula only involves f and g at nonnegative arguments. For $0\leq x<t$ it involves negative arguments, and oddness, in the form $f(x-t)=-f(t-x)$ and $\int_{x-t}^{t-x} g=0$, rewrites it as

$$u(t,x)=\frac{1}{2}\left(f(t+x)-f(t-x)\right)+\frac{1}{2}\int_{t-x}^{t+x} g(x')\,dx'$$

So for $t\geq 0$ the solution only depends on the values of f and g on $[0,\infty)$. The Neumann problem is solved in the same way with the even extensions $f(-x)=f(x)$ and $g(-x)=g(x)$. The resulting u is even in x, and when $f'(0)=g'(0)=0$ the extended data are C^2 and C^1, so u is a C^2 solution with $\partial_x u(t,0)=0$.

Example 2.2.1: Wave Equation Solutions

Consider the problem

$$\begin{cases} u_{tt}-9u_{xx}=0 & t\in\mathbb{R},\ x>-t \\ u|_{t=0}=4\cos(x) & x>0 \\ u_t|_{t=0}=18\sin(x) & x>0 \\ u_x|_{x=-t}=0 & t\in\mathbb{R} \end{cases}$$

on the region to the right of the line $x=-t$, which moves with speed 1, below the wave speed 3. Following d'Alembert's formula, write

$$u(t,x)=\varphi(x+3t)+\psi(x-3t)$$

The initial conditions give $\varphi+\psi=4\cos$ and $3(\varphi'-\psi')=18\sin$ on $(0,\infty)$, so, after moving a constant from ψ to φ,

$$\begin{aligned} \varphi(x)+\psi(x)&=4\cos(x) && x>0 \\ \varphi(x)-\psi(x)&=-6\cos(x) && x>0 \\ \Rightarrow \varphi(x)=-\cos(x)\qquad \psi(x)&=5\cos(x) && x>0 \end{aligned}$$

Both arguments $x\pm 3t$ are positive exactly when $x>3|t|$, and there

$$u(t,x)=-\cos(x+3t)+5\cos(x-3t)\qquad x>3|t|$$

For $t>0$ the argument $x+3t>2t$ stays positive, while $x-3t$ takes the negative values in $(-4t,0)$, where ψ is not yet known. The boundary condition determines it: $u_x=\varphi'(x+3t)+\psi'(x-3t)$, so at $x=-t$

$$\varphi'(2t)+\psi'(-4t)=0$$

With $\varphi'(2t)=\sin(2t)$ this gives $\psi'(-4t)=-\sin(2t)$, that is, $\psi'(y)=\sin(y/2)$ for $y<0$. So $\psi(y)=-2\cos(y/2)+C$, and continuity of ψ at 0, where $\psi(0)=5$, forces $C=7$. Hence for $t>0$ and $-t<x<3t$,

$$u(t,x)=-\cos(x+3t)-2\cos\left(\frac{x-3t}{2}\right)+7$$

For $t<0$ the roles are exchanged: $x-3t>0$ in the region, and φ is unknown on $(2t,0)$. The boundary condition $\varphi'(2t)=-\psi'(-4t)=-5\sin(4t)$ gives $\varphi'(y)=-5\sin(2y)$ for $y<0$, so $\varphi(y)=\frac{5}{2}\cos(2y)-\frac{7}{2}$ by continuity at 0, where $\varphi(0)=-1$. Hence for $t<0$ and $-t<x<-3t$,

$$u(t,x)=\frac{5}{2}\cos(2x+6t)-\frac{7}{2}+5\cos(x-3t)$$

The formulas in adjacent regions agree, together with their first derivatives, along the lines $x = 3t$ and $x = -3t$, while their second derivatives jump there. So u is C^1, and it is C^2 and solves the equation away from these two lines. Figure 2.2 shows the three regions.

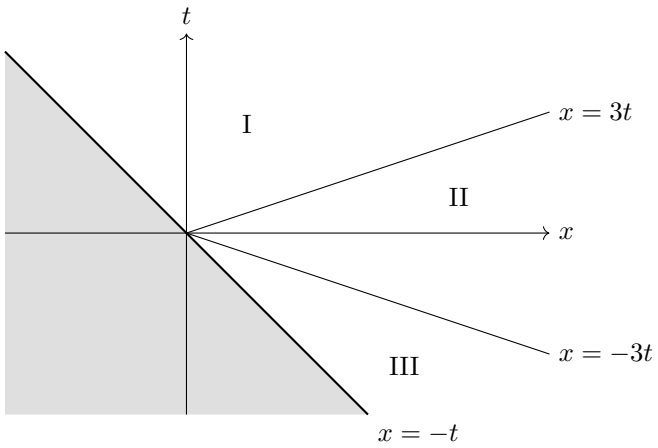


Figure 2.2: The regions of example 2.2.1: $u = -\cos(x+3t) - 2\cos(\frac{x-3t}{2}) + 7$ in I, $u = -\cos(x+3t) + 5\cos(x-3t)$ in II, and $u = \frac{5}{2}\cos(2x+6t) - \frac{7}{2} + 5\cos(x-3t)$ in III. The shaded half-plane $x < -t$ lies outside the domain

Animations of the reflection of a wave at the boundary under Dirichlet, Neumann and Robin conditions are on the half-line visualization page of Ivrii's online PDE textbook.

2.2.2 Separation of Variables

Consider the equation on the interval $(0, 2\pi)$

$$\partial_t^2 u - \partial_x^2 u = 0 \qquad x \in (0, 2\pi)$$

with Dirichlet boundary conditions

$$u(t, 0) = 0 = u(t, 2\pi)$$

We look for special solutions of the form $u(t, x) = v(t)w(x)$, in which the time and space variables separate. For such u the equation reads

$$w\,\partial_t^2 v = v\,\partial_x^2 w$$

and dividing by vw where it does not vanish gives

$$\frac{\partial_t^2 v}{v} = \frac{\partial_x^2 w}{w}$$

The left hand side depends only on t and the right hand side only on x, so both equal a constant λ:

$$\frac{\partial_t^2 v}{v} = \frac{\partial_x^2 w}{w} = \lambda$$

The problem splits into two ordinary differential equations. The *eigenvalue problem* for the pair (w, λ),

$$\partial_x^2 w = \lambda w \qquad w(0) = 0 = w(2\pi)$$

encodes the boundary conditions, and for each eigenvalue λ the *evolution problem*

$$\frac{d^2}{dt^2} v = \lambda v$$

with initial data $v(0) = a$ and $\dot{v}(0) = b$ encodes the initial conditions. For $\lambda \geq 0$ the boundary conditions force $w = 0$, and for $\lambda = -\mu^2$ with $\mu > 0$ they force $w(x) = B \sin(\mu x)$ with $\sin(2\pi\mu) = 0$, that is, $2\mu \in \mathbb{N}_{>0}$. So the eigenvalues and the normalized eigenfunctions are

$$\lambda_k = -\frac{k^2}{4} \qquad w_k(x) = \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2} k x\right) \qquad k \in \mathbb{N}_{>0}$$

For $\lambda = \lambda_k$ the evolution problem with data (a_k, b_k) has the solution

$$v_k(t) = a_k \cos(\omega_k t) + b_k \frac{\sin(\omega_k t)}{\omega_k}$$

with temporal frequency $\omega_k = \sqrt{-\lambda_k} = \frac{1}{2} k$.

Theorem 2.2.2: General Solution To Separation Of Variables

For $k \geq 1$ let $\varphi_k(x) = \frac{1}{\sqrt{\pi}} \sin\left(\frac{1}{2} k x\right)$ and $\omega_k = \frac{1}{2} k$.

1. The functions φ_k, $k \geq 1$, form an orthonormal basis of $L^2(0, 2\pi)$.
2. Let (a_k) and (b_k) be real sequences with $\sum_k k^2(|a_k| + |b_k|) < \infty$. Then the series

$$u(t, x) = \sum_{k=1}^{\infty} \left(a_k \cos(\omega_k t) + b_k \frac{\sin(\omega_k t)}{\omega_k} \right) \varphi_k(x)$$

converges uniformly on $\mathbb{R} \times [0, 2\pi]$ together with its partial derivatives of order at most 2, and $u \in C^2(\mathbb{R} \times [0, 2\pi])$ solves the initial-boundary value problem above with $u(0, \cdot) = \sum_k a_k \varphi_k$ and $\partial_t u(0, \cdot) = \sum_k b_k \varphi_k$.

***Proof*:**

1. For $k, l \geq 1$, $2 \sin(\frac{kx}{2}) \sin(\frac{lx}{2}) = \cos(\frac{(k-l)x}{2}) - \cos(\frac{(k+l)x}{2})$, and $\int_0^{2\pi} \cos(\frac{mx}{2})\, dx$ is 2π for $m = 0$ and $\frac{2}{m} \sin(m\pi) = 0$ for every other integer m, so $\langle \varphi_k, \varphi_l \rangle$ is 1 if $k = l$ and 0 otherwise. For completeness, let $f \in L^2(0, 2\pi)$ with $\langle f, \varphi_k \rangle = 0$ for all $k \geq 1$, extend f to an odd function F on $(-2\pi, 2\pi)$, and put $G(y) = F(4\pi y - 2\pi)$ for $0 \leq y < 1$, a function in $L^2(\mathbb{T})$. For $m \in \mathbb{Z}$,

$$\int_0^1 G(y) e^{-2\pi i m y}\, dy = \frac{e^{-i\pi m}}{4\pi} \int_{-2\pi}^{2\pi} F(x) e^{-imx/2}\, dx = \frac{-ie^{-i\pi m}}{2\pi} \int_0^{2\pi} f(x) \sin\left(\frac{mx}{2}\right) dx = 0$$

because F is odd and $\sin(\frac{mx}{2}) = \pm\sqrt{\pi}\varphi_{|m|}(x)$ for $m \neq 0$. The exponentials $e^{2\pi i m y}$ form

an orthonormal basis of $L^2(\mathbb{T})$ [9, Orthonormality Of Translation Functions], so $G = 0$, hence $f = 0$, and the orthonormal set $\{\varphi_k \mid k \geq 1\}$ is a basis.

2. Write T_k for the k-th term of the series. Its partial derivatives of order $j \leq 2$ are sums of a product of $a_k\omega_k^j$ and of a product of $b_k\omega_k^{j-1}$ with trigonometric factors bounded by $1/\sqrt{\pi}$, and $\omega_k = \frac{k}{2}$, so all of them, and T_k itself, are bounded on $\mathbb{R} \times [0, 2\pi]$ by $M_k = \frac{2}{\sqrt{\pi}}k^2(|a_k| + |b_k|)$, and $\sum_k M_k < \infty$. By the Weierstrass M-test [2, Weierstrass M-Test], the series of the T_k and the series of each of these partial derivatives converge uniformly, and their sums are continuous by [2, Uniform Convergence Preserve's Continuity]. Term-by-term differentiation [1, Term-by-Term Differentiation], in t on bounded intervals with x fixed and in x with t fixed, shows that the partial derivatives of u of order at most 2 exist and equal the sums of the differentiated series, so $u \in C^2(\mathbb{R} \times [0, 2\pi])$. Each T_k satisfies $\partial_t^2 T_k = -\omega_k^2 T_k = \partial_x^2 T_k$ and vanishes at $x = 0$ and $x = 2\pi$, so u solves the equation with the Dirichlet boundary conditions, and evaluating at $t = 0$ gives the data, as claimed.

Given data $f, g \in L^2(0, 2\pi)$, part (1) of theorem 2.2.2 and [7, Orthonormal Basis Condition] give the coefficients: $f = \sum_k a_k\varphi_k$ and $g = \sum_k b_k\varphi_k$ in $L^2(0, 2\pi)$, with

$$a_k = \langle f, \varphi_k\rangle \qquad b_k = \langle g, \varphi_k\rangle \qquad \langle f, g\rangle = \int_0^{2\pi} f(x)\overline{g(x)}\,dx$$

If f and g are continuous and $\sum_k k^2(|a_k| + |b_k|) < \infty$, part (2) gives a C^2 solution with $u(0, \cdot) = f$ and $\partial_t u(0, \cdot) = g$, because the uniformly convergent sums $\sum_k a_k\varphi_k$ and $\sum_k b_k\varphi_k$ are continuous and agree with f and g almost everywhere.

2.3 Wave Equation in Higher Dimensions

We now study the wave equation

$$\partial_t^2 u - \triangle u = 0$$

for functions u on $\mathbb{R}^{1+n}$, where $\triangle u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}$, written $\Box u = 0$ for short.

2.3.1 Fourier Transform

For data f, g in the Schwartz class $\mathcal{S}(\mathbb{R}^n)$, take the Fourier transform in the space variables,

$$\hat{u}(t, \xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix\cdot\xi} u(t, x)\,dx$$

The equation becomes the ordinary differential equation

$$\partial_t^2 \hat{u}(t, \xi) + |\xi|^2 \hat{u}(t, \xi) = 0$$

in t for each fixed ξ, whose solutions for $\xi \neq 0$ are the linear combinations of $e^{\pm i|\xi|t}$. With $\hat{u}(0, \xi) = \hat{f}(\xi)$ and $\partial_t \hat{u}(0, \xi) = \hat{g}(\xi)$ this gives

$$\hat{u}(t, \xi) = \cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)}{|\xi|}\hat{g}(\xi)$$

where $\frac{\sin(|\xi|t)}{|\xi|}$ is read as t at $\xi = 0$, and the inverse Fourier transform gives the solution

$$u(t,x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix\cdot\xi} \left(\cos(|\xi|t)\hat{f}(\xi) + \frac{\sin(|\xi|t)}{|\xi|}\hat{g}(\xi) \right) d\xi$$

The formula for $\hat{u}(t,\cdot)$ defines an element of $L^2(\mathbb{R}^n)$ for all $f, g \in L^2(\mathbb{R}^n)$, because $|\sin(|\xi|t)/|\xi|| \leq |t|$, and hence, through the Fourier transform on $L^2(\mathbb{R}^n)$, a function $u(t,\cdot) \in L^2(\mathbb{R}^n)$. The sense in which such a u solves the equation is the subject of chapter 3.

The wave operator is invariant under *Lorentz transformations*, the linear maps Λ of $\mathbb{R}^{1+n}$ that preserve the quadratic form $t^2 - |x|^2$: by the chain rule, $\Box(u \circ \Lambda) = (\Box u) \circ \Lambda$. The vector fields $\Omega_{0i} = t\partial_i + x_i\partial_t$ generate the Lorentz boosts, and their commutation with $\Box$ is one of the inputs of chapter 6.

2.3.2 Method of Spherical Means

Definition 2.3.1: Spherical Mean

Given a function $h \in C(\mathbb{R}^n)$, its *spherical mean* centered about the point $x \in \mathbb{R}^n$ is

$$M(h)(x,r) = \frac{1}{w_n r^{n-1}} \int_{|x-y|=r} h(y) dS_y$$

that is, the average of h over the sphere of radius r centered at x.

Here w_n is the surface area of the unit sphere $S^{n-1} \subseteq \mathbb{R}^n$, so $w_n r^{n-1}$ is the area of the sphere of radius r. For $n = 1$ the sphere of radius r about x consists of the two points $x \pm r$, and the spherical mean of a function f is

$$M(f)(x,r) = \frac{1}{2}\big(f(x+r) + f(x-r)\big)$$

the first term of d'Alembert's formula. The change of variables $y = x + r\xi$ with $|\xi| = 1$ gives a second expression for the spherical mean,

$$M(h)(x,r) = \frac{1}{w_n} \int_{|\xi|=1} h(x + r\xi)\, dS_\xi \tag{2.6}$$

which defines $M(h)(x,r)$ for every real r, and the substitution $\xi \mapsto -\xi$ shows that it is even in r:

$$M(h)(x,-r) = M(h)(x,r)$$

Lemma 2.3.2: Darboux Equation

Given $h \in C^2(\mathbb{R}^n)$, for $x \in \mathbb{R}^n$ and $r > 0$,

$$\triangle_x M(h)(x,r) = (\partial_r^2 + \frac{n-1}{r}\partial_r) M(h)(x,r)$$

a formula that relates the Laplace operator of $M(h)$ in the n-dimensional x-variables to an operator involving only one-dimensional r derivatives.

***Proof*:**
First, take one derivative:

$$\begin{aligned}\partial_r M(h)(x,r) &= \frac{1}{w_n}\int_{|\xi|=1} \partial_r h(x+r\xi)dS_\xi \\ &= \frac{1}{w_n}\int_{|\xi|=1}\sum_{j=1}^{n}\partial_{x_j}h(x+r\xi)\xi_j dS_\xi\end{aligned}$$

Notice that $\sum_{j=1}^{n}\partial_{x_j}h(x+r\xi)\xi_j = \nabla h\cdot N$, the outward normal derivative of h on the unit sphere. Thus, the divergence theorem gives:

$$\begin{aligned}\frac{1}{w_n}\int_{|\xi|=1}\sum_{j=1}^{n}\partial_{x_j}h(x+r\xi)\xi_j dS_\xi &= \frac{1}{w_n}\int_{|\xi|<1} r\triangle_x h(x+r\xi)d\xi \\ &= \frac{1}{w_n r^{n-1}}\triangle_x\int_{|x-y|<r}h(y)dy \\ &= \frac{1}{w_n r^{n-1}}\triangle_x\left(\int_0^r\int_{|x-y|=\rho}h(y)dS_y d\rho\right) \\ &= \frac{1}{r^{n-1}}\triangle_x\left(\int_0^r \rho^{n-1}M(h)(x,\rho)d\rho\right)\end{aligned}$$

Next, multiplying through by r^{n-1} and taking a second derivative:

$$\begin{aligned}\partial_r\left(r^{n-1}\partial_r M(h)(x,r)\right) &= \partial_r\left(\triangle_x\int_0^r\rho^{n-1}M(h)(x,\rho)d\rho\right) \\ &= \triangle_x\left(r^{n-1}M(h)(x,r)\right)\end{aligned}$$

Thus:

$$\begin{aligned}\triangle_x M(h)(x,r) &= \frac{1}{r^{n-1}}\partial_r\left(r^{n-1}\partial_r M(h)(x,r)\right) \\ &= \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right)M(h)(x,r)\end{aligned}$$

as we sought to show.

Proposition 2.3.3: Analysis Facts

1. For $h \in C(\mathbb{R}^n)$ and $x \in \mathbb{R}^n$, the value $h(x)$ is recovered from the spherical means:
$$h(x) = \lim_{r \to 0} M(h)(x, r) = M(h)(x, 0)$$
2. For $h \in C^2(\mathbb{R}^n)$, $x \in \mathbb{R}^n$ and $r > 0$,
$$\partial_r M(h)(x, r) = \frac{r}{w_n} \int_{|\xi|<1} \triangle h(x + r\xi)\, d\xi$$
so $\partial_r M(h)(x, r) \to 0$ as $r \to 0$.

***Proof*:**

1. By (2.6), $M(h)(x, r) - h(x) = \frac{1}{w_n} \int_{|\xi|=1} \big(h(x+r\xi) - h(x)\big)\, dS_\xi$, and h is uniformly continuous on the closed unit ball about x, so this tends to 0 as $r \to 0$.
2. The identity is the first two steps of the proof of lemma 2.3.2. Its right hand side is at most $\frac{r}{w_n} |B_1| \sup_{|y-x| \le r} |\triangle h(y)|$ in absolute value, where $|B_1|$ is the volume of the unit ball, and this tends to 0 as $r \to 0$ because $\triangle h$ is continuous, as claimed.

Spherical means give solutions of the wave equation for data that need not be of Schwartz class. Suppose that $u(t, x)$ is a C^2 solution of the Cauchy problem for the wave equation. Its spherical mean $M(u)(t, x, r)$ satisfies an auxiliary equation in the reduced space-time variables $(t, r) \in \mathbb{R}^2$. Specifically, define:
$$M(u)(t, x, r) = \frac{1}{w_n} \int_{|\xi|=1} u(t, x + r\xi) dS_\xi$$
Taking time derivatives and using the fact that $u(t, x)$ is a solution of the wave equation, we obtain:
$$\begin{aligned} \partial_t^2 M(u)(t, x, r) &= \frac{1}{w_n} \int_{|\xi|=1} \partial_t^2 u(t, x + r\xi) dS_\xi \\ &= \frac{1}{w_n} \int_{|\xi|=1} \triangle_x u(t, x + r\xi) dS_\xi \\ &= \triangle_x M(u)(t, x, r) \end{aligned}$$
Now using the Darboux equation (lemma 2.3.2), we get, for $r > 0$:
$$\partial_t^2 M(u)(t, x, r) = \left(\partial_r^2 + \frac{n-1}{r} \partial_r\right) M(u)(t, x, r) \tag{2.7}$$
a PDE in the two variables $(t, r) \in \mathbb{R}^2$. This is known as the *Euler-Poisson-Darboux* equation.

2.3.3 Spherical Means in Dimension 3

The solution of (2.7) depends on the dimension, and the case $n = 3$ is the simplest: there $r\big(\partial_r^2 + \frac{2}{r}\partial_r\big)M = \partial_r^2(rM)$, so $rM(u)$ solves the one-dimensional wave equation in (t, r), with x as a parameter,

and d'Alembert's formula applies.

Theorem 2.3.4: Kirchhoff's Formula

Let $n = 3$, and let $u \in C^2([0,\infty) \times \mathbb{R}^3)$ solve $\Box u = 0$ with $u(0,\cdot) = f$ and $\partial_t u(0,\cdot) = g$. Then for $t > 0$ and $x \in \mathbb{R}^3$,

$$u(t,x) = \partial_t \left(tM(f)(x,t)\right) + tM(g)(x,t) \tag{2.8}$$

The right hand side is defined whenever $f \in C^1(\mathbb{R}^3)$ and $g \in C(\mathbb{R}^3)$.

***Proof*:**
Fix $x \in \mathbb{R}^3$. For $t \geq 0$ and $r \in \mathbb{R}$ let $\mu(t,r) = M(u)(t,x,r)$, defined by (2.6) for all real r and even in r, and put $v(t,r) = r\mu(t,r)$. Differentiating under the integral over the unit sphere shows that μ, and hence v, is C^2, and that $\partial_t^2 \mu = \triangle_x M(u)$. For $r > 0$, lemma 2.3.2 with $n = 3$ gives $\partial_t^2 \mu = \partial_r^2 \mu + \frac{2}{r}\partial_r \mu$, that is, $\partial_t^2 v = \partial_r^2 v$. As v is odd in r, so are $\partial_t^2 v$ and $\partial_r^2 v$, so the equation also holds for $r < 0$, and for $r = 0$ by continuity. The data of v are $v(0,r) = \varphi(r)$ and $\partial_t v(0,r) = \gamma(r)$, where $\varphi(r) = rM(f)(x,r)$ and $\gamma(r) = rM(g)(x,r)$ are odd. The derivation of d'Alembert's formula (2.3) applies verbatim on the half-plane $t \geq 0$, which meets each line $r \pm t = c$ in an interval, and gives

$$v(t,r) = \frac{1}{2}\big(\varphi(r+t) + \varphi(r-t)\big) + \frac{1}{2}\int_{r-t}^{r+t} \gamma(\rho)\, d\rho$$

For $t > 0$ and $0 < r < t$, oddness gives $\varphi(r-t) = -\varphi(t-r)$ and $\int_{r-t}^{t-r} \gamma = 0$, so

$$\mu(t,r) = \frac{\varphi(t+r) - \varphi(t-r)}{2r} + \frac{1}{2r}\int_{t-r}^{t+r} \gamma(\rho)\, d\rho$$

As $r \to 0$, the left hand side tends to $u(t,x)$ by proposition 2.3.3(1), the first term tends to $\varphi'(t) = \partial_t\big(tM(f)(x,t)\big)$, and the second to $\gamma(t) = tM(g)(x,t)$, completing the proof.

Carrying out the differentiation in (2.8), with $\partial_t M(f)(x,t) = \frac{1}{4\pi}\int_{|\xi|=1} \nabla f(x+t\xi)\cdot\xi\, dS_\xi$, gives for $t > 0$

$$u(t,x) = \frac{1}{4\pi t^2}\int_{|y-x|=t} \left(tg(y) + f(y) + t\nabla f(y)\cdot\frac{y-x}{|y-x|}\right) dS_y \tag{2.9}$$

In terms of pointwise regularity, the solution is generally less smooth than the initial data, since the formula involves the derivative of f.

Corollary 2.3.5: Solution Space

Given initial data $g \in C^2(\mathbb{R}^3)$ and $f \in C^3(\mathbb{R}^3)$, the function u defined by (2.8) for all $t \in \mathbb{R}$ lies in $C^2(\mathbb{R}\times\mathbb{R}^3)$ and solves $\Box u = 0$ with $u(0,\cdot) = f$ and $\partial_t u(0,\cdot) = g$.

***Proof*:**
For $h \in C^k(\mathbb{R}^3)$, the function $(t,x) \mapsto M(h)(x,t)$, defined by (2.6) for all $t \in \mathbb{R}$ and even in t, is C^k, by differentiation under the integral over the unit sphere. Let $h \in C^k(\mathbb{R}^3)$ with $k \geq 2$ and $w(t,x) = tM(h)(x,t)$. Then $\partial_t^2 w = 2\partial_t M(h) + t\partial_t^2 M(h)$, while for $t \neq 0$, by lemma 2.3.2 with

$r = |t|$ and the evenness of $M(h)$ in t, $\triangle_x w = t\triangle_x M(h) = t\partial_t^2 M(h) + 2\partial_t M(h)$. So $\Box w = 0$ for $t \neq 0$, and at $t = 0$ by continuity. Moreover $w(0,\cdot) = 0$ and $\partial_t w(0,x) = M(h)(x,0) = h(x)$ by proposition 2.3.3(1). With $h = g$ this shows that $tM(g)(x,t)$ is a C^2 solution with data $(0,g)$. With $h = f \in C^3$, the function $z(t,x) = tM(f)(x,t)$ is C^3 and solves $\Box z = 0$ with data $(0,f)$, so $\partial_t z$ is C^2, $\Box\partial_t z = \partial_t \Box z = 0$, $\partial_t z(0,\cdot) = f$, and $\partial_t^2 z(0,\cdot) = \triangle z(0,\cdot) = 0$. Adding the two solutions gives the claim, as we sought to show.

In the Sobolev scale there is no such loss: for data in $H^s \times H^{s-1}$ the solution lies in $C([0,T],H^s) \cap C^1([0,T],H^{s-1})$, as theorem 5.1.1 will show.

2.3.4 Spherical Means in Odd Dimensions

For a C^2 solution u of the wave equation with data (f,g), the spherical means satisfy the Euler-Poisson-Darboux problem

$$\begin{aligned} \partial_t^2 M(u)(t,x,r) &= \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right) M(u)(t,x,r) \\ M(u)(0,x,r) &= M(f)(x,r) \\ \partial_t M(u)(0,x,r) &= M(g)(x,r) \end{aligned}$$

for $r > 0$. In odd dimensions $n = 2k+1 \geq 3$ an algebraic transformation reduces it to the one-dimensional wave equation in the variables (t,r), generalizing the passage from $M(u)$ to $rM(u)$ for $n = 3$.

Lemma 2.3.6: Useful Relations

Let $k \geq 1$ be an integer and $I \subseteq \mathbb{R} \setminus \{0\}$ an open interval.

1. For $\varphi \in C^{k+1}(I)$, on I,
$$\partial_r^2 \left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}\varphi\right) = \left(\frac{1}{r}\partial_r\right)^{k} \left(r^{2k}\partial_r\varphi\right)$$
2. There are constants β_j^k with $\beta_0^k = 1\cdot 3\cdots(2k-1) = (2k-1)!!$ such that for every $\varphi \in C^{k-1}(I)$, on I,
$$\left(\frac{1}{r}\partial_r\right)^{k-1} \left(r^{2k-1}\varphi\right) = \sum_{j=0}^{k-1} \beta_j^k r^{j+1}\partial_r^j \varphi$$

***Proof*:**
Write $D = \frac{1}{r}\partial_r$ and $T_k\varphi = D^{k-1}(r^{2k-1}\varphi)$. For every $k \geq 1$,

$$D\big(r^{2k+1}\varphi\big) = (2k+1)r^{2k-1}\varphi + r^{2k}\partial_r\varphi = r^{2k-1}\big((2k+1)\varphi + r\partial_r\varphi\big)$$

so that $T_{k+1}\varphi = D^{k-1}D(r^{2k+1}\varphi) = T_k\psi$ with $\psi = (2k+1)\varphi + r\partial_r\varphi$. Both parts follow by induction on k from this relation.

1. For $k = 1$ both sides equal $2\partial_r\varphi + r\partial_r^2\varphi$. Suppose that part (1) holds for some k and every

function in $C^{k+1}(I)$, and let $\varphi \in C^{k+2}(I)$. Then $\psi \in C^{k+1}(I)$, and part (1) for k gives

$$\partial_r^2 T_{k+1}\varphi = \partial_r^2 T_k \psi = D^k\big(r^{2k}\partial_r\psi\big)$$

Here $\partial_r\psi = (2k+2)\partial_r\varphi + r\partial_r^2\varphi$, so

$$r^{2k}\partial_r\psi = (2k+2)r^{2k}\partial_r\varphi + r^{2k+1}\partial_r^2\varphi = D\big(r^{2k+2}\partial_r\varphi\big)$$

and therefore $\partial_r^2 T_{k+1}\varphi = D^{k+1}(r^{2k+2}\partial_r\varphi)$, which is part (1) for $k+1$.

2. For $k=1$, $T_1\varphi = r\varphi$, so $\beta_0^1 = 1$. Suppose that part (2) holds for some k and every function in $C^{k-1}(I)$, and let $\varphi \in C^k(I)$. Then $\psi \in C^{k-1}(I)$, and since $\partial_r^j(r\partial_r\varphi) = r\partial_r^{j+1}\varphi + j\partial_r^j\varphi$,

$$T_{k+1}\varphi = T_k\psi = \sum_{j=0}^{k-1}\beta_j^k r^{j+1}\Big((2k{+}1{+}j)\partial_r^j\varphi + r\partial_r^{j+1}\varphi\Big) = \sum_{j=0}^{k}\big((2k{+}1{+}j)\beta_j^k + \beta_{j-1}^k\big)r^{j+1}\partial_r^j\varphi$$

with $\beta_{-1}^k = \beta_k^k = 0$. So part (2) holds for $k+1$ with $\beta_j^{k+1} = (2k+1+j)\beta_j^k + \beta_{j-1}^k$, and in particular $\beta_0^{k+1} = (2k+1)\beta_0^k$, which gives $\beta_0^k = (2k-1)!!$, as claimed.

Let $n = 2k{+}1$ and let u be a solution of class C^{k+1}. By lemma 2.3.6(1) and the Euler-Poisson-Darboux equation, whose coefficient is $n-1 = 2k$, the function

$$v(t,x,r) = \left(\frac{1}{r}\partial_r\right)^{k-1}\big(r^{2k-1}M(u)\big)(t,x,r)$$

solves the one-dimensional wave equation in (t,r) for $r>0$:

$$\begin{aligned}\partial_r^2 v &= \left(\frac{1}{r}\partial_r\right)^k\big(r^{2k}\partial_r M(u)\big) = \left(\frac{1}{r}\partial_r\right)^{k-1}\left(r^{2k-1}\left(\partial_r^2 M(u) + \frac{2k}{r}\partial_r M(u)\right)\right)\\ &= \left(\frac{1}{r}\partial_r\right)^{k-1}\big(r^{2k-1}\partial_t^2 M(u)\big) = \partial_t^2 v\end{aligned}$$

By lemma 2.3.6(2), $v(t,x,r)/(\beta_0^k r) \to M(u)(t,x,0) = u(t,x)$ as $r\to 0$. Computing v by d'Alembert's formula, as in the proof of theorem 2.3.4, and passing to this limit gives the formula of theorem 2.3.7(2).

Theorem 2.3.7: Solution For Odd Dimension

Let $n = 2k+1 \geq 3$ be odd. For $h \in C^{k-1}(\mathbb{R}^n)$ and $(t,x) \in \mathbb{R} \times \mathbb{R}^n$ put

$$(S_t h)(x) = \frac{1}{(2k-1)!!} \sum_{j=0}^{k-1} \beta_j^k t^{j+1} \partial_t^j M(h)(x,t)$$

with $M(h)(x,t)$ given by (2.6) for all real t. By lemma 2.3.6(2), $(S_t h)(x) = \frac{1}{(2k-1)!!} \left(\frac{1}{t}\partial_t\right)^{k-1} \left(t^{2k-1} M(h)(x,t)\right)$ for $t \neq 0$.

1. For $h \in C^{k+1}(\mathbb{R}^n)$, the function $(t,x) \mapsto (S_t h)(x)$ lies in $C^2(\mathbb{R} \times \mathbb{R}^n)$ and solves $\Box w = 0$ with $w(0,\cdot) = 0$ and $\partial_t w(0,\cdot) = h$.
2. For $f \in C^{k+2}(\mathbb{R}^n)$ and $g \in C^{k+1}(\mathbb{R}^n)$, the function

$$u(t,x) = \partial_t (S_t f)(x) + (S_t g)(x)$$

lies in $C^2(\mathbb{R} \times \mathbb{R}^n)$ and solves $\Box u = 0$ with $u(0,\cdot) = f$ and $\partial_t u(0,\cdot) = g$.

***Proof*:**

For $h \in C^m(\mathbb{R}^n)$, the function $(t,x) \mapsto M(h)(x,t)$ is C^m on $\mathbb{R} \times \mathbb{R}^n$, by differentiation under the integral over the unit sphere in (2.6). Each term of $S_t h$ involves $\partial_t^j M(h)$ with $j \leq k-1$, so $S_t h$ is C^2 for $h \in C^{k+1}(\mathbb{R}^n)$ and C^3 for $h \in C^{k+2}(\mathbb{R}^n)$.

1. *The equation.* Let $h \in C^{k+1}(\mathbb{R}^n)$, fix x, and write $H(t) = M(h)(x,t)$. For $t \neq 0$, lemma 2.3.2 with $r = |t|$, together with the evenness of $M(h)$ in t, gives $\partial_t^2 H + \frac{2k}{t}\partial_t H = \triangle_x M(h)(x,t)$. On each of the intervals $(0,\infty)$ and $(-\infty,0)$, lemma 2.3.6(1) gives, with $c = 1/(2k-1)!!$,

$$\begin{aligned}\partial_t^2 (S_t h)(x) &= c\left(\frac{1}{t}\partial_t\right)^k \left(t^{2k}\partial_t H\right) = c\left(\frac{1}{t}\partial_t\right)^{k-1} \left(t^{2k-1}\left(\partial_t^2 H + \frac{2k}{t}\partial_t H\right)\right) \\ &= c\left(\frac{1}{t}\partial_t\right)^{k-1} \left(t^{2k-1}\triangle_x M(h)(x,t)\right) = \triangle_x (S_t h)(x)\end{aligned}$$

   The last step moves $\triangle_x$ past at most $k-1$ derivatives in t, which is allowed because $M(h)$ is C^{k+1}. So $\Box S_t h = 0$ for $t \neq 0$, and for $t = 0$ by continuity of the second derivatives.

   The data. Every term of $S_t h$ contains the factor t^{j+1}, so $S_0 h = 0$. Differentiating once in t, every term with $j \geq 1$ still vanishes at $t = 0$, and the term with $j = 0$ contributes $c\beta_0^k M(h)(x,0) = h(x)$ by proposition 2.3.3(1), since $c\beta_0^k = 1$.

2. By part (1), $S_t g$ is a C^2 solution with data $(0,g)$. For $f \in C^{k+2}(\mathbb{R}^n)$ the function $z(t,x) = (S_t f)(x)$ is C^3 by the first paragraph and solves $\Box z = 0$ with data $(0,f)$ by part (1). Hence $\partial_t z$ is C^2, $\Box \partial_t z = \partial_t \Box z = 0$, $\partial_t z(0,\cdot) = f$, and $\partial_t^2 z(0,\cdot) = \triangle z(0,\cdot) = 0$. Adding the two solutions gives part (2), as claimed.

For $n = 3$, so $k = 1$, the theorem is corollary 2.3.5, since then $S_t h = tM(h)(x,t)$. Measured in the scale of data $(f,g) \in C^s \times C^{s-1}$, the solution loses $k = \frac{n-1}{2}$ derivatives: the formula differentiates the spherical means $k-1$ times in S_t and once more in $\partial_t S_t f$, and $f \in C^{k+2}$, $g \in C^{k+1}$ give a solution

that is only C^2.

2.3.5 Huygens' Principle

As in one dimension, information propagates with speed at most 1. In dimension 3 more is true: by Kirchhoff's formula, $u(t,x)$ only depends on the data near the sphere $\{y \mid |y-x| = |t|\}$, and not on the data inside it.

Theorem 2.3.8: Finite Propagation Speed And Strong Huygens' Principle

Let $u \in C^2(\mathbb{R}\times\mathbb{R}^3)$ solve $\Box u = 0$, and suppose that its data $f = u(0,\cdot)$ and $g = \partial_t u(0,\cdot)$ satisfy $\operatorname{supp}(f)\cup\operatorname{supp}(g) \subseteq B_R(0)$ for some $R > 0$. Then:

1. (*Finite propagation speed*) $\operatorname{supp}(u(t,\cdot)) \subseteq B_{R+|t|}(0)$ for every $t \in \mathbb{R}$.
2. (*Strong Huygens' principle*) $u(t,x) = 0$ whenever $|t| > R$ and $|x| \le |t| - R$.

***Proof*:**
For $t = 0$, part (1) is the hypothesis on f, and part (2) says nothing. The function $\tilde{u}(t,x) = u(-t,x)$ lies in $C^2(\mathbb{R}\times\mathbb{R}^3)$, solves $\Box\tilde{u} = 0$, and has data $(f,-g)$ with the same supports, so the claims for u at a time $t < 0$ are the claims for $\tilde{u}$ at the time $-t > 0$. Let $t > 0$.

By theorem 2.3.4, applied to the restriction of u to $[0,\infty)\times\mathbb{R}^3$, and by (2.9), $u(t,x) = 0$ whenever f, ∇f and g vanish on the sphere $\{y \mid |y-x| = t\}$. The set $K = \operatorname{supp}(f)\cup\operatorname{supp}(g)$ is closed and bounded, hence compact, and it lies in $B_R(0)$, so $K \subseteq \overline{B_{R'}(0)}$ for some $R' < R$. Then f, ∇f and g vanish on the open set $\{y \mid |y| > R'\}$.

1. If $|x| > R' + t$, every y with $|y-x| = t$ has $|y| \ge |x| - t > R'$, so $u(t,x) = 0$. Thus $u(t,\cdot)$ vanishes outside the closed ball $\overline{B_{R'+t}(0)}$, and $\operatorname{supp}(u(t,\cdot)) \subseteq \overline{B_{R'+t}(0)} \subseteq B_{R+t}(0)$, which is part (1).
2. If $|x| \le t - R$, every y with $|y-x| = t$ has $|y| \ge t - |x| \ge R > R'$, so $u(t,x) = 0$, as claimed.

Together, the two parts say that for $|t| > R$ the function $u(t,\cdot)$ vanishes outside the shell

$$\{x \mid |t| - R < |x| < |t| + R\}$$

The proof only uses that $u(t,x)$ is determined by the data near the sphere $\{y \mid |y-x| = |t|\}$. For odd $n \ge 3$ the solution of theorem 2.3.7(2) has this property, since $\partial_t^j M(h)(x,t) = \frac{1}{w_n}\int_{|\xi|=1}\big((\xi\cdot\nabla)^j h\big)(x+t\xi)\,dS_\xi$ only involves derivatives of h on that sphere, so both parts hold for it with the same proof. In even dimensions the solution formula, obtained from the next odd dimension by Hadamard's method of descent, involves the data on the whole ball $\{y \mid |y-x| \le |t|\}$, and only part (1), finite propagation speed, holds [10, Week 1, §1.4.3 and §1.5].

2.3.6 Duhamel's Principle in Higher Dimensions

Proposition 2.3.9: Duhamel's Principle

Let $F \in C(\mathbb{R} \times \mathbb{R}^n)$, and let $v(t, x; s)$, for $(t, x) \in \mathbb{R} \times \mathbb{R}^n$ and $s \in \mathbb{R}$, be a function such that v and its partial derivatives in (t, x) of order at most 2 are continuous in (t, x, s), and for every s the function $v(\cdot, \cdot\,; s)$ solves $\Box v = 0$ with $v(0, \cdot\,; s) = 0$ and $\partial_t v(0, \cdot\,; s) = F(s, \cdot)$. Then

$$u(t, x) = \int_0^t v(t - s, x; s)\, ds$$

lies in $C^2(\mathbb{R} \times \mathbb{R}^n)$ and solves $\Box u = F$ with $u(0, \cdot) = 0$ and $\partial_t u(0, \cdot) = 0$.

***Proof*:**

Step 1: Time derivatives. Fix x. The function $\psi(t, s) = v(t - s, x; s)$ and its derivative $\partial_t \psi(t, s) = (\partial_t v)(t - s, x; s)$ are continuous, and $\psi(t, t) = v(0, x; t) = 0$, so eq. (2.5) gives

$$\partial_t u(t, x) = \int_0^t (\partial_t v)(t - s, x; s)\, ds$$

Applying eq. (2.5) again, now to $\psi(t, s) = (\partial_t v)(t - s, x; s)$, with $\psi(t, t) = \partial_t v(0, x; t) = F(t, x)$, gives

$$\partial_t^2 u(t, x) = F(t, x) + \int_0^t (\partial_t^2 v)(t - s, x; s)\, ds$$

Step 2: Space derivatives. By [8, Leibniz Integral Rule], applied with x_i as the parameter, $\partial_{x_i} u$ and $\partial_{x_i} \partial_{x_j} u$ are the integrals over $[0, t]$ of the corresponding derivatives of $v(t - s, x; s)$. For $\partial_t \partial_{x_i} u$, apply eq. (2.5) to $\psi(t, s) = (\partial_{x_i} v)(t - s, x; s)$, whose value at $s = t$ is $\partial_{x_i} v(0, x; t) = 0$. Each of these integrals, and the two of Step 1, is continuous in (t, x) because its integrand is continuous in (t, x, s), so $u \in C^2(\mathbb{R} \times \mathbb{R}^n)$.

Step 3: Conclusion. By Steps 1 and 2,

$$\Box u(t, x) = F(t, x) + \int_0^t (\Box v)(t - s, x; s)\, ds = F(t, x)$$

and at $t = 0$ the integrals defining u and $\partial_t u$ run over $[0, 0]$, so both vanish, as claimed.

For $n = 3$ and F such that F and its partial derivatives in x of order at most 2 are continuous on $\mathbb{R} \times \mathbb{R}^3$, theorem 2.3.7(1) with $k = 1$ shows that $v(t, x; s) = tM(F(s, \cdot))(x, t)$ solves the homogeneous problem with data $(0, F(s, \cdot))$, and differentiation under the integral over the unit sphere shows that v satisfies the continuity hypothesis. For $t > 0$, the substitution $\rho = t - s$ and polar coordinates around x then give the *retarded potential*

$$u(t, x) = \int_0^t (t - s) M(F(s, \cdot))(x, t - s)\, ds = \frac{1}{4\pi} \int_{|y - x| \le t} \frac{F(t - |y - x|, y)}{|y - x|}\, dy$$

in which the forcing at (s, y) enters at the later time $t = s + |y - x|$. For odd $n = 2k + 1 \ge 5$, theorem 2.3.7(1) shows in the same way that $v(t, x; s) = S_t\big(F(s, \cdot)\big)(x)$ satisfies the hypotheses

of proposition 2.3.9 when the partial derivatives of F in x of order at most $k+1$ are continuous.

3

Weak Solutions and Distributions

We now deal with the case where solutions can be "weaker". As we saw, d'Alembert formula works perfectly well if $f, g \in L^1_{\text{loc}}(\mathbb{R})$. However, the initial conditions *require* that f, g be differentiable (afterall, they are slices of the function $u(t,x)$ and $\partial_t u(t,x)$, which must be at least twice and once differentiable respectively). So is there some way to circumnavigate this? The content of the first section of this chapter covers precisely this. In the next section, we shall generalize the results and define a whole new object that will allow us to greatly broaden our notion of function that will allow us to differentiate function that are simply locally integrable!

Let's start with the usual solution $u \in C^2$ on a time strip:

$$S_T = [0,T] \times \mathbb{R}^n$$

and assume that:

$$\Box u = F \qquad u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

We are going to use the idea of a test function φ. If you want to think of it in a Folland way, $\varphi \in C_c^\infty(S_T)$, and we are defining a functional on $C_c^\infty(S_T)$ taking $\varphi \mapsto \int \Box u \varphi d(x,t)$. We will take advantage of integration by parts to get the following:

$$\begin{aligned}
\int_{S_T} F\varphi dt dx &= \int_{S_T} (\Box u)\varphi \,\mathrm{d}t\mathrm{d}x \\
&= \int_{\mathbb{R}^n} \left(-\int_0^T \partial_t u \partial_t \varphi \,\mathrm{d}t - g(x)\varphi(0,x) \right) \mathrm{d}x - \int_{S_T} u \triangle \varphi \,\mathrm{d}t\mathrm{d}x \\
&= \int_{S_T} u \Box \varphi \mathrm{d}t\mathrm{d}x + \int_{\mathbb{R}^n} f(x)\partial_t\varphi(0,x)\,\mathrm{d}x - \int_{\mathbb{R}^n} g(x)\varphi(0,x)\mathrm{d}x
\end{aligned}$$

Which leads us to the following definition:

Definition 3.0.1: Weak Solution

Let $f, g \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $F \in L^1_{\text{loc}}(S_T)$. We say $u \in L^1_{\text{loc}}(S_T)$ is a *weak solution* of the wave equation if:

$$\int_{S_T} u \Box\varphi \, \mathrm{d}t\mathrm{d}x = \int_{S_T} F\varphi \, \mathrm{d}t\mathrm{d}x - \int_{\mathbb{R}^n} f(x)\partial_t\varphi(0,x) \, \mathrm{d}x + \int_{\mathbb{R}^n} g(x)\varphi(0,x) \, \mathrm{d}x \tag{3.1}$$

for all $\varphi \in C_c^\infty(\mathbb{R}^{1+n})$ supported in $(-\infty, T) \times \mathbb{R}^n$

Proposition 3.0.2: When Weak Solution Is Classical Solution

A weak solution u that belongs to $C^2(S_T)$ is a classical solution: after F, f and g are changed on sets of measure zero, $\Box u = F$ on S_T, $u(0,\cdot) = f$ and $\partial_t u(0,\cdot) = g$.

***Proof*:**

All the error terms below are bounded uniformly in $x \in \text{supp}\psi$, because u, $\partial_t u$, $\partial_t^2 u$ and (after Step 1) F are continuous, hence bounded on compact subsets of S_T.

Step 1: The equation. If φ is supported in $(0,T) \times \mathbb{R}^n$, the last two terms of (3.1) vanish, so $\int u\Box\varphi = \int F\varphi$. Integrating by parts twice in t and in each x_i, with no boundary terms because φ vanishes near the boundary of $(0,T) \times \mathbb{R}^n$, gives $\int u\Box\varphi = \int (\Box u)\varphi$. Hence

$$\int (\Box u - F)\varphi \, dtdx = 0$$

for all such φ. A function $h \in L^1_{\text{loc}}$ with $\int h\varphi = 0$ for all $\varphi \in C_c^\infty$ vanishes almost everywhere [7, The du Bois-Reymond Lemma]. So $\Box u = F$ almost everywhere on S_T, and after changing F on a set of measure zero, $F = \Box u \in C(S_T)$.

Step 2: The velocity. We show that $\partial_t u(0,x) = g(x)$ for almost every x, which is equivalent to

$$\int_{\mathbb{R}^n} \partial_t u(0,x)\psi(x) \, dx = \int_{\mathbb{R}^n} g(x)\psi(x) \, dx \quad \text{for all } \psi \in C_c^\infty(\mathbb{R}^n) \tag{3.2}$$

Fix such a ψ. Let $a \in C^\infty(\mathbb{R})$ satisfy $a(t) = 1$ for $t \le 0$ and $a(t) = 0$ for $t \ge 1$, and put $\theta_k(t) = a(kt)$ for $k > 1/T$, so that

$$\theta_k(0) = 1 \qquad \theta_k'(0) = 0 \qquad \theta_k(t) = 0 \quad \text{for } t \ge 1/k$$

Take $\varphi_k(t,x) = \theta_k(t)\psi(x)$ in (3.1). (To make it compactly supported, multiply it by a function of t in $C_c^\infty(\mathbb{R})$ equal to 1 on $[-1,T]$, which changes nothing on $[0,T] \times \mathbb{R}^n$.) As $\partial_t\varphi_k(0,x) = 0$ and $\varphi_k(0,x) = \psi(x)$, the right hand side of (3.1) reads

$$\int_{\mathbb{R}^n} \left(\int_0^{1/k} F(t,x)a(kt) \, \mathrm{d}t \right) \psi(x) \, dx + \int_{\mathbb{R}^n} g(x)\psi(x) \, dx \tag{3.3}$$

Since $F \in C(S_T)$ by Step 1, the first term is $O(1/k)$ as $k \to \infty$. We claim that the left hand side of (3.1) is equal to

$$\int_{\mathbb{R}^n} \partial_t u(0,x)\psi(x)\,dx + O(1/k) \tag{3.4}$$

If so, equating (3.3) with (3.4) and letting $k \to \infty$ gives (3.2). The left hand side of (3.1) is

$$\int_{S_T} u(t,x)\theta_k''(t)\psi(x)\,dtdx - \int_{\mathbb{R}^n} \left(\int_0^{1/k} u(t,x)a(kt)\,dt \right) \triangle\psi(x)\,dx$$

The second term is $O(1/k)$. For fixed x, integrating by parts in t, with no boundary terms because $\theta_k'(0) = \theta_k'(T) = 0$, turns $\int_0^T u\theta_k''\,dt$ into $-\int_0^T \partial_t u\,\theta_k'\,dt$, and a second integration by parts, using $\theta_k(0) = 1$ and $\theta_k(T) = 0$, turns this into

$$\int_0^{1/k} \partial_t^2 u(t,x)a(kt)\,dt + \partial_t u(0,x)$$

The first term is again $O(1/k)$, which proves the claim.

Step 3: The position. We show that $u(0,x) = f(x)$ for almost every x. Fix $\psi \in C_c^\infty(\mathbb{R}^n)$, let $b \in C_c^\infty(\mathbb{R})$ satisfy $b(0) = 0$ and $b'(0) = 1$, with $\operatorname{supp} b \subseteq [-R,R]$, and put $\theta_k(t) = \frac{1}{k}b(kt)$ for $k > R/T$. Then $\theta_k(0) = 0$, $\theta_k'(0) = 1$, $|\theta_k| \le \|b\|_{L^\infty}/k$, and θ_k vanishes outside $[-R/k, R/k]$. Take $\varphi_k(t,x) = \theta_k(t)\psi(x)$ in (3.1). Since $\partial_t\varphi_k(0,x) = \psi(x)$ and $\varphi_k(0,x) = 0$, the right hand side equals

$$\int_{S_T} F\theta_k\psi\,dtdx - \int_{\mathbb{R}^n} f(x)\psi(x)\,dx = -\int_{\mathbb{R}^n} f(x)\psi(x)\,dx + O(1/k^2)$$

because $\theta_k = O(1/k)$ on an interval of length R/k. On the left hand side, $\int_{S_T} u\theta_k\triangle\psi = O(1/k^2)$ for the same reason. For fixed x, integrating by parts with $\theta_k'(0) = 1$ and $\theta_k'(T) = 0$,

$$\int_0^T u\theta_k''\,dt = -u(0,x) - \int_0^T \partial_t u\,\theta_k'\,dt$$

and a second integration by parts, with $\theta_k(0) = \theta_k(T) = 0$, gives $-\int_0^T \partial_t u\,\theta_k'\,dt = \int_0^{R/k} \partial_t^2 u\,\theta_k\,dt = O(1/k^2)$. So the left hand side equals $-\int_{\mathbb{R}^n} u(0,x)\psi(x)\,dx + O(1/k^2)$. Letting $k \to \infty$ gives $\int u(0,x)\psi(x)\,dx = \int f(x)\psi(x)\,dx$ for every $\psi \in C_c^\infty(\mathbb{R}^n)$, so $u(0,\cdot) = f$ almost everywhere, completing the proof.

3.1 Distributions

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution. Locally integrable functions will be distributions (in the appropriate interpretation), but we will see that there are general solutions that do not even fall under the notion of "function".

The general theory of distributions, the space of test functions and its topology, the definition of a

distribution as a continuous linear functional, the distributional derivative, and the basic operations on distributions, is developed in the functional-analysis volume, [7]; the Fourier theory of tempered distributions is developed in the harmonic-analysis volume, [9]. This chapter recalls the definitions it needs and concentrates on the role distributions play in the weak solutions of wave equations. It does not re-derive the foundational theory, which those volumes own.

We recall from [7] the notation and facts used below, none of them re-derived here. A *distribution* on an open $U \subseteq \mathbb{R}^n$ is a continuous linear functional on $C_c^\infty(U)$; their space is $\mathcal{D}'(U)$, with $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$, under the weak* topology. We write $\langle F, \varphi\rangle$ for the pairing and identify $f \in L^1_{\mathrm{loc}}$ with $\varphi \mapsto \int f\varphi$; the point mass is $\langle \delta, \varphi\rangle = \varphi(0)$. Every $F \in \mathcal{D}'(U)$ has distributional derivatives $\langle \partial^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, \partial^\alpha \varphi\rangle$, may be multiplied by a smooth function and translated, and may be convolved with a test function ψ via $F * \psi(x) = \langle F, \tau_x \tilde{\psi}\rangle$ (so $\delta * \psi = \psi$). Finally $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$, and $\mathcal{D}'(U)$ is sequentially complete (a pointwise limit of distributions is again a distribution) by the uniform boundedness principle. Proofs and the continuity criterion are in [7, Distribution Continuity Check]; the inductive-limit topology on C_c^∞ itself is deliberately not constructed there (that book works with the seminorm test directly and defers the topology to its external references).

3.1.1 Time-dependent Distributions

Consider a map

$$\mathbb{R} \to \mathcal{D}'(\mathbb{R}^n), \qquad t \mapsto u_t$$

Since $\mathcal{D}'(\mathbb{R}^n)$ has a topology, the topology of point-wise convergence, the notion of continuity and differentiability can be defined. Thus, the above function is continuous at t_0 if and only if $t \to \langle u_t, \varphi\rangle$ is continuous for every $\varphi \in C_c^\infty(\mathbb{R}^n)$, and u is differentiable at t_0 if and only if there exists a $v \in \mathcal{D}'(\mathbb{R}^n)$ such that for every $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\frac{d}{dt}\,\langle u_t, \varphi\rangle|_{t=t_0} = \langle v, \varphi\rangle$$

and we write $u_t' = v$. This can naturally be repeated to form k-times differentiables maps in $C^k(\mathbb{R}, \mathcal{D}')$.

Proposition 3.1.1: Continuous Distributions Form Distributions

Every $u \in C(\mathbb{R}, \mathcal{D}')$ defines a distribution on $\mathbb{R}^{n+1}$ by

$$\langle u, \psi\rangle = \int_{\mathbb{R}} \langle u_t, \psi(t, \cdot)\rangle \qquad \text{for } \psi \in C_c^\infty(\mathbb{R}^{1+n})$$

namely, $u \in \mathcal{D}'(\mathbb{R}^{1+n})$

***Proof*:**
First, the integrand is continuous and compactly supported as a function of t, and so the integral exists. The functional u is in $\mathcal{D}'(\mathbb{R}^{1+n})$ if for every compact $K \subseteq \mathbb{R}^{1+n}$ there are C and N with $|\langle u, \psi\rangle| \le C \sum_{|\alpha|\le N} \|\partial^\alpha \psi\|_{L^\infty}$ for all $\psi \in C_c^\infty(K)$. Every compact set lies in some $I \times K'$ with $I \subseteq \mathbb{R}$ a compact interval and $K' \subseteq \mathbb{R}^n$ compact, so it suffices to take $K = I \times K'$. But then, by the uniform boundedness principle on the Fréchet space $C_c^\infty(K')$ ([7, Principle Of Uniform

Boundedness For Fréchet Spaces]), there exist $C_{I,K'} > 0$ and $N_{I,K'} \in \mathbb{N}$ such that

$$\sup_{t\in I} |\langle u(t), \varphi\rangle| \leq C_{I,K'} \sum_{|\alpha|\leq N_{I,K'}} \|\partial^\alpha \varphi\|_{L^\infty} \qquad \forall \varphi \in C_c^\infty(K')$$

Applying this to each $t \in I$ with $\varphi = \psi(t, \cdot)$, and then integrating with respect to t, we get the needed estimate to show it is bounded, showing it is indeed continuous, and so completing the proof.

With this build-up, we are now able to ask the following question: if $u, u' \in C^1(\mathbb{R}, \mathcal{D}')$, is it possible that:

$$\partial_t u = u'$$

with the derivative defined distributionally. The answer is yes! Unwarpping the definition, the above equation is equivalent to:

$$\int_{\mathbb{R}} \langle u(t), (-1)\partial_t \psi(t, \cdot)\rangle dt = \int_{\mathbb{R}} \langle u'(t), \psi(t, \cdot)\rangle dt$$

and since $u \in C^1$ (i.e. the derivative is continuous), then the derivative can be written in terms of distributions like so:

$$\frac{d}{dt}\langle u(t), \psi(t, \cdot)\rangle = \langle u'(t), \psi(t, \cdot)\rangle + \langle u(t), \partial_t \psi(t, \cdot)\rangle$$

To see this, note that if you take $f(t) = \langle u(t), \psi(t, \cdot)\rangle$, then

$$\begin{aligned}
&\frac{1}{h}\{f(t+h) - f(t)\} \\
&= \left\langle \frac{1}{h}\{u(t+h) - u(t)\}, \psi(t, \cdot) \right\rangle + \left\langle u(t+h), \frac{1}{h}\{\psi(t+h, \cdot) - \psi(t, \cdot)\} \right\rangle
\end{aligned}$$

and then apply the uniform boundedness principle on the last term.

3.1.2 Distributional Solution for $\Box u = 0$

We can now consider using distribution's to solve problems, namely:

$$\Box u = 0 \qquad \Leftrightarrow \qquad \langle u, \Box\varphi\rangle = 0 \qquad \forall \varphi \in C_c^\infty(U)$$

with the Cauchy problem:

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

In general, it does not make sense to restrict a distribution u to the plane at $t = 0$ (what would this mean for a measure?). However, this *does* make sense for a time-dependent distribution as discussed in the last section. In fact, given *any* two distributions f, g, we can find a solution to the wave equation using time-dependent distributions!!

Theorem 3.1.2: General Weak Solution To The Wave Equation

For all $f, g \in \mathcal{D}'(\mathbb{R}^n)$, there exists a time-dependent distribution $u \in C^\infty(\mathbb{R}, \mathcal{D}'(\mathbb{R}^n))$ which solves the Cauchy problem

***Proof Key ideas*:**
The solution is built from the *wave propagator*, a family

$$W \in C^\infty(\mathbb{R}, E'(\mathbb{R}^n)) \qquad W(0) = W''(0) = 0 \qquad W'(0) = \delta$$

of compactly supported distributions with $\mathrm{supp} W(t) \subseteq \{x \mid |x| \leq |t|\}$, read off from the representation formulas for smooth data (d'Alembert's formula and the spherical-mean formulas of the previous chapter). One sets

$$u(t) = W'(t) * f + W(t) * g \tag{3.5}$$

Since each $W(t)$ is compactly supported, these convolutions are defined for all $f, g \in \mathcal{D}'(\mathbb{R}^n)$, and the initial conditions hold because

$$u(0) = W'(0) * f + W(0) * g = \delta * f = f$$
$$u'(0) = W''(0) * f + W'(0) * g = \delta * g = g$$

To see that $\Box u = 0$, choose $f_j, g_j \in C_c^\infty(\mathbb{R}^n)$ with $f_j \to f$ and $g_j \to g$ in $\mathcal{D}'(\mathbb{R}^n)$, using the density of C_c^∞ in $\mathcal{D}'$. The functions $u_j(t) = W'(t) * f_j + W(t) * g_j$ are the classical solutions with data f_j, g_j, so $\Box u_j = 0$, and $u_j \to u$ in $\mathcal{D}'(\mathbb{R}^{1+n})$ gives $\Box u_j \to \Box u$ there, hence $\Box u = 0$.

Two ingredients of this outline are not developed in this book: the construction of $W(t)$ for general n with the stated smoothness in t, and the convolution $\mathcal{D}'(\mathbb{R}^n) \times E'(\mathbb{R}^n) \to \mathcal{D}'(\mathbb{R}^n)$ together with its continuity in each factor, which gives both $u \in C^\infty(\mathbb{R}, \mathcal{D}'(\mathbb{R}^n))$ and the convergence $u_j \to u$. The theorem is [10, Week 3, Theorem 13], where the argument above is given and these two ingredients are set as exercises.

The solution is also unique, which is covered in the next section. Besides the finite propagation speed $\mathrm{supp} W(t) \subseteq \{x \mid |x| \leq |t|\}$, the propagator satisfies the strong Huygens' principle

$$\mathrm{supp} W(t) \subseteq \{x \mid |x| = |t|\}$$

when the dimension n is odd and $n \geq 3$. The Fourier transform of $W(t)$ is computed in eq. (3.9) below.

3.2 More on Distributional Solutions

We now move on to non-homogeneous solutions: we want to find a $u \in C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$ that solves

$$\begin{cases} \Box u = F & \text{on} \qquad S_T = (0,T) \times \mathbb{R}^n \\ u|_{t=0} = f & \partial_t u|_{t=0} = g \end{cases}$$

where $F \in \mathcal{D}'(S_T)$, $f, g \in \mathcal{D}'(\mathbb{R}^n)$, and $\Box u = F$ is understood in the sense of distributions on S_T.

Proposition 3.2.1: Constant Weak Derivative

Let $u \in \mathcal{D}'(I)$, where $I \subseteq \mathbb{R}$ is an open interval. Then:

$$u' = 0 \Rightarrow u \text{ is constant}$$

***Proof*:**
Fix $\varphi \in C_c^\infty(I)$ with $\int \varphi = 1$ and put $c = \langle u, \varphi \rangle$. Let $\psi \in C_c^\infty(I)$. The function $\theta = \psi - \left(\int \psi\right) \varphi$ lies in $C_c^\infty(I)$ and $\int \theta = 0$, so $\eta(x) = \int_{-\infty}^{x} \theta$ vanishes to the left of suppθ and, since $\int \theta = 0$, also to the right of it. As I is an interval, $\eta \in C_c^\infty(I)$, and $\eta' = \theta$. Hence

$$\langle u, \psi \rangle - c \int \psi = \langle u, \theta \rangle = \langle u, \eta' \rangle = -\langle u', \eta \rangle = 0$$

so u is the constant function c, as claimed.

Proposition 3.2.2: Further Results About Weak Derivatives II

Let $\Omega \subseteq \mathbb{R}^n$ be open and $k \geq 1$. If $u \in C(\Omega)$ and the distributional derivatives $\partial^\alpha u$ are also in $C(\Omega)$ for all $|\alpha| \leq k$, then $u \in C^k(\Omega)$, and its classical and distributional derivatives of order at most k agree.

***Proof*:**
Step 1: One variable, $k = 1$. Let J be an open interval, fix $a \in J$, and let $u, v \in C(J)$ satisfy $u' = v$ in $\mathcal{D}'(J)$. Set $U(x) = u(a) + \int_a^x v$ for $x \in J$. Then $U \in C^1(J)$ with $U' = v$ classically, so integration by parts against test functions gives $(U - u)' = 0$ in $\mathcal{D}'(J)$. By proposition 3.2.1, $U - u$ is constant, and the constant is 0 because $U(a) = u(a)$ and both functions are continuous. Hence $u = U$ is C^1 with $u' = v$.

Step 2: $k = 1$. Write $v_j \in C(\Omega)$ for the distributional derivative $\partial_j u$. Fix $x_0 \in \Omega$ and an open box $Q = J \times Q' \subseteq \Omega$ around x_0, where J is an interval in the x_j direction and Q' a box in the remaining variables x'. For $\varphi_1 \in C_c^\infty(J)$ and $\varphi_2 \in C_c^\infty(Q')$, the identity $\int u\,(-\partial_j(\varphi_1\varphi_2)) = \int v_j \varphi_1 \varphi_2$ and Fubini's theorem give $\int_{Q'} G(x')\varphi_2(x')\,dx' = 0$, where

$$G(x') = \int_J u(x_j, x')\,(-\varphi_1'(x_j))\,dx_j - \int_J v_j(x_j, x')\varphi_1(x_j)\,dx_j$$

is continuous in x'. As φ_2 is arbitrary and G is continuous, $G = 0$ by [7, The du Bois-Reymond Lemma], that is, for each $x' \in Q'$ the function $x_j \mapsto u(x_j, x')$ has distributional derivative $x_j \mapsto v_j(x_j, x')$ on J. By Step 1, $\partial_j u$ exists classically on Q and equals v_j. So all first partial derivatives of u exist and are continuous, and $u \in C^1(\Omega)$.

Step 3: Induction on k. Suppose the claim holds for $k - 1 \geq 1$. By Step 2, $u \in C^1(\Omega)$ with classical derivatives $\partial_j u$ equal to the distributional ones. The distributional derivatives $\partial^\beta(\partial_j u) = \partial^{\beta + e_j} u$ with $|\beta| \leq k-1$ are continuous, so $\partial_j u \in C^{k-1}(\Omega)$ by the induction hypothesis. Hence $u \in C^k(\Omega)$, as claimed.

Proposition 3.2.3: Further Result about Weak Derivative I

Let $u, v \in C([0,T], \mathcal{D}'(\mathbb{R}^n))$ and set $S_T = (0,T) \times \mathbb{R}^n$. If

$$\partial_t u = v \qquad \text{in } \mathcal{D}'(S_T)$$

that is, v is the distributional time derivative of u, then $u \in C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$ and $u' = v$

***Proof*:**
The hypothesis says that

$$\int_0^T \langle u(t), -\partial_t \psi(t,\cdot)\rangle \, dt = \int_0^T \langle v(t), \psi(t,\cdot)\rangle \, dt \qquad \text{for all } \psi \in C_c^\infty(S_T)$$

Fix $\varphi \in C_c^\infty(\mathbb{R}^n)$, and let $a(t) = \langle u(t), \varphi\rangle$ and $b(t) = \langle v(t), \varphi\rangle$, which are continuous on $[0,T]$. Taking $\psi(t,x) = \theta(t)\varphi(x)$ with $\theta \in C_c^\infty((0,T))$ gives $\int_0^T (a\theta' + b\theta)\, dt = 0$, that is, $a' = b$ in $\mathcal{D}'((0,T))$. By proposition 3.2.2 with $n = 1$ and $k = 1$, $a \in C^1((0,T))$ with $a' = b$, so $a(t) = a(t_0) + \int_{t_0}^t b$ for $t, t_0 \in (0,T)$. By continuity this identity extends to $t, t_0 \in [0,T]$, so a is differentiable on $[0,T]$, one-sidedly at the endpoints, with $a' = b$. Since φ was arbitrary, $\frac{d}{dt}\langle u(t), \varphi\rangle = \langle v(t), \varphi\rangle$ for every φ and t, and v is continuous, so $u \in C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$ with $u' = v$, as claimed.

Theorem 3.2.4: Unique Solution to Distributional Solution

The above Cauchy problem has at most one solution in the class $C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$.

***Proof Key ideas*:**
By linearity it suffices to show that a solution u with $F = 0$ and $f = g = 0$ vanishes. First, $u \in C^2([0,T], \mathcal{D}'(\mathbb{R}^n))$: in $\mathcal{D}'(S_T)$ we have $\partial_t(u') = \partial_t^2 u = \triangle u$, and $\triangle u \in C([0,T], \mathcal{D}'(\mathbb{R}^n))$, so proposition 3.2.3 applied to u' and $\triangle u$ gives $u' \in C^1$. Next, for $\varphi \in C^\infty(\mathbb{R}^{1+n})$ whose slices $\varphi(t,\cdot)$, $t \in [0,T]$, are supported in a fixed compact set, two integrations by parts in t give

$$\int_0^T \langle u(t), \Box\varphi(t,\cdot)\rangle \, dt = \int_0^T \langle u''(t) - \triangle u(t), \varphi(t,\cdot)\rangle \, dt + \Big[\langle u(t), \partial_t\varphi(t,\cdot)\rangle - \langle u'(t), \varphi(t,\cdot)\rangle\Big]_0^T$$

Take for φ a solution of $\Box\varphi = 0$ with data $\varphi(T,\cdot) = \varphi_0$ and $\partial_t\varphi(T,\cdot) = \varphi_1$ at $t = T$, for arbitrary $\varphi_0, \varphi_1 \in C_c^\infty(\mathbb{R}^n)$. Every term except $\langle u(T), \varphi_1\rangle - \langle u'(T), \varphi_0\rangle$ vanishes, so $u(T) = u'(T) = 0$, and the same argument on $[0,T']$ gives $u(T') = 0$ for every $T' \in (0,T]$.

The step not available at this point, for even n, is the test function φ: a smooth solution of the backward Cauchy problem with C_c^∞ data whose slices stay in a fixed compact set. The representation formulas of the previous chapter give it only for odd n. The theorem is [10, Week 4, Theorem 14], where the argument above is set as Exercise 16.

3.2.1 Energy Identity

The energy identity gives a limits to how much the solution to a wave equation, or more generally a solution who is always on compact support which includes *inhomogeneous wave equations*, could have grown in a finite amount of time. In particular, we will bound $\|\nabla u(t,\cdot)\|_{L^2}$, by the "total" amount of potential energy that could have accumulated in the system. We first need to establish some notation and a lemma. Recall that the gradient of a differentiable function $u \in \mathbb{R}^{1+n}$ is:

$$\nabla u = (u_t, \nabla_x u)$$

which gives a basis for the tangent space of the codomain of u given the standard coordinates.

Lemma 3.2.5: Energy Identity in Homogeneous Case

Suppose $u \in C^2([0,T] \times \mathbb{R}^n)$ solves $\Box u = 0$, and that $u(t,\cdot)$ is compactly supported for every $t \in [0,T]$. Then:

$$\|\partial u(t,\cdot)\|_{L^2} = \|\partial u(0,\cdot)\|_{L^2}$$

***Proof*:**
Define the energy identity:

$$e(t) = \frac{1}{2}\int_{\mathbb{R}^n} |\partial u(t,x)|^2 \,\mathrm{d}x = \frac{1}{2}\|\partial u(t,\cdot)\|_{L^2}^2$$

Differentiating with respect to t and integrating by parts, we get:

$$e'(t) = \int_{\mathbb{R}^n} u_t \Box u dx$$

since $\Box u = 0$, we get that $e'(t) = 0$, and so $e(t) = c$ is a constant, and so $e(0) = e(t)$, which lets us conclude that:

$$\|\partial u(0,\cdot)\|_{L^2} = \|\partial u(t,\cdot)\|_{L^2}$$

as we sought to show.

We now move on to u which can be inhomogeneous. In fact, this proof works even more generally than solutions to wave equations, though as we will see it will naturally be applicable for wave equations

Theorem 3.2.6: Energy Inequality (Identity)

Let $u \in C^2([0,T] \times \mathbb{R}^n)$ and $u(t,\cdot)$ be compactly supported for every $t \in [0,T]$. Then:

$$\|\partial u(t,\cdot)\|_{L^2} \leq \|\partial u(0,\cdot)\|_{L^2} + \int_0^t \|\Box u(s,\cdot)\|_{L^2} ds$$

***Proof*:**
We start like how we have done in lemma 3.2.5: define the energy identity

$$e(t) = \frac{1}{2}\int_{\mathbb{R}^n} |\partial u(t,x)|^2 \,\mathrm{d}x = \frac{1}{2}\|\partial u(t,\cdot)\|_{L^2}^2 \tag{3.6}$$

Differentiating with respect to t and integrating by parts, we get:

$$e'(t) = \int_{\mathbb{R}^n} u_t \Box u \, dx \tag{3.7}$$

Now, apply the Cauchy-Schwarz inequality (equivalently Hölder's inequality) to eq. (3.7), we get:

$$e'(t) \le \|\partial u(t,\cdot)\|_{L^2}\|\Box u(t,\cdot)\|_{L^2} = \sqrt{2e(t)}\|\Box u(t,\cdot)\|_{L^2}$$

Fix $\epsilon > 0$. The function $e_\epsilon = e + \epsilon$ is positive, and $e'_\epsilon = e' \le \sqrt{2e_\epsilon}\,\|\Box u(t,\cdot)\|_{L^2}$, so

$$\frac{d}{dt}\sqrt{e_\epsilon(t)} = \frac{e'(t)}{2\sqrt{e_\epsilon(t)}} \le \frac{1}{\sqrt{2}}\|\Box u(t,\cdot)\|_{L^2}$$

Integrating over $[0,t]$ and letting $\epsilon \to 0$ gives

$$\sqrt{e(t)} \le \sqrt{e(0)} + \frac{1}{\sqrt{2}}\int_0^t \|\Box u(s,\cdot)\|_{L^2} ds$$

which, when substituting in what $e(t)$ equals to from eq. (3.6), we get the inequality from the theorem, completing the proof.

3.2.2 Fourier Transform and Tempered Distributions

The Schwartz space, tempered distributions, and the Fourier transform on them are developed in the harmonic-analysis volume, [9]. We recall only what the wave-equation applications below require.

We now generalize earlier results of solving PDEs using Fourier transformations but use fourier transforms of distributions. Let $\mathcal{S}$ be the Schwartz space and $\mathcal{S}'$ is the dual of $\mathcal{S}$. Then naturally,

$$E' \subseteq \mathcal{S}' \subseteq \mathcal{D}'$$

where E' is the set of all compactly supported distributions. The dual $\mathcal{S}'$ is the space of *tempered distributions*, developed in [9, Tempered Distribution]; we use it freely below.

If $u \in \mathcal{S}'$, it has a *Fourier transform*:

$$\langle \hat{u}, \varphi\rangle = \langle u, \hat{\varphi}\rangle$$

If $u \in E'$, then $\hat{u}$ is a C^∞ function given by:

$$\hat{u}(\xi) = (2\pi)^{-n/2}\langle u, E_\xi\rangle$$

where $E_\xi \in C^\infty(\mathbb{R}^n)$ is $E_\xi(x) = e^{-ix\xi}$. To make sure that function multiplication still remains of Schwartz class, we will need a $\psi \in C^\infty(\mathbb{R}^n)$ to be slowly increasing, that is for all partial derivatives to have a growth at most:

$$|\partial^\alpha \psi(x)| \le C_\alpha(1+|x|)^{N(\alpha)} \qquad \forall \alpha$$

Then $\psi\varphi \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$. Thus, in this case, we can define for $u \in \mathcal{S}'$:

$$\langle \psi u, \varphi \rangle = \langle u, \psi\varphi \rangle$$

With this, we are ready to use Fourier transforms to solve the wave equation: let u be the solution of:

$$\Box u = 0 \quad \text{on } \mathbb{R}^{1+n}, \quad u|_{t=0} = f \quad \partial_t u|_{t=0} = g$$

with $f, g \in \mathcal{S}(\mathbb{R}^n)$. Applying the Fourier transform to the space of variables:

$$\hat{u}(t,\xi) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{-ix\xi} u(t,x) dx$$

Then $\Box u = 0$ transforms into:

$$\partial_t^2 \hat{u}(t,\xi) + |\xi|^2 \hat{u}(t,\xi) = 0 \qquad \hat{u}(0,\xi) = \hat{f}(\xi) \qquad \partial_t \hat{u}(0,\xi) = \hat{g}(\xi)$$

But as before, this is just a second order ODE with respect to t with solution:

$$\hat{u}(t,\xi) = \cos(t|\xi|)\hat{f}(\xi) + \frac{\sin(t|\xi|)}{|\xi|}\hat{g}(\xi) \tag{3.8}$$

Now, we have seen earlier that:

$$u(t,\cdot) = W'(t) * f + W(t) * g$$

solve the wave equation. Since $W(t) \in E'$, we can apply the Fourier transform to the above equation and get:

$$\hat{u}(t,\xi) = (2\pi)^{n/2}\widehat{W'(t)}(\xi)\hat{f}(\xi) + (2\pi)^{n/2}\widehat{W(t)}(\xi)\hat{g}(\xi)$$

Comparing this with eq. (3.8), we see that:

$$\widehat{W(t)}(\xi) = (2\pi)^{-n/2}\frac{\sin(t|\xi|)}{|\xi|} \qquad \widehat{W'(t)}(\xi) = (2\pi)^{-n/2}\cos(t|\xi|) \tag{3.9}$$

Giving us the full solution when $\Box u = 0$. For the inhomogeneous case, $\Box u = F$, with vanishing data at $t = 0$, we have:

$$u(t,x) = \int_0^t W(t-s) * F(s,x) ds$$

Applying the Fourier transform and applying eq. (3.9), we get:

$$\hat{u}(t,\xi) = \int_0^t \frac{\sin((t-s)|\xi|)}{|\xi|}\hat{F}(s,\xi) ds$$

Thus, the full solution to the inhomogeneous case:

$$\Box u = F \quad \text{on } \mathbb{R}^{1+n}, \quad u|_{t=0} = f \quad \partial_t u|_{t=0} = g$$

is:

$$\hat{u}(t,\xi) = \cos(t|\xi|)\hat{f}(\xi) + \frac{\sin(t|\xi|)}{|\xi|}\hat{g}(\xi) + \int_0^t \frac{\sin((t-s)|\xi|)}{|\xi|}\hat{F}(s,\xi) ds$$

3.2.3 Sobolev Space H^s

So far, we found solutions to PDEs lie in $\mathcal{D}'$. This is great in finding general solutions, however the space $\mathcal{D}'$ doens't give much in the way of control of the regularity of functions (i.e. how much the derivative varries). In later studies of solutions to PDEs it will be quite important that we have some control over the derivative so that the solution "stays" nice. So, we can think of Sobolev spaces as spaces that

1. have sufficinetly many derivatives
2. the functions have sufficiently nice regularity properties.

The Bessel-potential construction of the $H^s(\mathbb{R}^n)$ scale (the operator $\Lambda^s = \mathcal{F}^{-1}(1+|\xi|^2)^{s/2}\mathcal{F}$, $H^s = \Lambda^{-s}L^2$ with $\|f\|_{H^s} = \|(1+|\xi|^2)^{s/2}\hat{f}\|_{L^2}$, the nesting $H^t \subseteq H^s$ for $s < t$, and the integer-order characterization $H^s = \left\{f \in L^2 \;\middle|\; \partial^\alpha f \in L^2,\ |\alpha| \le s\right\}$) is developed in [9, Sobolev Spaces (H^s), Properties of H^s]. We use these facts in the energy estimates of chapter 5.

Theorem 3.2.7: Sobolev's Lemma

Let $s > k + \frac{n}{2}$ where k is a non-negative integer. Then:

$$H^s(\mathbb{R}^n) \subseteq C^k(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$$

where the inclusion is continuous. Even better:

$$\sum_{|\alpha| \le k} \|\partial^\alpha f\|_{L^\infty} \le C_s \|f\|_{H^s}$$

where C_s is independent of f

***Proof*:**
This is [9, Sobolev Embedding Theorem], which embeds $H^s(\mathbb{R}^n)$ continuously in $C^k(\mathbb{R}^n)$ with the displayed estimate.

4

Littlewood-Paley Theory

Littlewood-Paley theory splits a function into pieces whose Fourier transforms lie in dyadic annuli, so that L^2 methods apply to each piece and their results carry over to L^p. The calculus inequality and the Moser inequality proved here are the estimates behind the local existence theory of chapter 5.

4.1 Littlewood-Paley Decomposition

We decompose an L^p function (indeed any tempered distribution) into pieces whose Fourier transforms lie in dyadic annuli. Fix a radial function $\chi \in C_c^\infty(\mathbb{R}^n)$ that is non-increasing in $|\xi|$ and satisfies

$$0 \le \chi \le 1 \qquad \chi(\xi) = \begin{cases} 1 & \text{if } |\xi| \le \frac{1}{2} \\ 0 & \text{if } |\xi| \ge 1 \end{cases}$$

Then for each integer $j \ge 0$ define $S_j : \mathcal{S}' \to \mathcal{S}'$ by

$$\widehat{S_j f} = \chi(2^{-j}\xi)\hat{f}$$

that is S_j takes only a certain ranges of frequencies. Equivalently, we can write

$$S_j = \mathcal{F}^{-1}\chi(2^{-j}\cdot)\mathcal{F}$$

Lemma 4.1.1: decomposing Schwartz Functions

For every $\varphi \in \mathcal{S}$, $\chi(2^{-j}\cdot)\varphi \to \varphi$ in $\mathcal{S}$ as $j \to \infty$

***Proof*:**
Convergence in $\mathcal{S}$ means convergence in each of the norms $\|\psi\|_{(N,\alpha)} = \sup_x (1+|x|)^N |\partial^\alpha \psi(x)|$ that define the Schwartz space [9, Schwartz Space]. Put $\chi_j(x) = \chi(2^{-j}x)$ and fix N and α. By the Leibniz rule,

$$\partial^\alpha(\chi_j\varphi - \varphi) = (\chi_j - 1)\partial^\alpha\varphi + \sum_{0\neq\gamma\leq\alpha} \binom{\alpha}{\gamma} (\partial^\gamma \chi_j)(\partial^{\alpha-\gamma}\varphi)$$

For $\gamma \neq 0$ we have $\partial^\gamma \chi_j(x) = 2^{-j|\gamma|}(\partial^\gamma\chi)(2^{-j}x)$, so $|\partial^\gamma\chi_j| \leq 2^{-j}\|\partial^\gamma\chi\|_{L^\infty}$, and the corresponding term has $\|\cdot\|_{(N,0)}$ at most $\binom{\alpha}{\gamma} 2^{-j}\|\partial^\gamma\chi\|_{L^\infty}\|\varphi\|_{(N,\alpha-\gamma)}$. For the first term, $\chi_j(x) = 1$ when $|x| \leq 2^{j-1}$ and $|\chi_j - 1| \leq 1$ everywhere, so

$$\sup_x (1+|x|)^N |(\chi_j - 1)\partial^\alpha\varphi(x)| \leq \sup_{|x|>2^{j-1}} (1+|x|)^{-1}(1+|x|)^{N+1}|\partial^\alpha\varphi(x)| \leq 2^{1-j}\|\varphi\|_{(N+1,\alpha)}$$

Each bound tends to 0 as $j \to \infty$, so $\|\chi_j\varphi - \varphi\|_{(N,\alpha)} \to 0$ for all N and α, completing the proof.

Since $\mathcal{F}$ is an automorphism on $\mathcal{S}$, we get:

$$S_j f \to f \qquad \text{in } \mathcal{S}$$

for every $f \in \mathcal{S}$. this then implies that $S_j f \to f$ in $\mathcal{S}'$ for every $f \in \mathcal{S}'$. Next, we define the following decomposition similar to martingale-difference representation used in the proof of Azuma's inequality: define

$$\triangle_0 = S_0 \qquad \triangle_j = S_j - S_{j-1} \qquad j \in \mathbb{N}$$

Thus, we get $f = \sum_0^\infty \triangle_j f$ in $\mathcal{S}$ (resp. $\mathcal{S}'$) for every $f \in \mathcal{S}$ (resp. $f \in \mathcal{S}'$)

Definition 4.1.2: Littlewood-Paley Decomposition

The decomposition $\{\triangle_j f\}_0^\infty$ is called the *Littlewood-Paley decomposition* of f.

Notice that $\widehat{\triangle_j f} = \beta_j(\xi)\hat{f}$ where

$$\beta_0(\xi) = \chi(\xi) \qquad \beta_j(\xi) = \chi(2^{-j}\xi) - \chi(2^{1-j}\xi) \qquad j \in \mathbb{N} \tag{4.1}$$

Since χ is radial, non-increasing in $|\xi|$ and takes values in $[0,1]$, we get $0 \leq \beta_j \leq 1$ for all j, and the partial sums $\sum_0^J \beta_j(\xi) = \chi(2^{-J}\xi)$ tend to 1, so

$$\sum_0^\infty \beta_j^2 \leq \sum_0^\infty \beta_j = 1$$

Proposition 4.1.3: Littlewood-Paley Decomposition

Let $f \in \mathcal{S}'$, let $\{\triangle_j f\}_0^\infty$ be its Littlewood-Paley decomposition, and let $\psi, \varphi \in \mathcal{S}$ be given by $\hat{\psi} = (2\pi)^{-n/2}\chi$ and $\hat{\varphi}(\xi) = (2\pi)^{-n/2}(\chi(\xi) - \chi(2\xi))$. Then:

1. $\text{supp}\widehat{S_j f} \subseteq \{|\xi| \leq 2^j\}$
2. $\text{supp}\widehat{\triangle_j f} \subseteq \{2^{j-2} \leq |\xi| \leq 2^j\}$ for $j \geq 1$
3. $S_j f = 2^{jn}\psi(2^j \cdot) * f$ if $f \in L^p$ for $1 \leq p \leq \infty$
4. $\triangle_j f = 2^{jn}\varphi(2^j \cdot) * f$ for $j \geq 1$ if $f \in L^p$ for $1 \leq p \leq \infty$
5. $\|S_j f\|_{L^p} \leq C\|f\|_{L^p}$ where $C = \|\psi\|_{L^1}$, if $f \in L^p$ for $1 \leq p \leq \infty$
6. $\|\triangle_j f\|_{L^p} \leq C\|f\|_{L^p}$ where $C = \max(\|\psi\|_{L^1}, \|\varphi\|_{L^1})$, if $f \in L^p$ for $1 \leq p \leq \infty$
7. $S_j f \in C^\infty$ if $f \in L^p$ for $1 \leq p \leq \infty$
8. $\triangle_j f \in C^\infty$ if $f \in L^p$ for $1 \leq p \leq \infty$

***Proof*:**
The functions ψ and φ lie in $\mathcal{S}$ because $\mathcal{F}$ maps $\mathcal{S}$ onto itself [9, Fourier isomorphism on Schwartz Space], a fact unaffected by the normalization of $\mathcal{F}$. Write $\psi_j = 2^{jn}\psi(2^j \cdot)$ and $\varphi_j = 2^{jn}\varphi(2^j \cdot)$. The substitution $z = 2^j x$ in the integral defining $\mathcal{F}$ gives $\hat{\psi}_j(\xi) = \hat{\psi}(2^{-j}\xi) = (2\pi)^{-n/2}\chi(2^{-j}\xi)$, and for $j \geq 1$ it gives $\hat{\varphi}_j = (2\pi)^{-n/2}\beta_j$. Also $\|\psi_j\|_{L^1} = \|\psi\|_{L^1}$ and $\|\varphi_j\|_{L^1} = \|\varphi\|_{L^1}$.

1. $\widehat{S_j f} = \chi(2^{-j}\xi)\hat{f}$, and $\chi(2^{-j}\xi) = 0$ when $|\xi| \geq 2^j$.
2. For $j \geq 1$, $\widehat{\triangle_j f} = \beta_j \hat{f}$. If $|\xi| \geq 2^j$, both terms of $\beta_j(\xi)$ vanish. If $|\xi| < 2^{j-2}$, then $|2^{-j}\xi| < \frac{1}{4}$ and $|2^{1-j}\xi| < \frac{1}{2}$, so both terms equal 1 and $\beta_j(\xi) = 0$.
3. Let p' be the exponent conjugate to p. Since $\psi_j \in \mathcal{S} \subseteq L^{p'}$, Hölder's inequality [7, Hölder's Inequality] shows that $g = \psi_j * f$ is defined everywhere and $|g| \leq \|\psi_j\|_{L^{p'}}\|f\|_{L^p}$, so $g \in L^\infty \subseteq \mathcal{S}'$. Let $\eta \in \mathcal{S}$. The double integral of $|\psi_j(x-y)f(y)\hat{\eta}(x)|$ is at most $\|f\|_{L^p}\,\||\tilde{\psi}_j| * |\hat{\eta}|\|_{L^{p'}}$, where $\tilde{\psi}_j(z) = \psi_j(-z)$, which is finite by Young's convolution inequality [7, Young's Convolution Inequality]. Fubini's theorem therefore gives

$$\langle g, \hat{\eta}\rangle = \int f(y)\left(\int \psi_j(x-y)\hat{\eta}(x)\,dx\right)dy$$

Inserting $\hat{\eta}(x) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi}\eta(\xi)\,d\xi$ and integrating first in x, which Fubini's theorem allows since $\psi_j, \eta \in \mathcal{S}$, and using $\int \psi_j(x-y)e^{-ix\cdot\xi}\,dx = (2\pi)^{n/2}e^{-iy\cdot\xi}\hat{\psi}_j(\xi)$, the inner integral becomes

$$(2\pi)^{-n/2}\int \eta(\xi)\int \psi_j(x-y)e^{-ix\cdot\xi}\,dx\,d\xi = \int \eta(\xi)\hat{\psi}_j(\xi)e^{-iy\cdot\xi}\,d\xi$$

that is, the Fourier transform of $(2\pi)^{n/2}\hat{\psi}_j\eta = \chi(2^{-j}\cdot)\eta$ evaluated at y. Hence $\langle \hat{g}, \eta\rangle = \langle g, \hat{\eta}\rangle = \langle f, \mathcal{F}(\chi(2^{-j}\cdot)\eta)\rangle = \langle \chi(2^{-j}\cdot)\hat{f}, \eta\rangle = \langle \widehat{S_j f}, \eta\rangle$. As $\mathcal{F}$ maps $\mathcal{S}$ onto itself, a tempered distribution with vanishing Fourier transform vanishes, so $g = S_j f$.

4. For $j \geq 1$, the computation of (3) with φ_j in place of ψ_j and β_j in place of $\chi(2^{-j}\cdot)$ gives $\varphi_j * f = \triangle_j f$.

5. By (3) and Young's convolution inequality with exponents $(1, p)$, $\|S_j f\|_{L^p} \leq \|\psi_j\|_{L^1}\|f\|_{L^p} = \|\psi\|_{L^1}\|f\|_{L^p}$.

6. For $j = 0$ this is (5), since $\triangle_0 = S_0$. For $j \geq 1$, (4) and Young's convolution inequality give $\|\triangle_j f\|_{L^p} \leq \|\varphi\|_{L^1}\|f\|_{L^p}$.

7. Let $\theta \in \mathcal{S}$, which will be ψ_j or one of its derivatives. For $N > n$ and $|x| \leq R$, the inequality $1 + |y| \leq (1 + |x - y|)(1 + |x|)$ gives $|\theta(x - y)| \leq \|\theta\|_{(N,0)}(1 + R)^N(1 + |y|)^{-N}$, and $(1 + |y|)^{-N}|f(y)|$ is integrable by Hölder's inequality, since $(1 + |y|)^{-N} \in L^{p'}$. By the mean value theorem the difference quotients of $x \mapsto \theta(x - y)$ in a coordinate direction, with increments of size at most 1, are bounded by the same kind of function, with $\partial_k\theta$ and $R+1$ in place of θ and R. The Dominated Convergence Theorem [8, Dominated Convergence Theorem (DCT)] then shows that $\theta * f$ is continuous and that $\partial_k(\theta * f) = (\partial_k\theta) * f$. Applying this to $\theta = \partial^\alpha\psi_j$ for every multi-index α, induction on $|\alpha|$ shows that $S_j f = \psi_j * f$ has continuous partial derivatives of all orders.

8. For $j = 0$ this is (7). For $j \geq 1$, $\triangle_j f = S_j f - S_{j-1} f$ is a difference of two C^∞ functions by (7).

This proves all eight assertions, as claimed.

We will apply this theory to Sobolev spaces H^s. Recall that

$$H^s = \left\{ f \in \mathcal{S}' \;\middle|\; \Lambda^s f \in L^2 \right\}$$

where $\Lambda^s = (I - \triangle)^{s/2}$. Then H^s is a Hilbert space with the inner product

$$\langle f, g\rangle_{H^s} = \langle \Lambda^s f, \Lambda^s g\rangle_{L^2} = \int (1 + |\xi|^2)^s \hat{f}(\xi)\overline{\hat{g}(\xi)}d\xi$$

giving us norm $\|f\|_{H^s} = \|\Lambda^s f\|_{L^2}$, and $\Lambda^s : H^s \to L^2$ is a unitary isomorphism. By this formula, if $\hat{f}$ and $\hat{g}$ have almost disjoint supports, then $f \perp g$ in H^s. H^s is self-dual via the map $g \mapsto \langle \cdot, g\rangle_{H^s}$. Moreover, the dual of H^s can be identified with H^{-s} via the pairing $\langle \cdot, \cdot \rangle \colon H^s \times \mathcal{S} \to \mathbb{C}$, through

$$|\langle f, g\rangle| = |\langle \Lambda^s f, \Lambda^{-s} g\rangle| \leq \|f\|_{H^s}\|g\|_{H^{-s}}$$

for all $f \in H^s$ and $g \in \mathcal{S}$. Thus, the map $g \mapsto \langle \cdot, g\rangle$ extends to a linear map from H^{-s} into $(H^s)^*$, which is a surjective isometry in view of the self-duality of L^2.

(O1) $S_j f \to f$ in H^s for every $f \in H^s$, since:

$$\|S_j f - f\|_{H^s}^2 = \int [1 - \chi(2^{-j}\xi)]^2(1 + |\xi|^2)^s|\hat{f}(\xi)|^2 d\xi \to 0$$

by the DCT

(O2) If $2^{j-2} \leq |\xi| \leq 2^j$, then

$$(1 + |\xi|^2)^{s/2} \approx 2^{js}$$

in the sense that there is a constant $C_s > 0$, independent of j, such that

$$C_s^{-1} 2^{js} \leq (1 + |\xi|^2)^{s/2} \leq C_s 2^{js}$$

and the same holds for $j = 0$ on the ball $|\xi| \leq 1$, which contains the spectrum of $\triangle_0 f$. By Plancherel's theorem and proposition 4.1.3(1) and (2), we then get that

$$\|\triangle_j f\|_{H^s} \approx 2^{js} \|\triangle_j f\|_{L^2}$$

(O3) By proposition 4.1.3(1) and (2), we have

$$\triangle_j f \perp \triangle_k f \qquad \text{if} \qquad |j - k| \geq 2$$

relative to the inner product on H^s

With these, we have the following estimate:

Proposition 4.1.4: Estimating Sobolev Norm

Let $s \in \mathbb{R}$. Then for every $f \in H^s$,

$$\|f\|_{H^s}^2 \approx \sum_0^\infty 2^{2js} \|\triangle_j f\|_{L^2}^2$$

***Proof*:**
By (O2), it suffices to show that

$$\|f\|_{H^s}^2 \approx \sum_0^\infty \|\triangle_j f\|_{H^s}^2$$

By (O1), $f = \sum_j \triangle_j f$ in H^s. Using (O3), with $\triangle_{-1} = 0$, and the Cauchy-Schwarz inequality, we get

$$\begin{aligned} \|f\|_{H^s}^2 = \langle f, f \rangle_{H^s} = \left\langle \sum_j \triangle_j f, \sum_k \triangle_k f \right\rangle_{H^s} &= \sum_{l=-1}^{1} \sum_{j=0}^{\infty} \langle \triangle_j f, \triangle_{j+l} f \rangle_{H^s} \\ \leq \sum_{l=-1}^{1} \left(\sum_{j=0}^{\infty} \|\triangle_j f\|_{H^s}^2 \right)^{1/2} \left(\sum_{j=0}^{\infty} \|\triangle_{j+l} f\|_{H^s}^2 \right)^{1/2} &\leq 3 \sum_{j=0}^{\infty} \|\triangle_j f\|_{H^s}^2 \end{aligned}$$

On the other hand, by (4.1) and $\sum_j \beta_j^2 \leq 1$,

$$\sum_0^\infty \|\triangle_j f\|_{H^s}^2 = \sum_0^\infty \int (1 + |\xi|^2)^s |\beta_j(\xi) \hat{f}(\xi)|^2 d\xi = \int \left[\sum_0^\infty \beta_j^2(\xi) \right] (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi \leq \|f\|_{H^s}^2$$

completing the proof.

Using this result, we can find what types of sequences in L^2 would converge in H^s. Suppose $\{f_j\}$ is a sequence in L^2 such that for some $R \geq 1$,

$$\text{supp}\hat{f}_j \subseteq \{R^{-1}2^j \leq |\xi| \leq R2^j\}$$

and $\sum 2^{2js}\|f_j\|_{L^2}^2 < \infty$. Then $f = \sum f_j$ converges in H^s, and

$$\|f\|_{H^s}^2 \leq C_{s,R} \sum 2^{2js}\|f_j\|_{L^2}^2$$

If $R < \sqrt{2}$ the annuli are pairwise disjoint, so the two sides are comparable. If $s > 0$, we can get essentially the same result if we only assume

$$\text{supp}\hat{f}_j \subseteq \{|\xi| \leq R2^j\}$$

Lemma 4.1.5: build up lemma

Let $s \geq 0$ and $R \geq 1$. If $f \in L^2$ and

$$\text{supp}\hat{f} \subseteq \{|\xi| \leq R\}$$

then $f \in H^s$ and

$$\|f\|_{H^s} \leq 2^s R^s \|f\|_{L^2}$$

***Proof*:**
For $|\xi| \leq R$ we have $(1 + |\xi|^2)^s \leq (1 + R^2)^s \leq (2R^2)^s$, because $R \geq 1$ and $s \geq 0$. Since $\mathcal{F}$ is unitary on L^2 [9, The Plancherel Theorem], a fact unaffected by the normalization of $\mathcal{F}$,

$$\|f\|_{H^s}^2 = \int (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 \, d\xi \leq (2R^2)^s \|\hat{f}\|_{L^2}^2 = (2R^2)^s \|f\|_{L^2}^2$$

so $\|f\|_{H^s} \leq 2^{s/2} R^s \|f\|_{L^2} \leq 2^s R^s \|f\|_{L^2}$, as claimed.

Proposition 4.1.6: Better Bounding Condition

Let $s > 0$. Suppose $f_j \in L^2$ satisfies

$$\text{supp}\hat{f}_j \subseteq \{|\xi| \leq R2^j\}$$

for some $R \geq 1$ and every $j \geq 0$, and that

$$\sum_0^\infty 2^{2js}\|f_j\|_{L^2}^2 < \infty$$

Then $f = \sum f_j$ converges in H^s, and

$$\|f\|_{H^s}^2 \leq C_{s,R} \sum_0^\infty 2^{2js}\|f_j\|_{L^2}^2$$

***Proof*:**
Let $q \geq 0$ be the least integer with $R \leq 2^q$, and put $a_j = 2^{js}\|f_j\|_{L^2}$. Each f_j lies in H^s by lemma 4.1.5. For $j \leq k$, the integrand of $\langle f_j, f_k\rangle_{H^s} = \int (1+|\xi|^2)^s \hat{f}_j(\xi)\overline{\hat{f}_k(\xi)}\, d\xi$ vanishes outside $\{|\xi| \leq 2^{j+q}\}$, where $(1+|\xi|^2)^s \leq (2\cdot 4^{j+q})^s$. By the Cauchy-Schwarz inequality and the unitarity of $\mathcal{F}$ on L^2 [9, The Plancherel Theorem], which is unaffected by the normalization of $\mathcal{F}$,

$$|\langle f_j, f_k\rangle_{H^s}| \leq 2^s 2^{2qs} 2^{2js}\|f_j\|_{L^2}\|f_k\|_{L^2} = 2^s 2^{2qs} 2^{-(k-j)s} a_j a_k$$

Hence, for $0 \leq M \leq N$, writing $l = k - j$ and using $a_j a_{j+l} \leq \frac{1}{2}(a_j^2 + a_{j+l}^2)$,

$$\left\| \sum_{j=M}^{N} f_j \right\|_{H^s}^2 \leq \sum_{j,k=M}^{N} |\langle f_j, f_k\rangle_{H^s}| \leq 2^s 2^{2qs} \sum_{l\in\mathbb{Z}} 2^{-|l|s} \sum_{j=M}^{N} a_j^2 = C_{s,R} \sum_{j=M}^{N} a_j^2$$

with $C_{s,R} = 2^s 2^{2qs}\frac{2^s+1}{2^s-1}$, which is finite because $s > 0$. Since $\sum a_j^2 < \infty$, the partial sums of $\sum f_j$ form a Cauchy sequence in the Hilbert space H^s, so the series converges to some $f \in H^s$. Taking $M = 0$ and letting $N \to \infty$ gives $\|f\|_{H^s}^2 \leq C_{s,R}\sum_0^\infty 2^{2js}\|f_j\|_{L^2}^2$, completing the proof.

Note that if $R = 2^q$, then the constant $C_{s,R}$ is of the form $C_s 2^{2qs}$, which the proof of Moser's inequality (theorem 4.3.2) uses.

4.2 Calculus Inequality

The calculus inequality, on which section 5.4 relies, rests on the behaviour of spectra under products.

Lemma 4.2.1: Spectrum Of A Product

Let $u, v \in L^2(\mathbb{R}^n)$ have spectra in the balls $\{\xi \mid |\xi| \leq a\}$ and $\{\xi \mid |\xi| \leq b\}$, and let v be bounded. Then $uv \in L^2$, and its spectrum lies in $\{\xi \mid |\xi| \leq a + b\}$.

***Proof*:**
Since v is bounded, $uv \in L^2$, and $uv \in L^1$ as a product of two functions in L^2, so $\widehat{uv}$ is the continuous function

$$\widehat{uv}(\xi) = (2\pi)^{-n/2}\int u(x)v(x)e^{-ix\cdot\xi}\, dx = (2\pi)^{-n/2}\langle u, w_\xi\rangle_{L^2} \qquad w_\xi(x) = \overline{v(x)}e^{ix\cdot\xi}$$

For $v \in \mathcal{S}$ a direct computation gives $\widehat{w_\xi}(\zeta) = \overline{\hat{v}(\xi-\zeta)}$, and this identity extends to $v \in L^2$ by the Plancherel theorem [9, The Plancherel Theorem], whose unitarity statement is unaffected by the normalization of $\mathcal{F}$. By the same theorem $\langle u, w_\xi\rangle_{L^2} = \langle \hat{u}, \widehat{w_\xi}\rangle_{L^2}$, so

$$\widehat{uv}(\xi) = (2\pi)^{-n/2}\int \hat{u}(\zeta)\hat{v}(\xi-\zeta)\, d\zeta$$

If $|\xi| > a + b$, then for every ζ either $|\zeta| > a$ or $|\xi - \zeta| > b$, so the integrand vanishes almost everywhere and $\widehat{uv}(\xi) = 0$, as claimed.

Theorem 4.2.2: Calculus Inequality

Let $s \geq 0$. There is a constant C_s such that for all $f, g \in H^s \cap L^\infty$, the product fg lies in H^s and

$$\|fg\|_{H^s} \leq C_s(\|f\|_{H^s}\|g\|_{L^\infty} + \|f\|_{L^\infty}\|g\|_{H^s})$$

***Proof*:**
For $s = 0$ this is $\|fg\|_{L^2} \leq \|f\|_{L^2}\|g\|_{L^\infty}$, so assume $s > 0$. Put $S_{-1} = 0$, $c = \|\psi\|_{L^1}$ with ψ as in proposition 4.1.3, and for $j \geq 0$

$$h_j = S_j f\, S_j g - S_{j-1} f\, S_{j-1} g = (\triangle_j f)(S_j g) + (S_{j-1} f)(\triangle_j g)$$

Step 1: The pieces. By proposition 4.1.3(1), (2) and (5), the functions $\triangle_j f$, $S_j g$, $S_{j-1} f$ and $\triangle_j g$ lie in $L^2 \cap L^\infty$ with spectra in $\{\xi \mid |\xi| \leq 2^j\}$, and $\|S_j g\|_{L^\infty} \leq c\|g\|_{L^\infty}$ and $\|S_{j-1} f\|_{L^\infty} \leq c\|f\|_{L^\infty}$. By lemma 4.2.1, $h_j \in L^2$ has spectrum in $\{\xi \mid |\xi| \leq 2 \cdot 2^j\}$, and

$$\|h_j\|_{L^2} \leq c\,(\|\triangle_j f\|_{L^2}\|g\|_{L^\infty} + \|f\|_{L^\infty}\|\triangle_j g\|_{L^2})$$

Step 2: Summing. By proposition 4.1.4 there is a constant C with $\sum_j 2^{2js}\|\triangle_j f\|_{L^2}^2 \leq C\|f\|_{H^s}^2$, and likewise for g, so

$$\sum_{j=0}^{\infty} 2^{2js}\|h_j\|_{L^2}^2 \leq 2c^2 C\left(\|f\|_{H^s}^2\|g\|_{L^\infty}^2 + \|f\|_{L^\infty}^2\|g\|_{H^s}^2\right) < \infty$$

By proposition 4.1.6 with $R = 2$, the series $\sum_j h_j$ converges in H^s to some h, with $\|h\|_{H^s}^2 \leq C_{s,2}\sum_j 2^{2js}\|h_j\|_{L^2}^2$.

Step 3: The limit is fg. The partial sums telescope, $\sum_{j=0}^{N} h_j = S_N f\, S_N g$, and

$$\|S_N f\, S_N g - fg\|_{L^2} \leq \|S_N f - f\|_{L^2}\, c\|g\|_{L^\infty} + \|f\|_{L^\infty}\|S_N g - g\|_{L^2} \to 0$$

by (O1) with $s = 0$. As convergence in H^s implies convergence in L^2 for $s > 0$, $h = fg$. Taking square roots in the bound of Step 2 gives the inequality with $C_s = (2c^2 C C_{s,2})^{1/2}$, as claimed.

4.3 Moser Inequality

The Moser inequality rests on a generalization of lemma 4.1.5 to L^p, in which the *spectrum* of f is the support of its Fourier transform. Throughout, ψ, φ and $\psi_j = 2^{jn}\psi(2^j\cdot)$ are as in proposition 4.1.3, and if a smooth function m equals 1 on a neighbourhood of the spectrum of f, then $m\hat{f} = \hat{f}$.

Lemma 4.3.1: Bernstein's Lemma

Let $j \geq 0$ be an integer, and assume that the spectrum of $f \in L^p$, $1 \leq p \leq \infty$, is contained in the ball $|\xi| \leq 2^j$. Then:

$$\|\partial^\alpha f\|_{L^p} \leq C_\alpha 2^{j|\alpha|}\|f\|_{L^p}$$

for any multi-index α. Moreover, if the spectrum is contained in $2^{j-2} \leq |\xi| \leq 2^j$, then

$$C_k^{-1}2^{jk}\|f\|_{L^p} \leq \sup_{|\alpha|=k}\|\partial^\alpha f\|_{L^p} \leq C_k 2^{jk}\|f\|_{L^p}$$

for any $k \in \mathbb{N}$

***Proof*:**

Step 1: The ball. Since $\chi(2^{-j-2}\xi) = 1$ for $|\xi| \leq 2^{j+1}$, a neighbourhood of the spectrum of f, we have $\widehat{S_{j+2}f} = \hat{f}$, so $f = S_{j+2}f = \psi_{j+2} * f$ by proposition 4.1.3(3) and the injectivity of $\mathcal{F}$ on $\mathcal{S}'$ [9, Properties Of The Fourier Transform On $\mathcal{S}'$]. As in the proof of proposition 4.1.3(7), $\partial^\alpha f = (\partial^\alpha \psi_{j+2}) * f$, and $\partial^\alpha \psi_{j+2}(x) = 2^{(j+2)|\alpha|}2^{(j+2)n}(\partial^\alpha\psi)(2^{j+2}x)$ has L^1 norm $2^{(j+2)|\alpha|}\|\partial^\alpha\psi\|_{L^1}$. Young's convolution inequality [7, Young's Convolution Inequality] gives the first claim with $C_\alpha = 4^{|\alpha|}\|\partial^\alpha\psi\|_{L^1}$. The upper bound in the second claim follows, with $C_k \geq \max_{|\alpha|=k} C_\alpha$.

Step 2: The lower bound. Fix $\theta \in C_c^\infty(\mathbb{R}^n)$ with $\theta = 1$ on $\left\{\eta \;\middle|\; \frac{1}{8} \leq |\eta| \leq 2\right\}$ and $\theta = 0$ on $\left\{\eta \;\middle|\; |\eta| \leq \frac{1}{16}\right\}$. For $|\alpha| = k$ put

$$m_\alpha(\eta) = \frac{k!}{\alpha!}\frac{\eta^\alpha\theta(\eta)}{|\eta|^{2k}}$$

which lies in $C_c^\infty(\mathbb{R}^n)$ because θ vanishes near 0. Let $K_\alpha \in \mathcal{S}$ have $\hat{K}_\alpha = (2\pi)^{-n/2}m_\alpha$, and $K_{\alpha,j} = 2^{jn}K_\alpha(2^j\cdot)$, so that $\hat{K}_{\alpha,j}(\xi) = (2\pi)^{-n/2}m_\alpha(2^{-j}\xi)$ and $\|K_{\alpha,j}\|_{L^1} = \|K_\alpha\|_{L^1}$. The multinomial identity $|\eta|^{2k} = \sum_{|\alpha|=k}\frac{k!}{\alpha!}\eta^{2\alpha}$ gives $\sum_{|\alpha|=k} m_\alpha(\eta)\eta^\alpha = \theta(\eta)$. In the normalization of $\mathcal{F}$, $\widehat{\partial^\alpha f} = (i\xi)^\alpha\hat{f}$ (the derivative rule of [9, Properties Of The Fourier Transform On $\mathcal{S}'$], which reads $(2\pi i\xi)^\alpha$ in the normalization $e^{-2\pi ix\cdot\xi}$ used there), so

$$\sum_{|\alpha|=k} m_\alpha(2^{-j}\xi)\,\widehat{\partial^\alpha f}(\xi) = i^k 2^{jk}\sum_{|\alpha|=k} m_\alpha(2^{-j}\xi)(2^{-j}\xi)^\alpha\hat{f}(\xi) = i^k 2^{jk}\theta(2^{-j}\xi)\hat{f}(\xi) = i^k 2^{jk}\hat{f}(\xi)$$

because $\theta(2^{-j}\cdot) = 1$ on a neighbourhood of the spectrum of f. Each $\partial^\alpha f$ lies in L^p by Step 1, and the computation in the proof of proposition 4.1.3(3), with $K_{\alpha,j}$ in place of ψ_j, gives $\mathcal{F}(K_{\alpha,j} * \partial^\alpha f) = m_\alpha(2^{-j}\cdot)\widehat{\partial^\alpha f}$. Hence

$$f = i^{-k}2^{-jk}\sum_{|\alpha|=k} K_{\alpha,j} * \partial^\alpha f$$

and Young's convolution inequality gives $\|f\|_{L^p} \leq 2^{-jk}\sum_{|\alpha|=k}\|K_\alpha\|_{L^1}\sup_{|\alpha|=k}\|\partial^\alpha f\|_{L^p}$. Enlarging C_k to at least $\sum_{|\alpha|=k}\|K_\alpha\|_{L^1}$ gives the lower bound, as claimed.

Theorem 4.3.2: Moser Inequality

Assume that $F : \mathbb{R}^N \to \mathbb{R}$ is C^∞ and $F(0) = 0$. Then for all $s \geq 0$, there is a continuous function $\gamma = \gamma_s : \mathbb{R} \to \mathbb{R}$ such that

$$\|F(f)\|_{H^s} \leq \gamma(\|f\|_{L^\infty})\|f\|_{H^s}$$

for all $\mathbb{R}^N$-valued $f \in H^s \cap L^\infty$

***Proof*:**
Here $DF(z)$ is the gradient of F at $z \in \mathbb{R}^N$, the operators S_j and $\triangle_j$ act on each component, and C denotes constants depending only on n, N, s and F. Fix f and put $r = \|f\|_{L^\infty}$.

Step 1: The case $s = 0$. By the mean value theorem $|F(f(x))| \leq \gamma_0(r)|f(x)|$, where $\gamma_0(r) = \sup_{|z|\leq r} |DF(z)|$ is continuous, and squaring and integrating gives the claim. From now on $s > 0$.

Step 2: Telescoping. Put $S_{-1} = 0$ and $c_1 = \max(\|\psi\|_{L^1}, \|\varphi\|_{L^1})$. By proposition 4.1.3(7) and (8), applied to each component, $S_j f$ and $\triangle_j f$ are smooth, and by the convolution formulas proposition 4.1.3(3) and (4), with $\triangle_0 f = S_0 f$, their sup norms are at most $c_1 r$. For $j \geq 0$ and $0 \leq \lambda \leq 1$ put $g_{j,\lambda} = S_{j-1}f + \lambda \triangle_j f$, so that $|g_{j,\lambda}| \leq 2c_1 r$ and, since $S_j f = S_{j-1}f + \triangle_j f$,

$$F(S_j f) - F(S_{j-1}f) = \int_0^1 \frac{d}{d\lambda} F(g_{j,\lambda})\, d\lambda = m_j \cdot \triangle_j f \qquad m_j = \int_0^1 DF(g_{j,\lambda})\, d\lambda$$

Summing, $F(S_p f) - F(S_q f) = \sum_{j=q+1}^{p} m_j \cdot \triangle_j f$ for $p > q \geq -1$.

Step 3: Derivatives of m_j. For $l \geq 0$ let $G_l(r) = \sup_{|z|\leq 2c_1 r} |D^{l+1}F(z)|$, a continuous increasing function. By the chain rule and the product rule, and induction on $|\alpha|$, for $\alpha \neq 0$ the derivative $\partial^\alpha[DF(g)]$ is a linear combination, with coefficients depending only on α, of terms $D^{l+1}F(g)(\partial^{\alpha_1}g, \ldots, \partial^{\alpha_l}g)$ with $1 \leq l \leq |\alpha|$ and $\alpha_1 + \cdots + \alpha_l = \alpha$. The spectra of $S_{j-1}f$ and $\triangle_j f$ lie in $|\xi| \leq 2^j$ by proposition 4.1.3(1) and (2), hence so does that of $g_{j,\lambda}$, and lemma 4.3.1 with $p = \infty$ gives $\|\partial^{\alpha_i} g_{j,\lambda}\|_{L^\infty} \leq C 2^{j|\alpha_i|} r$. Differentiating under the integral in λ, we get

$$\|\partial^\alpha m_j\|_{L^\infty} \leq c_\alpha(r) 2^{j|\alpha|} \qquad c_\alpha(r) = C \sum_{l \leq |\alpha|} G_l(r)(1+r)^l$$

with c_α continuous, and $c_0(r) = G_0(r)$.

Step 4: High frequencies of m_j. Let $k \geq 1$ and $M \geq 1$. The spectrum of $\triangle_{j+k} m_j$ lies in $2^{j+k-2} \leq |\xi| \leq 2^{j+k}$, so the second part of lemma 4.3.1 gives

$$\|\triangle_{j+k} m_j\|_{L^\infty} \leq C_M 2^{-M(j+k)} \sup_{|\alpha|=M} \|\partial^\alpha \triangle_{j+k} m_j\|_{L^\infty}$$

Differentiating under the integral sign in $\triangle_{j+k} m_j = \varphi_{j+k} * m_j$ gives $\partial^\alpha \triangle_{j+k} m_j = \varphi_{j+k} * \partial^\alpha m_j$, whose sup norm is at most $c_1 c_\alpha(r) 2^{jM}$ by Step 3. Hence, with $c'_M(r) = C_M c_1 \max_{|\alpha|=M} c_\alpha(r)$,

$$\|\triangle_{j+k} m_j\|_{L^\infty} \leq c'_M(r) 2^{-Mk}$$

Step 5: Splitting m_j. Write $m_j = DF(0) + m_j'$ with $m_j' = \int_0^1 (DF(g_{j,\lambda}) - DF(0))\,d\lambda$. As $|DF(z) - DF(0)| \le G_1(r)|z|$ for $|z| \le 2c_1 r$, we have $|m_j'| \le G_1(r)(|S_{j-1}f| + |\triangle_j f|)$, so $m_j' \in L^2 \cap L^\infty$, with the same derivatives as m_j. The constant $DF(0)$ has $S_l DF(0) = DF(0)$ and $\triangle_l DF(0) = 0$ for $l \ge 1$, since $\int \psi_l = \chi(0) = 1$ and $\int \varphi_l = 0$. By (O1) with $s = 0$, $m_j' = S_j m_j' + \sum_{k\ge 1} \triangle_{j+k} m_j'$ in L^2, and $\triangle_j f$ is bounded, so in L^2

$$F(S_p f) - F(S_q f) = A + \sum_{k\ge 1} B_k \qquad A = \sum_{j=q+1}^{p} (S_j m_j)\cdot \triangle_j f \qquad B_k = \sum_{j=q+1}^{p} (\triangle_{j+k} m_j)\cdot \triangle_j f$$

By lemma 4.2.1, applied to the components of $S_j m_j' = S_j m_j - DF(0)$, of $\triangle_{j+k} m_j' = \triangle_{j+k} m_j$ and of $\triangle_j f$, the spectrum of $(S_j m_j)\cdot \triangle_j f$ lies in $|\xi| \le 2 \cdot 2^j$, and that of $(\triangle_{j+k} m_j)\cdot \triangle_j f$ in $|\xi| \le 2^{k+1} 2^j$.

Step 6: The key sum. For $j \ge 1$ the spectrum of $\triangle_j f$ lies in $|\xi| \ge 2^{j-2}$, where $(1+|\xi|^2)^s \ge 2^{-4s} 2^{2js}$, and for $j = 0$ we have $(1+|\xi|^2)^s \ge 1$. So $2^{2js}\|\triangle_j f\|_{L^2}^2 \le 2^{4s}\|\triangle_j f\|_{H^s}^2$. Since $\beta_j \ge 0$ and $\widehat{S_p f - S_q f} = (\sum_{j=q+1}^p \beta_j)\hat f$,

$$\sum_{j=q+1}^{p} 2^{2js}\|\triangle_j f\|_{L^2}^2 \le 2^{4s}\int (1+|\xi|^2)^s \sum_{j=q+1}^{p} \beta_j^2 |\hat f|^2\,d\xi \le 2^{4s}\|S_p f - S_q f\|_{H^s}^2$$

Step 7: Estimates. By proposition 4.1.3(5), $\|(S_j m_j)\cdot \triangle_j f\|_{L^2} \le c_1 c_0(r)\|\triangle_j f\|_{L^2}$, so proposition 4.1.6 with $R = 2$ and Step 6 give

$$\|A\|_{H^s} \le C_{s,2}^{1/2} c_1 c_0(r) 2^{2s}\|S_p f - S_q f\|_{H^s}$$

By Step 4 and proposition 4.1.6 with $R = 2^{k+1}$, whose constant is $C_s 2^{2(k+1)s}$ by the remark after its proof,

$$\|B_k\|_{H^s} \le C_s^{1/2} 2^{(k+1)s} c_M'(r) 2^{-Mk} 2^{2s}\|S_p f - S_q f\|_{H^s}$$

Take M to be the least integer greater than s. Then $\sum_k \|B_k\|_{H^s}$ converges, the series $\sum_k B_k$ converges in H^s to its L^2 sum, and

$$\|F(S_p f) - F(S_q f)\|_{H^s} \le \gamma(r)\|S_p f - S_q f\|_{H^s} \qquad p > q \ge -1 \tag{4.2}$$

where $\gamma(r) = 2^{2s}\left(C_{s,2}^{1/2} c_1 c_0(r) + C_s^{1/2} 2^s c_M'(r)\sum_{k\ge 1} 2^{(s-M)k}\right)$ is continuous.

Step 8: The limit. Taking $q = -1$ in (4.2), and using $F(0) = 0$ and $\|S_p f\|_{H^s} \le \|f\|_{H^s}$, gives $\|F(S_p f)\|_{H^s} \le \gamma(r)\|f\|_{H^s}$. Since $S_p f \to f$ in H^s by (O1), (4.2) shows that $(F(S_p f))_p$ is Cauchy in H^s, with a limit G. A subsequence of $S_p f$ converges to f almost everywhere, so $F(S_p f) \to F(f)$ almost everywhere along it, and a further subsequence converges to G almost everywhere, because convergence in H^s implies convergence in L^2. Hence $G = F(f)$, and $\|F(f)\|_{H^s} = \lim_p \|F(S_p f)\|_{H^s} \le \gamma(\|f\|_{L^\infty})\|f\|_{H^s}$, completing the proof.

A map between Banach spaces is differentiable at a point if it has a bounded linear approximation there with an error that is o of the increment, and *smooth* if all its iterated derivatives exist and are continuous.

Corollary 4.3.3: Moser Inequality Consequence

Let F be as in theorem 4.3.2. If $s > \frac{n}{2}$, the map $f \mapsto F(f)$, $H^s \to H^s$ is smooth

***Proof*:**
For a smooth $G\colon \mathbb{R}^N \to \mathbb{R}^K$ write $\Psi_G(f) = G(f) - G(0)$. Below, H^s denotes $H^s(\mathbb{R}^n;\mathbb{R}^d)$ for the relevant d.

Step 1: An algebra. By theorem 4.2.2 and Sobolev's lemma (theorem 3.2.7 with $k = 0$), $\|ab\|_{H^s} \le C_s(\|a\|_{H^s}\|b\|_{L^\infty} + \|a\|_{L^\infty}\|b\|_{H^s}) \le C\|a\|_{H^s}\|b\|_{H^s}$ for scalar $a, b \in H^s$, and the same holds for products of matrix-valued and vector-valued functions, componentwise. Also $\|f\|_{L^\infty} \le C\|f\|_{H^s}$.

Step 2: Ψ_G maps H^s into H^s. Theorem 4.3.2, applied to each component of $G - G(0)$, gives $\|\Psi_G(f)\|_{H^s} \le \gamma(\|f\|_{L^\infty})\|f\|_{H^s}$, which is bounded on bounded subsets of H^s.

Step 3: Differentiability. Let $f, h \in H^s$ with $\|h\|_{H^s} \le 1$. Taylor's formula with integral remainder gives, pointwise,

$$G(f+h) - G(f) - DG(f)h = \sum_{a,b=1}^{N} R_{ab}(f,h)\, h_a h_b \qquad R_{ab}(y,z) = \int_0^1 (1-\lambda)\, \partial_a\partial_b G(y+\lambda z)\, d\lambda$$

where R_{ab} is a smooth function of $(y,z) \in \mathbb{R}^{2N}$. Writing $R_{ab}(f,h) = R_{ab}(0,0) + \Psi_{R_{ab}}(f,h)$ and using Steps 1 and 2 for the $\mathbb{R}^{2N}$-valued function (f,h),

$$\|R_{ab}(f,h)h_a h_b\|_{H^s} \le C\left(|R_{ab}(0,0)| + \|\Psi_{R_{ab}}(f,h)\|_{H^s}\right)\|h\|_{H^s}^2 = O(\|h\|_{H^s}^2)$$

Similarly $DG(f)h = DG(0)h + \Psi_{DG}(f)h$, so the linear map $L_f\colon h \mapsto DG(f)h$ is bounded on H^s. Hence Ψ_G is differentiable at f, with derivative L_f.

Step 4: Smoothness. The map $f \mapsto L_f$ is the constant $DG(0)$ plus the composition of Ψ_{DG} with the bounded linear map $A \mapsto (h \mapsto Ah)$, from $H^s(\mathbb{R}^n;\mathbb{R}^{K\times N})$ to the bounded operators $H^s(\mathbb{R}^n;\mathbb{R}^N) \to H^s(\mathbb{R}^n;\mathbb{R}^K)$, which is bounded by Step 1. By Step 3 applied to the smooth function DG, Ψ_{DG} is differentiable, hence continuous, so Ψ_G has a continuous derivative. If Ψ_G has continuous derivatives up to order m for every smooth G, then so does Ψ_{DG}, and therefore the derivative of Ψ_G has continuous derivatives up to order m, that is, Ψ_G has them up to order $m+1$. By induction Ψ_G is smooth for every smooth G, and $\Psi_F(f) = F(f)$ because $F(0) = 0$, as claimed.

4.4 Further Applications of Littlewood-Paley Theory

The L^p behaviour of the Littlewood-Paley pieces gives the embedding of H^s into $L^{2n/(n-2s)}$ for $0 \le s < \frac{n}{2}$, and Sobolev's lemma gives the embedding into L^∞ for $s > \frac{n}{2}$. The second embedding fails at $s = \frac{n}{2}$, and as a consequence $H^{n/2}$ is not an algebra.

Theorem 4.4.1: Generalizing to L^p

Let $1 < p < \infty$. There is a constant C such that for every $f \in L^p$ the following hold. If $2 \leq p$, then

$$\|f\|_{L^p}^2 \leq C \sum_{j=0}^{\infty} \|\triangle_j f\|_{L^p}^2$$

and if $p \leq 2$, then

$$\sum_{j=0}^{\infty} \|\triangle_j f\|_{L^p}^2 \leq C \|f\|_{L^p}^2$$

***Proof Key ideas*:**
Both inequalities follow from the Littlewood-Paley square function theorem, which asserts for $1 < p < \infty$ and $f \in L^p$ that

$$\|f\|_{L^p} \approx \Big\| \Big(\sum_j |\triangle_j f|^2 \Big)^{1/2} \Big\|_{L^p}$$

For $p \geq 2$, Minkowski's inequality in $L^{p/2}$ gives $\| \sum_j |\triangle_j f|^2 \|_{L^{p/2}} \leq \sum_j \|\triangle_j f\|_{L^p}^2$, and for $p \leq 2$ the reverse Minkowski inequality for nonnegative functions in $L^{p/2}$ gives the opposite inequality. The standard proof of the square function theorem applies the Calderón-Zygmund theorem to the operator $f \mapsto (\triangle_j f)_{j \geq 0}$, which takes values in the Hilbert space ℓ^2. The prerequisites state only the scalar form of that theorem [9, Calderón–Zygmund Theorem], and this book does not develop the vector-valued form. The statement is [10, Week 6, Theorem 27], where it is also stated without proof. Duoandikoetxea proves the square function bound for $(\triangle_j f)_{j \geq 1}$ by this route [4, Theorem 8.6], and the reverse bound follows from it by duality, as in the proof of [4, Theorem 8.4].

Theorem 4.4.2: Embeddings For H^s

Let $s \geq 0$. If $s < \frac{n}{2}$, then

$$H^s \hookrightarrow L^{\frac{2n}{n-2s}} \tag{4.3}$$

and if $s > \frac{n}{2}$, then

$$H^s \hookrightarrow L^{\infty} \tag{4.4}$$

Here $X \hookrightarrow Y$ means that $X \subseteq Y$ and that the inclusion is continuous.

***Proof*:**
The embedding (4.4) is the case $k = 0$ of theorem 3.2.7, which gives $\|f\|_{L^\infty} \leq C_s \|f\|_{H^s}$ for all $f \in H^s$ when $s > \frac{n}{2}$.

For (4.3), put $q = \frac{2n}{n-2s}$, so that $2 \leq q < \infty$ and $\frac{s}{n} = \frac{1}{2} - \frac{1}{q}$, and define $r \in [1, 2)$ by $\frac{1}{r} = \frac{1}{2} + \frac{1}{q}$. With ψ and φ as in proposition 4.1.3, put $\theta_0 = \psi$ and $\theta_j = 2^{jn} \varphi(2^j \cdot)$ for $j \geq 1$, and $K = \max(\|\psi\|_{L^r}, \|\varphi\|_{L^r})$, which is finite because $\psi, \varphi \in \mathcal{S}$.

Step 1: One piece. For $j \geq 1$ the substitution $y = 2^j x$ gives $\|\theta_j\|_{L^r} = 2^{jn(1-1/r)}\|\varphi\|_{L^r} = 2^{js}\|\varphi\|_{L^r}$, so $\|\theta_j\|_{L^r} \leq K2^{js}$ for all $j \geq 0$. For $h \in L^2$, proposition 4.1.3(3) and (4) give $\triangle_j h = \theta_j * h$ (recall $\triangle_0 = S_0$), and Young's convolution inequality [7, Young's Convolution Inequality], with exponents r, 2 and q, gives

$$\|\triangle_j h\|_{L^q} \leq K2^{js}\|h\|_{L^2} \qquad j \geq 0$$

Step 2: The bound on $H^s \cap L^q$. Let $g \in H^s \cap L^q$, so that $g \in L^2$ because $s \geq 0$. Put $\triangle_{-1} = 0$ and $h_j = \triangle_{j-1}g + \triangle_j g + \triangle_{j+1}g \in L^2$. By (4.1) the partial sums are $\sum_{k=0}^{J} \beta_k = \chi(2^{-J}\cdot)$, so $\beta_{j-1} + \beta_j + \beta_{j+1}$ equals $\chi(2^{-j-1}\cdot) - \chi(2^{2-j}\cdot)$ for $j \geq 2$ and $\chi(2^{-j-1}\cdot)$ for $j \in \{0, 1\}$. As $\chi = 1$ on $\left\{\eta \;\middle|\; |\eta| \leq \frac{1}{2}\right\}$ and $\chi = 0$ on $\{\eta \mid |\eta| \geq 1\}$, this sum equals 1 on $\left\{\xi \;\middle|\; 2^{j-2} \leq |\xi| \leq 2^j\right\}$ for $j \geq 1$ and on $\{\xi \mid |\xi| \leq 1\}$ for $j = 0$, and β_j vanishes outside these sets, as in the proof of proposition 4.1.3(2). Hence $\beta_j(\beta_{j-1} + \beta_j + \beta_{j+1}) = \beta_j$ and $\triangle_j h_j = \triangle_j g$. By Step 1 and $2^{js} \leq 2^s 2^{ks}$ for $k \geq j - 1$,

$$\|\triangle_j g\|_{L^q} = \|\triangle_j h_j\|_{L^q} \leq K2^s \sum_{k=j-1}^{j+1} 2^{ks}\|\triangle_k g\|_{L^2}$$

Squaring, using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$ and summing over $j \geq 0$, where each k occurs for at most three values of j, and then applying proposition 4.1.4, which gives a constant C_1 with $\sum_k 2^{2ks}\|\triangle_k g\|_{L^2}^2 \leq C_1\|g\|_{H^s}^2$, we get

$$\sum_{j=0}^{\infty} \|\triangle_j g\|_{L^q}^2 \leq 9K^2 4^s \sum_{k=0}^{\infty} 2^{2ks}\|\triangle_k g\|_{L^2}^2 \leq 9K^2 4^s C_1\|g\|_{H^s}^2$$

Since $g \in L^q$ and $2 \leq q < \infty$, theorem 4.4.1 bounds $\|g\|_{L^q}^2$ by a constant times the left hand side, so $\|g\|_{L^q} \leq C_2\|g\|_{H^s}$ with C_2 independent of g.

Step 3: All of H^s. Let $f \in H^s$, so that $f \in L^2$. For $N \geq 0$, $S_N f = \psi_N * f$ by proposition 4.1.3(3), where $\psi_N = 2^{Nn}\psi(2^N\cdot) \in L^r$, so $S_N f \in L^q$ by Young's convolution inequality, and $S_N f \in H^s$ because $|\chi| \leq 1$. By (O1), $S_N f \to f$ in H^s, and hence in L^2. Step 2 applied to $S_N f - S_M f \in H^s \cap L^q$ shows that $(S_N f)_N$ is a Cauchy sequence in L^q, so it converges in L^q to some g. A subsequence converges almost everywhere both to g and to f, so $f = g \in L^q$ and

$$\|f\|_{L^q} = \lim_{N\to\infty} \|S_N f\|_{L^q} \leq C_2 \lim_{N\to\infty} \|S_N f\|_{H^s} = C_2\|f\|_{H^s}$$

which is (4.3), completing the proof.

The embedding (4.4) fails for every $s \leq \frac{n}{2}$. Since $H^{n/2} \subseteq H^s$ for $s \leq \frac{n}{2}$, it suffices to treat $s = \frac{n}{2}$.

Example 4.4.3: Fails If $s \leq \frac{n}{2}$

The space $H^{n/2}$ is not contained in L^∞. Suppose it were. The inclusion map $H^{n/2} \to L^\infty$ has closed graph: if $f_k \to f$ in $H^{n/2}$ and $f_k \to g$ in L^∞, then $f_k \to f$ in L^2, so a subsequence converges to f almost everywhere, and $g = f$. By the closed graph theorem [7, Closed Graph Theorem] there is a constant C with $\|f\|_{L^\infty} \leq C\|f\|_{H^{n/2}}$ for all $f \in H^{n/2}$. For $R > 1$ let f_R have Fourier

transform $\hat{f_R}(\xi) = (1+|\xi|^2)^{-n/2}$ for $|\xi| \le R$ and $\hat{f_R}(\xi) = 0$ otherwise, and put

$$I_R = \int_{|\xi|\le R} (1+|\xi|^2)^{-n/2}\, d\xi$$

Since $\hat{f_R} \in L^1 \cap L^2$, the Fourier inversion formula shows that f_R is continuous and bounded with $f_R(0) = (2\pi)^{-n/2} \int \hat{f_R} = (2\pi)^{-n/2} I_R$, while $\|f_R\|_{H^{n/2}}^2 = \int (1+|\xi|^2)^{n/2} |\hat{f_R}(\xi)|^2\, d\xi = I_R$. Hence

$$(2\pi)^{-n/2} I_R \le \|f_R\|_{L^\infty} \le C\|f_R\|_{H^{n/2}} = C I_R^{1/2}$$

so $I_R \le (2\pi)^n C^2$ for every R. But for $|\xi| \ge 1$ the integrand is at least $2^{-n/2}|\xi|^{-n}$, so in polar coordinates $I_R \ge 2^{-n/2}|S^{n-1}| \int_1^R r^{-1}\, dr = 2^{-n/2}|S^{n-1}| \log R$, which is unbounded. This contradiction shows that $H^{n/2} \not\subseteq L^\infty$.

Corollary 4.4.4: Not All Sobolev Spaces Form Algebras

$H^{n/2}$ is not an algebra: there are $f, g \in H^{n/2}$ with $fg \notin H^{n/2}$.

***Proof*:**
Suppose that $fg \in H^{n/2}$ for all $f, g \in H^{n/2}$, and write $\|\cdot\|$ for the norm of $H^{n/2}$.

Step 1: A product bound. Fix $g \in H^{n/2}$. The linear map $M_g \colon f \mapsto fg$ from $H^{n/2}$ to itself has closed graph: if $f_k \to f$ and $f_k g \to h$ in $H^{n/2}$, then both convergences hold in L^2, and $\|f_k g - fg\|_{L^1} \le \|f_k - f\|_{L^2}\|g\|_{L^2} \to 0$ by the Cauchy-Schwarz inequality, so a subsequence of $(f_k g)$ converges almost everywhere both to h and to fg, and $h = fg$. By the closed graph theorem [7, Closed Graph Theorem], each M_g is bounded. For each f, $\sup_{\|g\|\le 1} \|M_g f\| = \sup_{\|g\|\le 1} \|M_f g\| \le \|M_f\| < \infty$, so the principle of uniform boundedness [7, Principle Of Uniform Boundedness] gives $C = \sup_{\|g\|\le 1} \|M_g\| < \infty$. Scaling g then gives $\|fg\| \le C\|f\|\|g\|$ for all $f, g \in H^{n/2}$, and we may take $C \ge 1$.

Step 2: A contradiction. By induction, $\|f^k\| \le C^{k-1}\|f\|^k$ for $k \ge 1$. If $\|f\| < 1/C$, then $f^k \to 0$ in $H^{n/2}$, hence in L^2, as $k \to \infty$, so a subsequence of (f^k) converges to 0 almost everywhere. As $|f^k| \ge 1$ wherever $|f| \ge 1$, this forces $|f| < 1$ almost everywhere. Applying this to $f/(2C\|f\|)$ for $f \ne 0$ gives $\|f\|_{L^\infty} \le 2C\|f\|$ for every $f \in H^{n/2}$, so $H^{n/2} \subseteq L^\infty$, contradicting example 4.4.3. Hence some product fg with $f, g \in H^{n/2}$ lies outside $H^{n/2}$, as claimed.

5

Non-Linear Equations

This chapter solves wave equations in the *inhomogeneous* and *non-linear* settings. We first take f and g in H^s and H^{s-1} respectively, which gives enough regularity to solve the *inhomogeneous* problem. We then turn to the *non-linear* case, which is much harder: in contrast to the linear case, a global solution need not exist.

For the non-linear case we construct *local solutions*, in Sobolev spaces, which therefore play a prominent role in this chapter.

5.1 Existence and Uniqueness Inhomogeneous PDE

We first solve the Cauchy problem in the nicest of Sobolev spaces. Take the inhomogenous PDE:

$$\Box u = F \tag{5.1}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g \tag{5.2}$$

Theorem 5.1.1: Unique Solution In Sobolev Space

Let $s \in \mathbb{R}$. Let $f \in H^s$ and $g \in H^{s-1}$ and $F \in L^1([0,T], H^{s-1})$. Then for every $T > 0$, there exists a unique solution u which belongs to

$$C([0,T], H^s) \cap C^1([0,T], H^{s-1}) \tag{5.3}$$

and solves eq. (5.1) on $S_T = (0,T) \times \mathbb{R}^n$ with the data eq. (5.2). Moreover, u satisfies:

$$\|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}} \leq C_T \left(\|f\|_{H^s} + \|g\|_{H^{s-1}} + \int_0^t \|F(t')\|_{H^{s-1}} dt' \right)$$

for all $0 \leq t \leq T$ where $C_T = C(1+T)$ and C only depends on s

***Proof Key ideas*:**
Uniqueness. The inclusions $C_c^\infty \subseteq \mathcal{S} \subseteq H^s \subseteq \mathcal{S}' \subseteq \mathcal{D}'$ are sequentially continuous, so a solution in (5.3) lies in $C^1([0,T], \mathcal{D}'(\mathbb{R}^n))$, and two solutions with the same data coincide by theorem 3.2.4.

Existence. For $f, g \in \mathcal{S}$ and $F \in C^\infty([0,T], \mathcal{S})$ the solution is given on the Fourier side by

$$\hat{u}(t,\xi) = \cos(t|\xi|)\hat{f}(\xi) + \frac{\sin(t|\xi|)}{|\xi|}\hat{g}(\xi) + \int_0^t \frac{\sin((t-t')|\xi|)}{|\xi|}\hat{F}(t',\xi)\, dt'$$

and the estimate follows from Plancherel's theorem, because $|\cos(t|\xi|)| \leq 1$, $|\xi \sin(t|\xi|)| \leq (1+|\xi|^2)^{1/2}$ and

$$\left| \frac{\sin(t|\xi|)}{|\xi|} \right|^2 \leq \frac{2(1+t^2)}{1+|\xi|^2}$$

(use the bound t when $|\xi| < 1$ and $1/|\xi|$ when $|\xi| \geq 1$). For general data one approximates f, g and F by such data, and the estimate shows that the solutions form a Cauchy sequence in the Banach space $X_T = C([0,T], H^s) \cap C^1([0,T], H^{s-1})$, whose limit solves the problem in $\mathcal{D}'(S_T)$.

The step not developed here is the approximation of $F \in L^1([0,T], H^{s-1})$ by functions $F_j \in C^\infty([0,T], \mathcal{S})$ in the norm $\int_0^T \|F(t)\|_{H^{s-1}}\, dt$, together with the sequential continuity of the inclusion $L^1([0,T], H^{s-1}) \subseteq \mathcal{D}'(S_T)$ needed to pass to the limit in $\Box u_j = F_j$. The theorem and this argument are [10, Week 5, Theorem 18], where these two facts are set as Exercises 21 and 22.

5.2 Non-linear Cauchy Problem

A non-linear wave equation need not have a global solution. Let

$$\Box u = (\partial_t u)^2 \tag{5.4}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g \tag{5.5}$$

Inside a backward light cone, a C^2 solution of (5.4) is determined by its data on the base of the cone.

The difference of two solutions solves a *linear* equation, and for linear equations uniqueness in the cone follows from the energy method on shrinking balls.

Lemma 5.2.1: Uniqueness In Cones For Linear Equations

Let $T > 0$ and $x_0 \in \mathbb{R}^n$, and let

$$\Omega = \{(t, x) \mid 0 \le t < T,\ |x - x_0| < T - t\}$$

be the solid backward light cone with base $B_0 = \{x \mid |x - x_0| < T\}$. Let $a \in C(\Omega)$ and $b \in C(\Omega, \mathbb{R}^{1+n})$. If $u \in C^2(\Omega)$ solves

$$\Box u = au + b \cdot \partial u \qquad \text{in } \Omega$$

and $u(0, \cdot) = \partial_t u(0, \cdot) = 0$ on B_0, then $u = 0$ in Ω.

***Proof*:**
Fix $0 < \epsilon < T$, put $r(t) = T - \epsilon - t$, and let $D_t = \{x \mid |x - x_0| \le r(t)\}$ for $0 \le t \le T - \epsilon$. The set $K = \{(t, x) \mid 0 \le t \le T - \epsilon,\ x \in D_t\}$ is a compact subset of Ω, so there is a constant C with $1 + |a| + 2|b| \le C$ on K. Write $|\partial u|^2 = (\partial_t u)^2 + |\nabla u|^2$.

Step 1: The energy. Set

$$E(t) = \frac{1}{2} \int_{D_t} \left(u^2 + |\partial u|^2\right) dx \qquad 0 \le t \le T - \epsilon$$

Writing the integral in polar coordinates centred at x_0 and differentiating in t, with $r'(t) = -1$, gives

$$E'(t) = \int_{D_t} \left(u \partial_t u + \partial_t u\, \partial_t^2 u + \nabla u \cdot \nabla \partial_t u\right) dx - \frac{1}{2} \int_{\partial D_t} \left(u^2 + |\partial u|^2\right) d\sigma$$

Step 2: Using the equation. Substituting $\partial_t^2 u = \triangle u + au + b \cdot \partial u$ and $\nabla u \cdot \nabla \partial_t u + \partial_t u \triangle u = \operatorname{div}(\partial_t u \nabla u)$ gives $E'(t) = I(t) + II(t)$, where

$$I(t) = \int_{D_t} \partial_t u \left((1 + a)u + b \cdot \partial u\right) dx$$
$$II(t) = \int_{D_t} \operatorname{div}(\partial_t u \nabla u)\, dx - \frac{1}{2} \int_{\partial D_t} \left(u^2 + |\partial u|^2\right) d\sigma$$

By the divergence theorem, and since $|\partial_t u \nabla u \cdot \nu| \le \frac{1}{2}|\partial u|^2$ for the outward unit normal ν,

$$II(t) = \int_{\partial D_t} \partial_t u \nabla u \cdot \nu\, d\sigma - \frac{1}{2} \int_{\partial D_t} \left(u^2 + |\partial u|^2\right) d\sigma \le -\frac{1}{2} \int_{\partial D_t} u^2\, d\sigma \le 0$$

Since $|u\, \partial_t u| \le \frac{1}{2}(u^2 + |\partial u|^2)$ and $|\partial_t u\, b \cdot \partial u| \le |b|\, |\partial u|^2$, we have $I(t) \le \int_{D_t} \frac{1}{2}(1 + |a| + 2|b|)(u^2 + |\partial u|^2)\, dx \le CE(t)$. Hence $E'(t) \le CE(t)$.

Step 3: Conclusion. The function $E(t)e^{-Ct}$ is non-increasing, so $E(t) \le E(0)e^{Ct}$. Since

$u(0,\cdot) = 0$ on the open ball B_0, also $\nabla u(0,\cdot) = 0$ there, and $\partial_t u(0,\cdot) = 0$, so $E(0) = 0$. Hence $E = 0$ on $[0, T-\epsilon]$, and as u is continuous, $u = 0$ on K. Every point of Ω lies in K once ϵ is small enough, so $u = 0$ in Ω, as claimed.

Theorem 5.2.2: Uniqueness In Backward Light Cones

Let $F\colon \mathbb{R}^{n+2} \to \mathbb{R}$ be continuously differentiable, and let Ω and B_0 be as in lemma 5.2.1. If $u, v \in C^2(\Omega)$ both solve

$$\Box w = F(w, \partial w) \qquad \text{in } \Omega$$

and $u(0,\cdot) = v(0,\cdot)$ and $\partial_t u(0,\cdot) = \partial_t v(0,\cdot)$ on B_0, then $u = v$ in Ω. In particular, a C^2 solution of (5.4) in Ω is determined by its data on B_0.

***Proof*:**
For $z, z' \in \mathbb{R}^{n+2}$, the fundamental theorem of calculus along the segment from z' to z gives

$$F(z) - F(z') = \int_0^1 \frac{d}{d\lambda} F(z' + \lambda(z - z'))\, d\lambda = G(z, z') \cdot (z - z')$$

with

$$G(z, z') = \int_0^1 \nabla F(z' + \lambda(z - z'))\, d\lambda$$

where G is continuous on $\mathbb{R}^{2(n+2)}$. Hence $w = u - v$ satisfies

$$\Box w = F(u, \partial u) - F(v, \partial v) = G(u, \partial u, v, \partial v) \cdot (w, \partial w) = aw + b \cdot \partial w$$

where a is the first component of $G(u, \partial u, v, \partial v)$ and b collects the other $n+1$ components. Both are continuous on Ω, and $w(0,\cdot) = \partial_t w(0,\cdot) = 0$ on B_0, so $w = 0$ in Ω by lemma 5.2.1. For (5.4) the coefficients are explicit, since $(\partial_t u)^2 - (\partial_t v)^2 = (\partial_t u + \partial_t v)\, \partial_t w$, so that $a = 0$ and $b = (\partial_t u + \partial_t v, 0, \ldots, 0)$. This completes the proof.

Theorem 5.2.3: Non-Global Solution to Wave Equation

For every $T > 0$ there exists $g \in C_c^\infty(\mathbb{R}^n)$ such that (5.4) with the data $f = 0$ and $\partial_t u|_{t=0} = g$ has no solution in $C^2([0,T] \times \mathbb{R}^n)$.

***Proof*:**
Step 1: An explicit solution. For $c > 0$, the function $y(t) = c/(1 - ct)$ solves the separable problem $y' = y^2$, $y(0) = c$, on $[0, 1/c)$, and blows up as $t \to 1/c$. Hence $w(t,x) = -\log(1 - ct)$, which has $\partial_t w = y$ and $\triangle w = 0$, solves (5.4) on $[0, 1/c) \times \mathbb{R}^n$ with $w(0,\cdot) = 0$ and $\partial_t w(0,\cdot) = c$.

Step 2: Comparison in a cone. Fix $x_0 \in \mathbb{R}^n$, put $c = 1/T$, and choose $g \in C_c^\infty(\mathbb{R}^n)$ with $g = c$ on the closed ball $|x - x_0| \le T$, for instance a smooth cutoff function equal to 1 there, multiplied by c. Suppose $u \in C^2([0,T] \times \mathbb{R}^n)$ solves (5.4) with $u(0,\cdot) = 0$ and $\partial_t u(0,\cdot) = g$. Let Ω be the backward cone with vertex (T, x_0) and base $B_0 = \{x \mid |x - x_0| < T\}$. Then u and w are C^2 solutions in $\Omega \subseteq [0,T) \times \mathbb{R}^n$ with the same data on B_0, so $u = w$ in Ω by theorem 5.2.2.

Step 3: Contradiction. The points (t, x_0) with $0 \leq t < T$ lie in Ω, so $u(t, x_0) = -\log(1 - t/T) \to \infty$ as $t \to T$. This contradicts the continuity of u at (T, x_0), so no such u exists, as claimed.

5.3 Local Existence and Uniqueness

We now study the general non-linear case. Global solutions need not exist, but local solutions do. Consider the Cauchy Problem:

$$\Box u = F(u, \partial u) \qquad F(0) = 0 \tag{5.6}$$

$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g \tag{5.7}$$

Here $F \colon \mathbb{R}^{n+2} \to \mathbb{R}$ is a given smooth function. In the following theorems, we will consider u and F as real-valued, but the results carry over, with the same proofs, to system valued functions (i.e. $\mathbb{R}^N$-valued), which we omit for presentation clarity.

For the model equation $\Box u = (\partial_t u)^2$ the three theorems below are proved in the next section (theorem 5.4.1, theorem 5.4.2 and theorem 5.4.3). For a general nonlinearity $F(u, \partial u)$ the same arguments go through with the Moser inequality (theorem 4.3.2) in place of the calculus inequality. Both were proved with Littlewood-Paley theory in chapter 4, and for general F we record the method and a source.

Theorem 5.3.1: Local Existence And Uniqueness

Let $s > \frac{n}{2} + 1$. Then for all $(f, g) \in H^s \times H^{s-1}$, there exists $T > 0$ and a unique

$$u \in C([0, T], H^s) \cap C^1([0, T], H^{s-1})$$

solving the IVP in eq. (5.6) on $S_T = (0, T) \times \mathbb{R}^n$ with the data eq. (5.7). Furthermore, the time T can be chosen to depend continuously on $\|f\|_{H^s} + \|g\|_{H^{s-1}}$. Moreover, for every $T' > 0$ the problem has at most one solution in $C([0, T'], H^s) \cap C^1([0, T'], H^{s-1})$.

***Proof Key ideas*:**
Existence comes from a Picard iteration in the Banach space $X_T = C([0, T], H^s) \cap C^1([0, T], H^{s-1})$: one solves the linear problems $\Box u_i = F(u_{i-1}, \partial u_{i-1})$ with the given data, using theorem 5.1.1, and shows with the energy estimate of that theorem that for T small, in terms of $\|f\|_{H^s} + \|g\|_{H^{s-1}}$ only, the iterates stay in a fixed ball of X_T and contract there. Uniqueness, in this class for every $T' > 0$, comes from the same estimate and Gronwall's lemma applied to the difference of two solutions. The remaining step is the tame estimate that keeps $F(u, \partial u)$ in $C([0, T], H^{s-1})$ and controls its differences. The Moser inequality (theorem 4.3.2) gives this estimate, and the details are [10, Week 5, Theorem 22 and the remark after it].

We next might want to see when we can extend a solution to a time $T' > T$:

Theorem 5.3.2: Continuation Theorem

Let $s > \frac{n}{2} + 1$. Fix f, g as in theorem 5.3.1. Let $T^* = T^*(f, g)$ be the supremum of all $T > 0$ such that (5.6) has a solution $u \in C([0,T], H^s) \cap C^1([0,T], H^{s-1})$ on S_T satisfying (5.7). Then either $T^* = \infty$ or:

$$\partial u \notin L^\infty(S_{T^*})$$

***Proof Key ideas*:**
Suppose $T^* < \infty$ and $\partial u \in L^\infty(S_{T^*})$, where u is the solution on S_{T^*} given by theorem 5.3.1 and uniqueness. Then u is bounded too, since $u(t) = f + \int_0^t \partial_t u$ and $f \in H^s \subseteq L^\infty$. The energy estimate of theorem 5.1.1, applied with right-hand side $F(u, \partial u)$, together with the Moser inequality bounds $A(t) = \|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}}$ by $C + C\int_0^t A(t')\,dt'$, with C depending on the bounds for u and ∂u, so Gronwall's lemma gives $\sup_{t<T^*} A(t) < \infty$. Since the existence time in theorem 5.3.1 depends only on the size of the data, the solution then extends beyond T^*, contradicting the definition of T^*. As for the previous theorem, the remaining input is the Moser inequality (theorem 4.3.2), and the details are [10, Week 5, Theorem 23].

Using these two, we can restrict the IVP functions to smooth compact functions and always find a solution:

Theorem 5.3.3: Smooth Solution

Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exist a $T > 0$ and a unique

$$u \in C^\infty([0,T] \times \mathbb{R}^n)$$

solving eq. (5.6) on S_T with the data eq. (5.7).

***Proof Key ideas*:**
The data lie in H^s for every s. Theorem 5.3.1 at a fixed $s_0 > \frac{n}{2} + 1$ gives a solution u on S_T, and at each $s > s_0$ a solution on some S_{T_s}, which agrees with u by uniqueness. Sobolev's lemma $H^{s_0-1} \subseteq L^\infty$ bounds ∂u on S_T, so theorem 5.3.2 at level s shows that the lifespan at level s exceeds T, and $u \in C([0,T], H^s)$ for every s. By Sobolev's lemma again every spatial derivative of u and of $\partial_t u$ is continuous on $[0,T] \times \mathbb{R}^n$, and differentiating the equation $\partial_t^2 u = \triangle u + F(u, \partial u)$ in t converts each higher time derivative into spatial derivatives, so $u \in C^\infty([0,T] \times \mathbb{R}^n)$.

For uniqueness, let $v \in C^\infty([0,T] \times \mathbb{R}^n)$ be a second solution, and choose R with $\operatorname{supp} f \cup \operatorname{supp} g \subseteq \{x \mid |x| \le R\}$. If $t_0 < T$ and $|x_0| > R + t_0$, the backward cone with vertex (T', x_0), where $t_0 < T' < \min(T, |x_0| - R)$, contains (t_0, x_0) and has its base outside $\{x \mid |x| \le R\}$. There v and the zero solution (recall $F(0) = 0$) have the same data, so $v(t_0, x_0) = 0$ by theorem 5.2.2, and by continuity this extends to $t_0 = T$. Thus every $v(t, \cdot)$ is supported in $\{x \mid |x| \le R + T\}$, so $v \in C([0,T], H^s) \cap C^1([0,T], H^{s-1})$, and $v = u$ by the uniqueness in theorem 5.3.1.

The argument rests on the two preceding theorems, whose proofs need the Moser inequality (theorem 4.3.2), and on Sobolev's lemma (theorem 3.2.7). The theorem is [10, Week 5, Theorem 24].

By the uniqueness of solutions in backward light cones, the previous result extends as follows.

Corollary 5.3.4: Global Smooth Solution

Let $f, g \in C^\infty(\mathbb{R}^n)$. Then there exists a set

$$A = \{(t, x) \mid 0 \le t \le T(x)\}$$

where T is a continuous and strictly positive function on $\mathbb{R}^n$, and a unique solution $u \in C^\infty(A)$ of (5.6) with the data (5.7).

***Proof*:**
Step 1: Local solutions. Choose $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ on the unit ball, and for $j \ge 1$ put $f_j = \chi(\cdot/j)f$ and $g_j = \chi(\cdot/j)g$, so that $f_j = f$ and $g_j = g$ on $\{x \mid |x| \le j\}$. By theorem 5.3.3 there are $T_j > 0$ and solutions $u_j \in C^\infty([0, T_j] \times \mathbb{R}^n)$ of (5.6) with data (f_j, g_j). Replacing T_j by $\min(T_1, \dots, T_j)$, we may assume $T_{j+1} \le T_j$. Let

$$K_j = \{(t, x) \mid 0 \le t \le T_j,\ |x| \le j - t\}$$

Step 2: The u_j agree. Let $(t, x) \in K_j \cap K_k$ with $t > 0$. The backward cone with vertex (t, x) has base $\{y \mid |y - x| < t\}$, which lies in $\{y \mid |y| < j\}$ and in $\{y \mid |y| < k\}$ because $|x| + t \le \min(j, k)$. On this base $f_j = f_k = f$ and $g_j = g_k = g$, and $t \le \min(T_j, T_k)$, so $u_j = u_k$ in the cone by theorem 5.2.2, and at (t, x) by continuity. At $t = 0$ both equal $f(x)$. So $u = u_j$ on K_j defines a function on $\bigcup_j K_j$.

Step 3: The set A. For integers $m \ge 0$ put $c_m = \min(1, T_{m+2})$, which decrease and lie in $(0, 1]$. Let $h\colon [0, \infty) \to (0, 1]$ be the piecewise linear function with $h(m) = c_m$, and put $T(x) = \frac{1}{2}h(|x|)$. Consecutive values c_m differ by less than 1, so h is 1-Lipschitz and T is continuous, positive and $\frac{1}{2}$-Lipschitz. Given x, let j be the least integer with $j \ge |x| + 1$. For $|x| \in [m, m + 1]$ we have $j \le m + 2$, so $T(x) \le \frac{1}{2}c_m \le \frac{1}{2}T_{m+2} \le \frac{1}{2}T_j$ because the T_j decrease, and also $T(x) \le \frac{1}{2}$. Hence the points (t, x) with $0 \le t \le T(x)$ satisfy $t < T_j$ and $|x| \le j - 1 < j - t$, so a neighbourhood of each of them in $[0, \infty) \times \mathbb{R}^n$ lies in K_j, where $u = u_j$ is smooth. So $u \in C^\infty(A)$ solves the problem.

Step 4: Uniqueness. Let $\tilde{u} \in C^\infty(A)$ be another solution and $(t, x) \in A$ with $t > 0$. If $0 \le t' < t - |y - x|$, then $t' < T(x) - |y - x| \le T(y) - \frac{1}{2}|y - x| \le T(y)$ because T is $\frac{1}{2}$-Lipschitz, so the backward cone with vertex (t, x) lies in A. In it, u and $\tilde{u}$ are C^2 solutions with the same data, so $\tilde{u}(t, x) = u(t, x)$ by theorem 5.2.2 and continuity. At $t = 0$ both equal $f(x)$, as claimed.

5.4 The model equation $\Box u = (\partial_t u)^2$

This section proves the three theorems of the previous section in the special case

$$\Box u = (\partial_t u)^2$$

We will need the following identities and inequalities:

1. The *energy inequality*, that is, the estimate of theorem 5.1.1
2. *Sobolev's Lemma* on $\mathbb{R}^n$, in particular the inequality:
$$\|f\|_{L^\infty} \leq C_s \|f\|_{H^s} \qquad \forall f \in H^s \quad s > \frac{n}{2}$$
3. The *calculus inequality*
$$\|fg\|_{H^s} \leq C_s(\|f\|_{H^s}\|g\|_{L^\infty} + \|f\|_{L^\infty}\|g\|_{H^s})$$
for all $s \geq 0$ and all $f, g \in H^s \cap L^\infty$. For a general nonlinearity it is replaced by the *Moser inequality* (theorem 4.3.2).
4. A special case of *Gronwall's Lemma*: If $C_1, C_2 \geq 0$ are given constants, and A is a continuous non-negative function on $[0, T]$ such that
$$A(t) \leq C_1 + C_2 \int_0^t A(\tau) d\tau \qquad 0 \leq t \leq T$$
Then:
$$A(t) \leq C_1 e^{C_2 t} \qquad 0 \leq t \leq T$$
This is the case of constant $b = C_2$ and $a = C_1$, on the interval $[0, T]$ with $t_0 = 0$, of [3, Grönwall's Inequality].

The calculus inequality is theorem 4.2.2, proved in section 4.2 from Littlewood-Paley theory.

As a consequence:
$$|f(x)| \leq C_s \|f\|_{H^s} \qquad s > \frac{n}{2}$$
which, by the integer-order characterization of the Sobolev norm, means that for every integer $m > \frac{n}{2}$:
$$|f(x)| \leq C \sum_{|\alpha| \leq m} \|\partial^\alpha f\|_{L^2}$$

The model Cauchy problem is
$$\Box u = (\partial_t u)^2 \tag{5.8}$$
$$u|_{t=0} = f \qquad \partial_t u|_{t=0} = g$$

Theorem 5.4.1: Local Existence And Uniqueness for Model non-linear PDE

Let $s > \frac{n}{2} + 1$. Then for all $(f, g) \in H^s \times H^{s-1}$, there exists $T > 0$ and a unique
$$u \in C([0, T], H^s) \cap C^1([0, T], H^{s-1})$$
solving the IVP in eq. (5.8) on $S_T = (0, T) \times \mathbb{R}^n$. Furthermore, the time T can be chosen to depend continuously on $\|f\|_{H^s} + \|g\|_{H^{s-1}}$. Moreover, for every $T' > 0$ the problem has at most one solution in $C([0, T'], H^s) \cap C^1([0, T'], H^{s-1})$.

***Proof*:**
Existence. The solution is the limit of a sequence of iterates. Set

$$u_{-1} = 0$$

Then define $u_0, u_1, ...$ inductively by:

$$\Box u_i = (\partial_t u_{i-1})^2 \quad u_i|_{t=0} = f,\ \partial_t u_i|_{t=0} = g \tag{5.9}$$

Then for $T > 0$ define $X_T = C([0,T], H^s) \cap C^1([0,T], H^{s-1})$. Note that X_T is a Banach space with the norm:

$$\|u\|_{X_T} = \sup_{0\le t\le T} (\|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}})$$

Now, notice that:

$$u_{i-1} \in X_T \Rightarrow u_i \in X_T$$

Indeed, since $s - 1 > \frac{n}{2}$, Sobolev's lemma and the calculus inequality give us:

$$\|vw\|_{H^{s-1}} \le C_s \|v\|_{H^{s-1}} \|w\|_{H^{s-1}} \tag{5.10}$$

for all $v, w \in H^{s-1}(\mathbb{R}^n)$. Thus:

$$u_{i-1} \in X_T \Rightarrow (\partial_t u_{i-1})^2 \in C([0,T], H^{s-1})$$

and by theorem 5.1.1 each problem in (5.9) has a unique solution in X_T. Thus, the sequence of iterates is well-defined in X_T for any $T > 0$. If it is Cauchy, its limit u solves (5.8) on $S_T = (0,T) \times \mathbb{R}^n$, because

$$\begin{aligned} u_i \to u \in X_T \quad &\Rightarrow \quad u_i \to u \quad \text{in } \mathcal{D}'(S_T) \\ &\Rightarrow \Box u_i \to \Box u \quad \text{in } \mathcal{D}'(S_T) \end{aligned}$$

and by eq. (5.10):

$$\begin{aligned} u_i \to u \in X_T \quad &\Rightarrow \quad (\partial_t u_i)^2 \to (\partial_t u)^2 \quad \text{in } C([0,T], H^{s-1}) \\ &\Rightarrow \Box u_{i+1} = (\partial_t u_i)^2 \to (\partial_t u)^2 \quad \text{in } \mathcal{D}'(S_T) \end{aligned}$$

so that $\Box u = (\partial_t u)^2$, while $u(0) = f$ and $\partial_t u(0) = g$ since $u_i \to u$ in X_T.
It remains to show that the sequence is Cauchy for small T.

Step 1: The sequence is bounded:

$$\|u_i\|_{X_T} \le 2CE_s \qquad \text{for} \qquad i = 0, 1, 2, ... \tag{5.11}$$

provided that $T \le 1$ is so small that:

$$T \le \frac{1}{8C^2 E_s}$$

where

$$E_s = \|f\|_{H^s} + \|g\|_{H^{s-1}}$$

and $C > 0$ is a constant which only depends on s and n. Suppose the bound in (5.11) holds for $i-1$. Applying the energy inequality (the estimate of theorem 5.1.1) to the problems (5.9) and using the estimate (5.10), we get:

$$\|u_i\|_{X_T} \leq CE_s + CT\|u_{i-1}\|_{X_T}^2 \leq CE_s + CT(2CE_s)^2$$

where C only depends on s and n because $T \leq 1$ (C comes from the energy inequality and the estimate (5.10)). If $T \leq \frac{1}{8C^2E_s}$, the right hand side is bounded by $CE_s + \frac{1}{2}CE_s \leq 2CE_s$, and so we obtain the bound (5.11) for all i by induction, the case $i = -1$ being trivial.

Step 2: The sequence satisfies

$$\|u_{i+1} - u_i\|_{X_T} \leq \frac{1}{2}\|u_i - u_{i-1}\|_{X_T}$$

which will show it is Cauchy. To see this, note that:

$$\Box(u_{i+1} - u_i) = (\partial_t u_i)^2 - (\partial_t u_{i-1})^2 = \partial_t(u_i - u_{i-1})\,\partial_t(u_i + u_{i-1})$$

with vanishing initial data at $t = 0$. Thus, if we apply the energy inequality and use eq. (5.10) and the uniform bound of eq. (5.11) we get:

$$\begin{aligned}\|u_{i+1} - u_i\|_{X_T} &\leq CT\left(\|u_i\|_{X_T} + \|u_{i-1}\|_{X_T}\right)\|u_i - u_{i-1}\|_{X_T} \\ &\leq CT \cdot 4CE_s\|u_i - u_{i-1}\|_{X_T}\end{aligned}$$

Thus, we get the desired bound using the fact that $T \leq \frac{1}{8C^2E_s}$, completing the proof for existence. The time $T = \min(1, \frac{1}{8C^2E_s})$ depends continuously on E_s.

Uniqueness. Now suppose that $u, v \in X_T$ for some $T > 0$ solve the Cauchy problem on $S_T = (0,T) \times \mathbb{R}^n$ with the identical data at $t = 0$. Then:

$$\Box(u - v) = \partial_t(u + v)\partial_t(u - v)$$

and then setting

$$A(t) = \|(u - v)(t)\|_{H^s} + \|\partial_t(u - v)(t)\|_{H^{s-1}}$$

we get by the energy inequality and (5.10):

$$A(t) \leq C\int_0^t A(t')\,dt' \qquad \text{for} \qquad 0 \leq t \leq T$$

for some constant C independent of t (note that C depends on the norms $\|u\|_{X_T}$ and $\|v\|_{X_T}$, but this is not a problem since u and v are considered fixed for this argument). Then by Gronwall's Lemma with $C_1 = 0$, $A(t) = 0$ for $0 \leq t \leq T$, and so $u = v$ in S_T, showing the solution is also unique and so completing the proof.

Theorem 5.4.2: Continuation Theorem for Model non-linear PDE

Let $s > \frac{n}{2} + 1$. Fix f, g as in theorem 5.4.1. Let $T^* = T^*(f, g)$ be the supremum of all $T > 0$ such that (5.8) has a solution $u \in C([0, T], H^s) \cap C^1([0, T], H^{s-1})$ on S_T with $u|_{t=0} = f$ and $\partial_t u|_{t=0} = g$. Then either $T^* = \infty$ or:

$$\partial u \notin L^\infty(S_{T^*})$$

***Proof*:**
By the uniqueness in theorem 5.4.1, the solutions on the strips S_T, $T < T^*$, agree where they overlap, and define a solution u on S_{T^*}. Suppose $T^* < \infty$ and $\partial u \in L^\infty(S_{T^*})$. Define:

$$A(t) = \|u(t)\|_{H^s} + \|\partial_t u(t)\|_{H^{s-1}} \qquad 0 \leq t < T^* \tag{5.12}$$

By the energy inequality and the calculus inequality, applied with $s - 1 \geq 0$ to $(\partial_t u)^2$, we get for $0 \leq t < T^*$:

$$\begin{aligned} A(t) &\leq C_{s,T^*} \left(\|f\|_{H^s} + \|g\|_{H^{s-1}} + \int_0^t \|\partial_t u(t')\|_{L^\infty} \|\partial_t u(t')\|_{H^{s-1}} \, dt' \right) \\ &\leq C_{s,T^*} \left(\|f\|_{H^s} + \|g\|_{H^{s-1}} + \|\partial u\|_{L^\infty(S_{T^*})} \int_0^t A(t') \, dt' \right) \end{aligned}$$

and Gronwall's lemma gives $A(t) \leq M$ for $0 \leq t < T^*$, for a constant M. By theorem 5.4.1, data of size at most M at a time t_0 yield a solution on $[t_0, t_0 + \tau]$, with $\tau > 0$ depending only on M. Starting at $t_0 = \max(0, T^* - \tau/2)$ with the data $(u(t_0), \partial_t u(t_0))$ and gluing by uniqueness gives a solution on $S_{T^* + \tau/2}$, contradicting the definition of T^*. This completes the proof.

Theorem 5.4.3: Smooth Solution for Model non-linear PDE

Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exist a $T > 0$ and a unique

$$u \in C^\infty([0, T] \times \mathbb{R}^n)$$

solving (5.8) on S_T with $u|_{t=0} = f$ and $\partial_t u|_{t=0} = g$

***Proof*:**
Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then *a priori* we have that $f, g \in H^s$ for every $s \in \mathbb{R}$. Fix $s_0 > \frac{n}{2} + 1$. Then by theorem 5.4.1, there exists a $T > 0$ and a unique solution:

$$u \in C([0, T], H^{s_0}) \cap C^1([0, T], H^{s_0 - 1})$$

solving (5.8) on $S_T = (0, T) \times \mathbb{R}^n$. We also have from the same theorem that for every $s > s_0$ there exists a $T_s > 0$ such that

$$u \in C([0, T_s], H^s) \cap C^1([0, T_s], H^{s-1})$$

We claim that we can take $T_s = T$. Indeed, this follows from theorem 5.4.2 since by Sobolev's lemma and by the fact that u is in $C([0, T], H^{s_0}) \cap C^1([0, T], H^{s_0 - 1})$:

$$\partial u \in L^\infty(S_T)$$

Thus

$$u \in C([0,T], H^s) \cap C^1([0,T], H^{s-1})$$

for every $s \geq s_0$. By Sobolev's lemma again, we get:

$$\partial_t^j \partial_x^\alpha u \in C([0,T] \times \mathbb{R}^n) \tag{5.13}$$

for $j \in \{0,1\}$ and all α. Since u solves the non-linear PDE, we have that:

$$\partial_t^2 u = \triangle u + (\partial_t u)^2 \tag{5.14}$$

Thus, using this and eq. (5.13), we have:

$$\partial_t^2 \partial_x^\alpha u \in C([0,T] \times \mathbb{R}^n)$$

so eq. (5.13) holds for $j \in \{0,1,2\}$ and all α. Applying ∂_t to both sides of eq. (5.14), we get:

$$\partial_t^3 u = \triangle \partial_t u + 2\partial_t u \partial_t^2 u$$

and now from this we see that eq. (5.13) holds for $j \in \{0,1,2,3\}$. If we keep taking successive time derivatives of the equation, we obtain eq. (5.13) for all j by induction. These functions are the distributional derivatives of u on S_T, and they are continuous on $[0,T] \times \mathbb{R}^n$, so proposition 3.2.2, applied on S_T for every order k, shows that they are the classical derivatives of u. Thus, we have that:

$$u \in C^\infty([0,T] \times \mathbb{R}^n)$$

For uniqueness, let $v \in C^\infty([0,T] \times \mathbb{R}^n)$ be a second solution, and choose R with $\text{supp} f \cup \text{supp} g \subseteq \{x \mid |x| \leq R\}$. The zero function solves (5.8), so for $t_0 < T$ and $|x_0| > R + t_0$, theorem 5.2.2 applied to v and 0 on the backward cone with vertex (T', x_0), where $t_0 < T' < \min(T, |x_0| - R)$, gives $v(t_0, x_0) = 0$, and by continuity this extends to $t_0 = T$. So every $v(t, \cdot)$ is supported in $\{x \mid |x| \leq R + T\}$, and $v \in C([0,T], H^{s_0}) \cap C^1([0,T], H^{s_0 - 1})$. Hence $v = u$ by the uniqueness in theorem 5.4.1, as we sought to show.

6

Global Existence for Non-Linear Wave Equation

This chapter studies global existence for nonlinear wave equations whose nonlinearity vanishes to second order at the origin, and dimensions below 4 are harder. A global solution need not exist (theorem 5.2.3), but when the data f and g are replaced by ϵf and ϵg, the lifespan becomes as long as desired as $\epsilon \to 0$, and infinite in dimension $n \geq 4$.

The local theory controlled solutions through Sobolev's inequality, which bounds $|u(t,x)|$ by L^2 norms of derivatives of $u(t,\cdot)$. The Klainerman-Sobolev inequality replaces it by a bound that also decays in t, at the cost of using derivatives along the invariant vector fields.

6.1 Goal

In $\mathbb{R}^{1+n}$, consider the nonlinear Cauchy problem

$$\Box u = F(\partial u) \tag{6.1}$$

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} = \epsilon g \tag{6.2}$$

where $F\colon \mathbb{R}^{1+n} \to \mathbb{R}$ is a given smooth function that vanishes to second order at the origin:

$$F(0) = 0 \qquad DF(0) = 0$$

Theorem 6.1.1: Global Existence Of Non-Linaer Wave Equation

Let $n \geq 4$. Let $f, g \in C_c^\infty(\mathbb{R}^n)$. Then there exists $\epsilon_0 > 0$ such that the two above equations have a solution

$$u \in C^\infty([0,\infty) \times \mathbb{R}^n)$$

provided $0 < \epsilon \leq \epsilon_0$.

The proof occupies section 6.4, after the Klainerman-Sobolev inequality, where the theorem is restated as theorem 6.4.1. Note that ϵ_0 depends on f and g. Furthermore, the solution is unique due to the material in chapter 5.

For dimension $n \in \{1, 2, 3\}$, the proof gives lower bounds, as $\epsilon \to 0$, on the lifespan T_ϵ, the supremum of the $T > 0$ for which the two equations above have a solution $u \in C^\infty([0, T] \times \mathbb{R}^n)$. Specifically, there is $c > 0$, depending on f and g, with

$$\begin{aligned} T_\epsilon &\geq e^{c/\epsilon} && \text{if } n = 3 \\ T_\epsilon &\geq c/\epsilon^2 && \text{if } n = 2 \\ T_\epsilon &\geq c/\epsilon && \text{if } n = 1 \end{aligned}$$

for all sufficiently small $\epsilon > 0$.

6.2 The Invariant Vector Fields

For an integer $m > \frac{n}{2}$, Sobolev's lemma (theorem 3.2.7) and the integer-order characterization of the Sobolev norm give $|f(x)| \leq C \sum_{|\alpha| \leq m} \|\partial^\alpha f\|_{L^2}$. Taking for m the least integer greater than $\frac{n}{2}$, this reads

$$|f(x)| \leq C \sum_{|\alpha| \leq \frac{n+2}{2}} \|\partial^\alpha f\|_{L^2} \qquad f \in C_c^\infty(\mathbb{R}^n)$$

a bound that drove the local theory of section 5.4. Global existence needs a similar bound for $|u(t,x)|$ in terms of L^2 norms in x of spacetime derivatives of u, with a factor that decays in t. The derivatives are taken along the *invariant vector fields*

$$\partial_t, \partial_1, \ldots, \partial_n \tag{6.3}$$

$$\Omega_{ij} = x_j \partial_i - x_i \partial_j \tag{6.4}$$

$$\Omega_{0i} = t\partial_i + x_i \partial_t \tag{6.5}$$

$$L_0 = t\partial_t + \sum_{i=1}^{n} x_i \partial_i \tag{6.6}$$

where $1 \leq i, j \leq n$. With $x_0 = t$ and $\partial_0 = \partial_t$ the notation is uniform. Since $\Omega_{ji} = -\Omega_{ij}$ and $\Omega_{ii} = 0$, the fields (6.4) with $1 \leq i < j \leq n$ give all of them up to sign, so there are

$$(n+1) + \frac{n(n-1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2} + 1$$

different vector fields, enumerated as

$$\Gamma = (\Gamma_0, \Gamma_1, \dots, \Gamma_m) \qquad m = \frac{(n+1)(n+2)}{2}$$

For a finite sequence $\alpha = (i_1, \dots, i_k)$ of indices in $\{0, \dots, m\}$, write $\Gamma^\alpha = \Gamma_{i_1} \cdots \Gamma_{i_k}$ and $|\alpha| = k$, with Γ^α the identity when $k = 0$. A sum over $|\alpha| \le k$ runs over all such sequences, so it contains every ordering of the factors, while derivatives ∂^α keep the usual multi-index notation. The commutator of any two of the fields Γ_i is a linear combination of the Γ_k with constant coefficients, so each Γ^α with $|\alpha| = k$ is a constant-coefficient combination of ordered products $\Gamma_0^{a_0} \cdots \Gamma_m^{a_m}$ with $a_0 + \cdots + a_m \le k$. Hence sums over all sequences with $|\alpha| \le k$ and over these ordered products bound each other up to constants, and estimates stated for ordered products, as in [10], hold in the present convention. The fields (6.4), (6.5) and (6.6) are called *homogeneous*, as their coefficients are linear in (t, x).

6.3 The Klainerman-Sobolev Inequality

The following will replace the Sobolev inequality:

Theorem 6.3.1: Klainerman-Sobolev Inequality

There is a constant C such that

$$(1 + t + |x|)^{\frac{n-1}{2}} |u(t,x)| \le C \sum_{|\alpha| \le \frac{n+2}{2}} \|\Gamma^\alpha u(t, \cdot)\|_{L^2} \qquad \text{for} \qquad t > 0, x \in \mathbb{R}^n$$

whenever $u \in C^\infty([0, \infty) \times \mathbb{R}^n)$ and $\operatorname{supp}(u(t, \cdot))$ is compact for every t

The same estimate holds, with the same constant C, for $0 < t < T$ when $u \in C^\infty([0, T) \times \mathbb{R}^n)$ and $\operatorname{supp}(u(t, \cdot))$ is compact for every $t \in [0, T)$. The homogeneous vector fields (6.4), (6.5) and (6.6) span the tangent space of $\mathbb{R}^{1+n}$ at every point outside the light cone.

Lemma 6.3.2: Klainerman-Sobolev Inequality Lemma I

For any multi-index $\alpha \neq 0$ and $(t, x) \notin \Lambda = \{(t, x) \mid |t| = |x|\}$, that is, outside the light cone,

$$\partial^\alpha = \sum_{1 \le |\beta| \le |\alpha|} c_{\alpha\beta}(t, x) \Gamma^\beta$$

where the $c_{\alpha\beta}$ are C^∞ and homogeneous of degree $-|\alpha|$ outside Λ. In fact, the sum on the right hand side only involves the homogeneous vector fields.

***Proof*:**

Here a function c on $\mathbb{R}^{1+n} \setminus \Lambda$ is homogeneous of degree d if $c(\lambda t, \lambda x) = \lambda^d c(t, x)$ for $\lambda > 0$. Then $\partial_\mu c$ is homogeneous of degree $d - 1$, and a product of homogeneous functions has the sum of their degrees as its degree. Put $\rho = t^2 - |x|^2$, which does not vanish outside Λ.

Step 1: First derivatives. Expanding the definitions, $tL_0 = t^2 \partial_t + t \sum_i x_i \partial_i$ and $\sum_i x_i \Omega_{0i} =$

$t\sum_i x_i\partial_i + |x|^2\partial_t$, while for $1 \le j \le n$, $t\Omega_{0j} = t^2\partial_j + tx_j\partial_t$, $\sum_i x_i\Omega_{ij} = x_j\sum_i x_i\partial_i - |x|^2\partial_j$ and $x_jL_0 = tx_j\partial_t + x_j\sum_i x_i\partial_i$, all sums running over $1 \le i \le n$. Hence

$$\rho\,\partial_t = tL_0 - \sum_{i=1}^{n} x_i\Omega_{0i} \qquad \rho\,\partial_j = t\Omega_{0j} + \sum_{i=1}^{n} x_i\Omega_{ij} - x_jL_0$$

Dividing by ρ writes each ∂_μ, $0 \le \mu \le n$, as $\sum_\gamma d_{\mu\gamma}\Gamma_\gamma$, a sum over the homogeneous fields (with $\Omega_{ij} = -\Omega_{ji}$ for $i > j$) whose coefficients are among $\pm t/\rho$ and $\pm x_i/\rho$, hence C^∞ outside Λ and homogeneous of degree -1. This is the claim for $|\alpha| = 1$.

Step 2: Induction. Suppose the claim holds for a multi-index $\alpha' \neq 0$, and let $\alpha = \alpha' + e_\mu$, so that $\partial^\alpha = \partial_\mu\partial^{\alpha'}$. For each term $c\,\Gamma^\beta$ of the expansion of $\partial^{\alpha'}$, where c is homogeneous of degree $-|\alpha'|$ and Γ^β is a product of $1 \le |\beta| \le |\alpha'|$ homogeneous fields, the product rule and Step 1 give

$$\partial_\mu \circ c\,\Gamma^\beta = (\partial_\mu c)\Gamma^\beta + \sum_\gamma c\,d_{\mu\gamma}\Gamma_\gamma\Gamma^\beta$$

The coefficients $\partial_\mu c$ and $c\,d_{\mu\gamma}$ are C^∞ outside Λ and homogeneous of degree $-|\alpha'| - 1 = -|\alpha|$, and $\Gamma_\gamma\Gamma^\beta$ is a product of $|\beta| + 1 \le |\alpha|$ homogeneous fields. Summing over the terms gives the claim for α, completing the proof.

Lemma 6.3.3: Klainerman-Sobolev Inequality Lemma II

Given $\delta > 0$, there exists a constant C_δ such that

$$|f(0)|^2 \le C_\delta \sum_{|\alpha| \le \frac{n+2}{2}} \int_{|y|<\delta} |\partial^\alpha f(y)|^2 dy$$

for all $f \in C^\infty(\mathbb{R}^n)$. Moreover, $\sup_{\delta \ge \delta_0} C_\delta < \infty$ for every $\delta_0 > 0$.

***Proof*:**
Fix a cutoff $\chi \in C_c^\infty(\mathbb{R}^n)$ supported in the unit ball with $\chi(0) = 1$, and apply

$$|f(x)| \le C \sum_{|\alpha| \le \frac{n+2}{2}} \|\partial^\alpha f\|_{L^2}$$

at $x = 0$ to the function $g(y) = \chi(y/\delta)f(y)$, which lies in $C_c^\infty(\mathbb{R}^n)$ and has $g(0) = f(0)$. By the Leibniz rule, $\partial^\alpha g$ is a sum of terms $\delta^{-|\alpha-\beta|}(\partial^{\alpha-\beta}\chi)(y/\delta)\partial^\beta f(y)$ with $\beta \le \alpha$, each supported in $\{y \mid |y| < \delta\}$. Since $2|\alpha - \beta| \le n + 2$ for $|\alpha| \le \frac{n+2}{2}$, squaring and integrating give the inequality with $C_\delta \le C(1+\delta^{-n-2})$, where C depends only on n and χ. This bound is at most $C(1+\delta_0^{-n-2})$ for $\delta \ge \delta_0$, which gives the final statement and completes the proof.

Consider a smooth function $v(q, \omega)$ with $q \in \mathbb{R}$ and $\omega \in S^{n-1}$. The vector fields Ω_{ij}, $1 \le i < j \le n$, are tangent to S^{n-1}, so they define vector fields on S^{n-1}. Listing them in lexicographic order, write

$$\partial_\omega^\alpha = \Omega_{1,2}^{\alpha_1} \cdots \Omega_{n-1,n}^{\alpha_m}$$

at any point $\omega \in S^{n-1}$, where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ and $m = n(n-1)/2$.

Lemma 6.3.4: Klainerman-Sobolev Inequality Lemma III

Given $\delta > 0$, there exists a constant C_δ such that

$$|v(q,\omega)|^2 \leq C_\delta \sum_{j+|\alpha|\leq \frac{n+2}{2}} \int_{|p|<\delta} \int_{\eta\in S^{n-1}} |\partial_q^j \partial_\omega^\alpha v(q+p,\eta)|^2 d\sigma(\eta) dp$$

for all $v \in C^\infty(\mathbb{R}\times S^{n-1})$ and all $(q,\omega)\in\mathbb{R}\times S^{n-1}$. Moreover, $\sup_{\delta\geq\delta_0} C_\delta < \infty$ for every $\delta_0 > 0$.

***Proof Key ideas*:**
Cover S^{n-1} by finitely many coordinate charts with a subordinate partition of unity. In a chart, the points $(q+p,\eta)$ become points of $\mathbb{R}\times\mathbb{R}^{n-1} = \mathbb{R}^n$, and lemma 6.3.3 applies to v written in these coordinates, with δ replaced by a fixed fraction of $\min(\delta,1)$, which keeps the constants bounded for $\delta \geq \delta_0$. Since the fields Ω_{ij}, $1 \leq i < j \leq n$, span $T_\eta S^{n-1}$ at every η, each coordinate derivative is a combination of them with coefficients bounded on the chart domains, and their commutators are again combinations of them, so the derivatives in lemma 6.3.3 become the operators $\partial_q^j \partial_\omega^\alpha$. In each chart the surface measure is comparable to Lebesgue measure. This book develops neither coordinate charts on spheres nor the commutation relations of the Ω_{ij}, so the computation is not carried out here. The lemma is [10, Week 7, Lemma 7], where the same argument is sketched.

***Proof of theorem 6.3.1*:**
Fix $t > 0$ and $x \in \mathbb{R}^n$, and write $\sigma = \frac{n+2}{2}$.

Step 1: The case $t + |x| \leq 1$. The bound $|f(x)| \leq C\sum_{|\alpha|\leq\sigma} \|\partial^\alpha f\|_{L^2}$ for $f \in C_c^\infty(\mathbb{R}^n)$ recorded in section 6.2, applied to $u(t,\cdot)$, gives the inequality, because the derivatives ∂_x^α are among the Γ^α and $(1+t+|x|)^{\frac{n-1}{2}} \leq 2^{\frac{n-1}{2}}$. From now on $R = t + |x| > 1$.

Step 2: Away from the light cone. Suppose $|x| \leq \frac{t}{2}$ or $|x| \geq \frac{3t}{2}$, and let $|y| \leq \frac{R}{8}$. If $|x| \leq \frac{t}{2}$, then $t \geq \frac{2R}{3}$ and $t - |x+y| \geq \frac{t}{2} - \frac{R}{8} \geq \frac{5R}{24}$. If $|x| \geq \frac{3t}{2}$, then $|x| \geq \frac{3R}{5}$ and $|x+y| - t \geq \frac{|x|}{3} - \frac{R}{8} \geq \frac{3R}{40}$. Also $\big|t + |x+y| - R\big| = \big||x+y| - |x|\big| \leq \frac{R}{8}$. So the point $R^{-1}(t, x+y)$ lies in the compact set

$$K = \big\{(s,z) \in \mathbb{R}^{1+n} \;\big|\; s \geq 0,\ \tfrac{7}{8} \leq s + |z| \leq \tfrac{9}{8},\ \big|s - |z|\big| \geq \tfrac{3}{40}\big\}$$

which does not meet the light cone Λ. The coefficients $c_{\alpha\beta}$ of lemma 6.3.2 are continuous on K and homogeneous of degree $-|\alpha|$, so $|c_{\alpha\beta}(t,x+y)| = R^{-|\alpha|}|c_{\alpha\beta}(R^{-1}(t,x+y))| \leq CR^{-|\alpha|}$ for $1 \leq |\beta| \leq |\alpha| \leq \sigma$. Now lemma 6.3.3 with $\delta = \frac{1}{8}$, applied to $z \mapsto u(t, x+Rz)$, and the substitution $y = Rz$ give

$$|u(t,x)|^2 \leq C \sum_{|\alpha|\leq\sigma} \int_{|z|<\frac{1}{8}} \big|R^{|\alpha|}\partial_x^\alpha u(t,x+Rz)\big|^2 dz = CR^{-n} \sum_{|\alpha|\leq\sigma} \int_{|y|<\frac{R}{8}} \big|R^{|\alpha|}\partial_x^\alpha u(t,x+y)\big|^2 dy$$

For $\alpha \neq 0$, lemma 6.3.2 writes $\partial_x^\alpha u(t,x+y)$ as $\sum_{1\leq|\beta|\leq|\alpha|} c_{\alpha\beta}(t,x+y)\Gamma^\beta u(t,x+y)$, so $R^{|\alpha|}|\partial_x^\alpha u(t,x+y)| \leq C\sum_{|\beta|\leq\sigma} |\Gamma^\beta u(t,x+y)|$ for $|y| \leq \frac{R}{8}$, and this also holds for $\alpha = 0$. Hence

$R^n|u(t,x)|^2 \le C\sum_{|\beta|\le\sigma}\|\Gamma^\beta u(t,\cdot)\|_{L^2}^2$. As $1+t+|x|\le 2R$ and $R>1$, $(1+t+|x|)^{n-1}|u(t,x)|^2 \le 2^{n-1}R^n|u(t,x)|^2$, and taking square roots gives the inequality.

Step 3: Near the light cone. Suppose $\frac{t}{2}\le|x|\le\frac{3t}{2}$, so that $1<t+|x|\le\frac{5t}{2}$ and $t>\frac{2}{5}$. Write $x=r\omega$ with $r=|x|$ and $\omega\in S^{n-1}$, put $q=r-t$, and let $v(q',\eta)=u(t,(t+q')\eta)$ for $q'\in\mathbb{R}$ and $\eta\in S^{n-1}$, a smooth function on $\mathbb{R}\times S^{n-1}$. For fixed η, $\partial_{q'}$ differentiates along the line through η, so

$$\partial_{q'}^j v(q',\eta)=\sum_{|\gamma|=j}\frac{j!}{\gamma!}\eta^\gamma(\partial_x^\gamma u)(t,(t+q')\eta)$$

and the rotation fields commute with dilations, so $\partial_\omega^\alpha v(q',\eta)=(\Omega^\alpha u)(t,(t+q')\eta)$, where Ω^α is the product of rotation fields defining ∂_ω^α. Hence $|\partial_{q'}^j\partial_\omega^\alpha v(q',\eta)|\le C\sum_{|\beta|\le j+|\alpha|}|\Gamma^\beta u(t,(t+q')\eta)|$. By lemma 6.3.4 with $\delta=\frac{t}{4}\ge\frac{1}{10}$, whose constant is bounded for $\delta\ge\frac{1}{10}$,

$$|u(t,x)|^2=|v(q,\omega)|^2\le C\sum_{|\beta|\le\sigma}\int_{|p|<\frac{t}{4}}\int_{S^{n-1}}|\Gamma^\beta u(t,(t+q+p)\eta)|^2\,d\sigma(\eta)\,dp$$

As $|q|\le\frac{t}{2}$, the radius $\rho=t+q+p$ lies in $[\frac{t}{4},2t]$, where $\rho^{n-1}\ge(\frac{t}{4})^{n-1}$, so changing variables to ρ and using polar coordinates,

$$|u(t,x)|^2\le C\Big(\frac{4}{t}\Big)^{n-1}\sum_{|\beta|\le\sigma}\int_{\frac{t}{4}}^{2t}\int_{S^{n-1}}|\Gamma^\beta u(t,\rho\eta)|^2\rho^{n-1}\,d\sigma(\eta)\,d\rho\le Ct^{1-n}\sum_{|\beta|\le\sigma}\|\Gamma^\beta u(t,\cdot)\|_{L^2}^2$$

Since $1<\frac{5t}{2}$, we have $1+t+|x|\le 5t$, and multiplying by $(1+t+|x|)^{n-1}$ and taking square roots gives the inequality, completing the proof.

6.4 Proof of main theorem

Theorem 6.4.1: Global Existence Of Non-Linaer Wave Equation Reminder

Let $n\ge 4$. Let $f,g\in C_c^\infty(\mathbb{R}^n)$. Then there exists $\epsilon_0>0$ such that the two equations

$$\Box u=F(\partial u) \tag{6.7}$$

$$u|_{t=0}=\epsilon f\qquad \partial_t u|_{t=0}=\epsilon g \tag{6.8}$$

where $F\colon\mathbb{R}^{1+n}\to\mathbb{R}$ is a given smooth function that vanishes to second order at the origin,

$$F(0)=0\qquad DF(0)=0$$

has a solution

$$u\in C^\infty([0,\infty)\times\mathbb{R}^n)$$

provided $0<\epsilon\le\epsilon_0$.

This restates theorem 6.1.1.

Lemma 6.4.2: Build-Up For Existence Theorem

Let F be as in theorem 6.4.1, and let $u \in C^\infty([0,T)\times\mathbb{R}^n)$ for some $T>0$. Below, D^mF stands for any partial derivative $\partial^\gamma F$ with $|\gamma|=m$.

1. $|F(z)|\le G_2(|z|)|z|^2$, $|DF(z)|\le G_2(|z|)|z|$ and $|D^mF(z)|\le G_m(|z|)$ for $m\ge 2$, for all $z\in\mathbb{R}^{1+n}$, where $G_2, G_3, \dots$ are continuous, increasing functions on $[0,\infty)$
2. $\Box\Gamma^\alpha = \sum_{|\beta|\le|\alpha|} c_{\alpha\beta}\Gamma^\beta\Box$ where the $c_{\alpha\beta}$ are constants
3. For $|\alpha|\ge 1$, $\Gamma^\alpha[F(\partial u)]$ is a linear combination, with constant coefficients, of terms
$$[D^mF](\partial u)\,\Gamma^{\beta_1}\partial_{\mu_1}u\cdots\Gamma^{\beta_m}\partial_{\mu_m}u \tag{6.9}$$
where $1\le m\le|\alpha|$, $0\le\mu_1,\dots,\mu_m\le n$ and $\sum_1^m|\beta_i|=|\alpha|$
4. In (6.9), at most one β_i can have order $|\beta_i|>|\alpha|/2$. If the β_i are ordered so that $|\beta_m|=\max_{1\le i\le m}|\beta_i|$, then
$$|\beta_i|\le\frac{|\alpha|}{2}\qquad\text{for}\qquad 1\le i\le m-1$$
5. Let $N=n+4$. If $|\alpha|\le N$ and $|\beta_i|\le|\alpha|/2$, then
$$|\beta_i|+1+\frac{n+2}{2}\le N$$

***Proof*:**

1. For $m\ge 2$ put $M_m(r)=\max_{|\gamma|=m}\sup_{|z|\le r}|\partial^\gamma F(z)|$, which is finite, increasing, and continuous in r by the uniform continuity of each $\partial^\gamma F$ on compact balls. Let $G_2=(n+1)^2M_2$ and $G_m=M_m$ for $m\ge 3$, so that $|D^mF(z)|\le G_m(|z|)$ for $m\ge 2$. Since $DF(0)=0$, the fundamental theorem of calculus along the segment from 0 to z gives
$$\partial_jF(z)=\int_0^1\frac{d}{d\lambda}\partial_jF(\lambda z)\,d\lambda=\sum_{\mu=0}^n z_\mu\int_0^1\partial_\mu\partial_jF(\lambda z)\,d\lambda$$
As $|z_\mu|\le|z|$ and $|\lambda z|\le|z|$, this gives $|\partial_jF(z)|\le(n+1)M_2(|z|)|z|\le G_2(|z|)|z|$. In the same way, since $F(0)=0$, $|z_j|\le|z|$ and $|\partial_jF(\lambda z)|\le(n+1)M_2(|z|)|z|$ for $0\le\lambda\le 1$,
$$|F(z)|=\Big|\sum_{j=0}^n z_j\int_0^1\partial_jF(\lambda z)\,d\lambda\Big|\le(n+1)^2M_2(|z|)|z|^2=G_2(|z|)|z|^2$$
2. Here $\Box=\partial_t^2-\triangle=\sum_{\mu=0}^n\epsilon_\mu\partial_\mu^2$, with $\epsilon_0=1$ and $\epsilon_1=\cdots=\epsilon_n=-1$. For $0\le\mu,\nu\le n$, the Leibniz rule gives $\Box(x_\mu w)=x_\mu\Box w+2\epsilon_\mu\partial_\mu w$ for smooth w, and with $w=\partial_\nu v$,
$$\Box(x_\mu\partial_\nu v)=x_\mu\partial_\nu\Box v+2\epsilon_\mu\partial_\mu\partial_\nu v$$
For $1\le i,j\le n$ the correction terms cancel in $\Box(\Omega_{ij}v)=\Box(x_j\partial_iv)-\Box(x_i\partial_jv)$, because $\epsilon_i=\epsilon_j$, and in $\Box(\Omega_{0i}v)=\Box(x_0\partial_iv)+\Box(x_i\partial_0v)$, because $\epsilon_0=-\epsilon_i$, while summing over

$\mu = \nu$ gives $\Box(L_0 v) = L_0 \Box v + 2\sum_\mu \epsilon_\mu \partial_\mu^2 v = L_0 \Box v + 2\Box v$. Since also $\Box \partial_\mu = \partial_\mu \Box$, every field satisfies $\Box \Gamma_k = (\Gamma_k + c_k)\Box$, with $c_k = 2$ if $\Gamma_k = L_0$ and $c_k = 0$ otherwise. For $\alpha = (i_1, \dots, i_l)$, moving $\Box$ to the right one factor at a time gives

$$\Box \Gamma^\alpha = (\Gamma_{i_1} + c_{i_1}) \cdots (\Gamma_{i_l} + c_{i_l}) \Box$$

and expanding the product writes $\Box \Gamma^\alpha$ as a linear combination, with constant coefficients, of the operators $\Gamma^\beta \Box$ with β a subsequence of α, so that $|\beta| \leq |\alpha|$.

3. Each Γ_k is a first-order differential operator without zeroth-order term, so it satisfies the Leibniz rule, and the chain rule gives $\Gamma_k[G(\partial u)] = \sum_{\mu=0}^n (\partial_\mu G)(\partial u)\, \Gamma_k \partial_\mu u$ for smooth $G \colon \mathbb{R}^{1+n} \to \mathbb{R}$, where $\partial_\mu G$ is the derivative of G in its μ-th variable. We argue by induction on $|\alpha|$. For $\alpha = (k)$ this is the claim with $m = 1$ and $\beta_1 = (k)$. If $\alpha = (k, i_1, \dots, i_r)$ with $r \geq 1$, then $\Gamma^\alpha = \Gamma_k \Gamma^{\alpha'}$ with $\alpha' = (i_1, \dots, i_r)$, and by the induction hypothesis it suffices to apply Γ_k to one term (6.9) for α'. By the Leibniz rule this gives

$$\sum_{\mu=0}^n [\partial_\mu D^m F](\partial u)\, \Gamma_k \partial_\mu u \prod_{i=1}^m \Gamma^{\beta_i} \partial_{\mu_i} u + \sum_{i=1}^m [D^m F](\partial u)\, \Gamma_k \Gamma^{\beta_i} \partial_{\mu_i} u \prod_{l \neq i} \Gamma^{\beta_l} \partial_{\mu_l} u$$

In the first sum $\partial_\mu D^m F$ is a derivative of order $m+1 \leq |\alpha|$, and the orders $1, |\beta_1|, \dots, |\beta_m|$ add up to $1 + |\alpha'| = |\alpha|$. In the second sum $\Gamma_k \Gamma^{\beta_i}$ is $\Gamma^{\beta_i'}$, where β_i' is β_i with k put in front, so $|\beta_i'| = |\beta_i| + 1$ and again the orders add up to $|\alpha|$, with $m \leq |\alpha|$ factors. All these are terms of the form (6.9) for α.

4. If two of the β_i had order greater than $|\alpha|/2$, their orders alone would add up to more than $|\alpha| = \sum_1^m |\beta_i|$. The factors of a term commute, so the pairs (β_i, μ_i) may be relabelled so that $|\beta_m|$ is the largest order. If then $|\beta_i| > |\alpha|/2$ for some $i \leq m-1$, also $|\beta_m| \geq |\beta_i| > |\alpha|/2$, which is impossible.

5. From $|\beta_i| \leq |\alpha|/2 \leq N/2$ and $N = n + 4$,

$$|\beta_i| + 1 + \frac{n+2}{2} \leq \frac{N}{2} + \frac{n+4}{2} = N$$

This proves the five assertions, as claimed.

Lemma 6.4.3: Lifespan Of Smooth Solutions

Let $F \colon \mathbb{R}^{n+2} \to \mathbb{R}$ be smooth with $F(0) = 0$, let $f, g \in C_c^\infty(\mathbb{R}^n)$ with $\mathrm{supp} f \cup \mathrm{supp} g \subseteq \{x \mid |x| \leq R\}$, and let T_* be the supremum of the $T > 0$ for which (5.6) with the data (5.7) has a solution in $C^\infty([0,T] \times \mathbb{R}^n)$. These solutions agree where they overlap and define $u \in C^\infty([0, T_*) \times \mathbb{R}^n)$, with $\mathrm{supp} u(t, \cdot) \subseteq \{x \mid |x| \leq R + t\}$. If $T_* < \infty$, then ∂u is unbounded on $[0, T_*) \times \mathbb{R}^n$.

***Proof*:**
By theorem 5.3.3, $T_* > 0$.

Step 1: Agreement and support. Let $v \in C^\infty([0,T] \times \mathbb{R}^n)$ and $w \in C^\infty([0,T'] \times \mathbb{R}^n)$ be

solutions, and let $0 < t_0 \leq \min(T, T')$ and $x_0 \in \mathbb{R}^n$. In the backward cone with vertex (t_0, x_0) and base $\{y \mid |y - x_0| < t_0\}$, both are C^2 solutions with the same data, so they agree there by theorem 5.2.2, and at (t_0, x_0) by continuity. This defines u. If $|x_0| > R + t_0$, the base of that cone does not meet $\{y \mid |y| \leq R\}$, so u agrees in the cone with the zero solution, which solves (5.6) because $F(0) = 0$, and $u(t_0, x_0) = 0$.

Step 2: A bounded gradient extends the H^s solutions. Suppose that $T_* < \infty$ and that ∂u is bounded on $[0, T_*) \times \mathbb{R}^n$, and fix $s > \frac{n}{2} + 1$. For $T < T_*$ the slices $u(t, \cdot)$, $t \leq T$, are smooth and supported in a fixed ball by Step 1, so $u \in C([0, T], H^s) \cap C^1([0, T], H^{s-1})$, and u is the solution of theorem 5.3.1 on S_T. Hence the supremum T^* of theorem 5.3.2 satisfies $T^* \geq T_*$. If $T^* = T_*$, the solution on S_{T^*} in that theorem is u, and $\partial u \in L^\infty(S_{T^*})$ contradicts it. So $T^* > T_*$, and u extends to a solution in $C([0, T_s], H^s) \cap C^1([0, T_s], H^{s-1})$ for some $T_s > T_*$.

Step 3: A contradiction. By Step 2, $u \in C([0, T_*], H^s) \cap C^1([0, T_*], H^{s-1})$ for every s. By theorem 3.2.7, every spatial derivative of u and of $\partial_t u$ is continuous on $[0, T_*] \times \mathbb{R}^n$, and differentiating $\partial_t^2 u = \triangle u + F(u, \partial u)$ in t expresses every higher time derivative through these, as in the proof of theorem 5.4.3. So every derivative of u extends continuously to $[0, T_*] \times \mathbb{R}^n$, and $u(T_*, \cdot)$ and $\partial_t u(T_*, \cdot)$ are smooth and, by Step 1, supported in $\{x \mid |x| \leq R + T_*\}$. The equation does not depend on t, so theorem 5.3.3 with these data at the initial time T_* gives a solution $w \in C^\infty([T_*, T_* + \tau] \times \mathbb{R}^n)$ for some $\tau > 0$. At $t = T_*$ all derivatives of u and of w agree, because the equation determines the time derivatives from the data, so u and w together form a solution in $C^\infty([0, T_* + \tau] \times \mathbb{R}^n)$, against the definition of T_*. Hence ∂u is unbounded when $T_* < \infty$, as claimed.

***Proof of theorem 6.4.1*:**
Fix R with $\text{supp} f \cup \text{supp} g \subseteq \{x \mid |x| \leq R\}$ and set $N = n + 4$. For $0 < \epsilon \leq 1$, let T_ϵ be the lifespan and $u \in C^\infty([0, T_\epsilon) \times \mathbb{R}^n)$ the solution given by lemma 6.4.3, applied with the nonlinearity $(u, \partial u) \mapsto F(\partial u)$, so that $\text{supp} u(t, \cdot) \subseteq \{x \mid |x| \leq R + t\}$, and put

$$A(t) = \sum_{|\alpha| \leq N} \|\Gamma^\alpha \partial u(t, \cdot)\|_{L^2} \qquad 0 \leq t < T_\epsilon$$

where $\Gamma^\alpha \partial u$ is the vector $(\Gamma^\alpha \partial_0 u, \ldots, \Gamma^\alpha \partial_n u)$. Each $\Gamma^\alpha \partial_\mu u$ is smooth, and for t in a compact subinterval of $[0, T_\epsilon)$ it is supported in a fixed ball, so A is continuous. The constants below depend only on n, F, f and g.

Step 1: Commuting ∂ and Γ. Write $\Gamma_i = \sum_\nu a_{i\nu} \partial_\nu$. The coefficients $a_{i\nu}$ are polynomials of degree at most one in (t, x), so $[\Gamma_i, \partial_\mu] = -\sum_\nu (\partial_\mu a_{i\nu}) \partial_\nu$ has constant coefficients. By induction on $|\alpha|$, every $\Gamma^\alpha \partial_\mu$ is a linear combination with constant coefficients of operators $\partial_\nu \Gamma^\beta$ with $|\beta| \leq |\alpha|$, and every $\partial_\mu \Gamma^\alpha$ is one of operators $\Gamma^\beta \partial_\nu$ with $|\beta| \leq |\alpha|$. So, with $B(t) = \sum_{|\alpha| \leq N} \|\partial \Gamma^\alpha u(t, \cdot)\|_{L^2}$, there is $C_0 \geq 1$ with $A \leq C_0 B$ and $B \leq C_0 A$.

Step 2: Initial size and pointwise bounds. At $t = 0$ each $\Gamma^\alpha \partial_\mu u$ is a combination, with coefficients that are polynomials in x, of derivatives of u of order at most $N + 1$. Using $\partial_t^2 u = \triangle u + F(\partial u)$ repeatedly, and induction on the order, each such derivative at $t = 0$ is bounded by a constant times ϵ and supported in $\{x \mid |x| \leq R\}$: the derivatives of ϵf and ϵg carry the factor ϵ, $|F(\partial u(0, x))| \leq M |\partial u(0, x)|$ with $M = \sup\{|DF(z)| \mid |z| \leq \|\nabla f\|_{L^\infty} + \|g\|_{L^\infty}\}$

because $F(0) = 0$, and every derivative of $F(\partial u)$ contains a derivative of ∂u as a factor. Hence there is c with $A(0) \leq c\epsilon$. Fix $A = 2C_0^2 c$ (or any $A > 0$ if $c = 0$), so that $C_0^2 A(0) \leq A\epsilon/2$. By theorem 3.2.7 with $k = 0$ and s the least integer above $\frac{n}{2}$, and the integer-order characterization of the Sobolev norm, applied to $\partial_\mu u(t, \cdot)$, there is C_1 with

$$\|\partial u(t, \cdot)\|_{L^\infty} \leq C_1 A(t) \qquad 0 \leq t < T_\epsilon$$

because the derivatives ∂_x^α are among the Γ^α. By theorem 6.3.1, in its form for $0 < t < T$, applied to $\Gamma^\beta \partial_\mu u$, there is C_2 with

$$(1+t)^{\frac{n-1}{2}} \|\Gamma^\beta \partial u(t, \cdot)\|_{L^\infty} \leq C_2 \sum_{|\delta| \leq \frac{n+2}{2}} \|\Gamma^\delta \Gamma^\beta \partial u(t, \cdot)\|_{L^2} \leq C_2 A(t) \qquad 0 < t < T_\epsilon$$

whenever $|\beta| + \frac{n+2}{2} \leq N$.

Step 3: The integral inequality. Suppose that $0 < T' < T_\epsilon$ and $A(t) \leq 2A\epsilon$ for $0 \leq t \leq T'$. We show that there is C_A, depending only on A, n, F, f and g, with

$$A(t) \leq \frac{A\epsilon}{2} + C_A \epsilon \int_0^t \frac{A(s)}{(1+s)^{\frac{n-1}{2}}}\, ds \qquad 0 \leq t \leq T' \tag{6.10}$$

This step does not use $n \geq 4$. By theorem 3.2.6, applied to each $\Gamma^\beta u$, which is smooth with compactly supported slices, by lemma 6.4.2(2) and by $\Box u = F(\partial u)$, there is C_3 with

$$B(t) \leq B(0) + \sum_{|\beta| \leq N} \int_0^t \|\Box \Gamma^\beta u(s, \cdot)\|_{L^2}\, ds \leq C_0 A(0) + C_3 \sum_{|\gamma| \leq N} \int_0^t \|\Gamma^\gamma [F(\partial u)](s, \cdot)\|_{L^2}\, ds$$

Let $0 < s \leq T'$. By Step 2, $\|\partial u(s, \cdot)\|_{L^\infty} \leq 2C_1 A$, so lemma 6.4.2(1) bounds $|DF(\partial u)|$ by $G_2(2C_1 A)|\partial u|$ and $|D^m F(\partial u)|$ by $G_m(2C_1 A)$ for $m \geq 2$. Also $\|\Gamma^\beta \partial u(s, \cdot)\|_{L^2} \leq A(s) \leq 2A\epsilon$ for $|\beta| \leq N$. If γ is empty, $|F(z)| \leq G_2(|z|)|z|^2$ and Step 2 give

$$\|F(\partial u)(s, \cdot)\|_{L^2} \leq G_2(2C_1 A)\, \|\partial u(s, \cdot)\|_{L^\infty} \|\partial u(s, \cdot)\|_{L^2} \leq G_2(2C_1 A)\, C_2 \frac{A(s)}{(1+s)^{\frac{n-1}{2}}}\, 2A\epsilon$$

If $|\gamma| \geq 1$, parts (3) and (4) of lemma 6.4.2 write $\Gamma^\gamma [F(\partial u)]$ as a combination of terms $[D^m F](\partial u)\, \Gamma^{\beta_1} \partial_{\mu_1} u \cdots \Gamma^{\beta_m} \partial_{\mu_m} u$ with $|\beta_m| \leq |\gamma| \leq N$ and $|\beta_i| \leq |\gamma|/2$ for $i < m$. By part (5), $|\beta_i| + \frac{n+2}{2} \leq N$ for $i < m$, so Step 2 bounds $\|\Gamma^{\beta_i} \partial u(s, \cdot)\|_{L^\infty}$ by $C_2 A(s)(1+s)^{-\frac{n-1}{2}} \leq 2C_2 A$. The L^2 norm of the term is at most

$$\|[D^m F](\partial u)\|_{L^\infty} \prod_{i<m} \|\Gamma^{\beta_i} \partial u(s, \cdot)\|_{L^\infty}\, \|\Gamma^{\beta_m} \partial u(s, \cdot)\|_{L^2}$$

For $m \geq 2$ this is at most $G_m(2C_1 A)(2C_2 A)^{m-2} C_2 A(s)(1+s)^{-\frac{n-1}{2}} \cdot 2A\epsilon$, bounding all but one factor of the product by $2C_2 A$. For $m = 1$ it is at most $G_2(2C_1 A)\|\partial u(s, \cdot)\|_{L^\infty} \cdot 2A\epsilon \leq G_2(2C_1 A) C_2 A(s)(1+s)^{-\frac{n-1}{2}} \cdot 2A\epsilon$. The number of terms depends only on $|\gamma|$ and n, so in all cases $\|\Gamma^\gamma [F(\partial u)](s, \cdot)\|_{L^2} \leq C_4 \epsilon A(s)(1+s)^{-\frac{n-1}{2}}$ with C_4 depending only on A, n and F. Since $A \leq C_0 B$ and $C_0^2 A(0) \leq A\epsilon/2$, this gives (6.10) with C_A equal to $C_0 C_3 C_4$ times the number of sequences γ with $|\gamma| \leq N$.

Step 4: The continuity argument. The integral $I = \int_0^\infty (1+s)^{-\frac{n-1}{2}}\,ds$ is finite because $n \geq 4$. Choose $\epsilon_0 \in (0,1]$ with $e^{C_A\epsilon_0 I} \leq 2$, let $0 < \epsilon \leq \epsilon_0$, and put

$$E = \{t \in [0, T_\epsilon) \mid A(s) \leq A\epsilon \text{ for } 0 \leq s \leq t\}$$

As $A(0) \leq C_0^2 A(0) \leq A\epsilon/2$, $0 \in E$, and E is closed in $[0, T_\epsilon)$ because A is continuous. Let $t_0 \in E$. By continuity there is $t_1 \in (t_0, T_\epsilon)$ with $A(t) \leq 2A\epsilon$ for $0 \leq t \leq t_1$. Step 3 applies with $T' = t_1$, and [3, Grönwall's Inequality], with $a = A\epsilon/2$ and $b(s) = C_A\epsilon(1+s)^{-\frac{n-1}{2}}$, gives

$$A(t) \leq \frac{A\epsilon}{2} \exp\Big[C_A\epsilon \int_0^t \frac{ds}{(1+s)^{\frac{n-1}{2}}}\Big] \leq \frac{A\epsilon}{2} e^{C_A\epsilon_0 I} \leq A\epsilon \qquad 0 \leq t \leq t_1$$

So $[0, t_1] \subseteq E$, and E is open in $[0, T_\epsilon)$. As $[0, T_\epsilon)$ is connected, $E = [0, T_\epsilon)$.

Step 5: Conclusion. By Steps 2 and 4, $\|\partial u\|_{L^\infty([0,T_\epsilon)\times\mathbb{R}^n)} \leq C_1 A\epsilon$, so $T_\epsilon = \infty$ by lemma 6.4.3. Thus for every $0 < \epsilon \leq \epsilon_0$ the solution u lies in $C^\infty([0,\infty)\times\mathbb{R}^n)$, completing the proof.

6.5 Looking at Low Dimensions Solutions

For $n \in \{1,2,3\}$ global existence can fail, and theorem 6.5.1 below gives a three-dimensional example. The proof for $n \geq 4$ still gives the lower bounds on the lifespan stated after theorem 6.1.1.

6.5.1 Proof of lower bounds

Let $n \in \{1,2,3\}$, let $0 < \epsilon \leq 1$, and let T_ϵ, u, $N = n+4$, $A(t)$, the constant A and the constants C_1 and C_A be as in the proof of theorem 6.4.1. Steps 1 to 3 of that proof do not use $n \geq 4$, so whenever $A(t) \leq 2A\epsilon$ for $0 \leq t \leq T' < T_\epsilon$, the inequality (6.10) holds on $[0, T']$, and [3, Grönwall's Inequality] gives

$$A(t) \leq \frac{A\epsilon}{2} \exp\big[C_A\epsilon\, I_n(t)\big] \qquad I_n(t) = \int_0^t \frac{ds}{(1+s)^{\frac{n-1}{2}}} \qquad 0 \leq t \leq T'$$

Here $I_3(t) = \log(1+t)$, $I_2(t) = 2\sqrt{1+t} - 2 \leq 2\sqrt{t}$ and $I_1(t) = t$. Let $T(\epsilon)$ be $e^{\log 2/(C_A\epsilon)} - 1$, $\big(\log 2/(2C_A\epsilon)\big)^2$ and $\log 2/(C_A\epsilon)$ for $n = 3$, 2 and 1, so that $C_A\epsilon I_n(t) \leq \log 2$ for $0 \leq t \leq T(\epsilon)$. Running Step 4 of that proof on $[0, \min(T_\epsilon, T(\epsilon)))$, where the exponential factor is at most 2, gives $A(t) \leq A\epsilon$ there. If $T_\epsilon < T(\epsilon)$, then $\|\partial u\|_{L^\infty} \leq C_1 A\epsilon$ on $[0, T_\epsilon)\times\mathbb{R}^n$ by Step 2, and lemma 6.4.3 gives $T_\epsilon = \infty$, which is absurd. Hence $T_\epsilon \geq T(\epsilon)$. With $c_0 = \log 2/(2C_A)$, which depends only on f, g, F and n, this reads $T_\epsilon \geq 2c_0/\epsilon$ for $n = 1$ and $T_\epsilon \geq c_0^2/\epsilon^2$ for $n = 2$, and for $n = 3$ it gives $T_\epsilon \geq e^{2c_0/\epsilon} - 1 \geq e^{c_0/\epsilon}$ once $e^{c_0/\epsilon} \geq 2$. These are the stated lower bounds, with $c = 2c_0$, $c = c_0^2$ and $c = c_0$ respectively.

6.5.2 The Null Condition And Global Existence For Dimension 3

In dimension $n = 3$, global existence of smooth solutions for small data can fail for equations of the type $\Box u = F(\partial u)$. The standard example is due to F. John.

Theorem 6.5.1: Non-Existence In Three Dimensions

Let $u \in C^2([0,\infty)\times\mathbb{R}^3)$ solve

$$\Box u = (\partial_t u)^2$$

with $u(0,\cdot), \partial_t u(0,\cdot) \in C_c^\infty(\mathbb{R}^3)$. Then $u = 0$. In particular, for data in $C_c^\infty(\mathbb{R}^3)$ that are not both zero, the lifespan of lemma 6.4.3 is finite.

***Proof Key ideas*:**
The proof works with spherical means, as in the derivation of Kirchhoff's formula. With $\psi = rM(u)(t,x_0,r)$ for a fixed x_0, the identity $(\partial_t^2 - \partial_r^2)\psi = rM(\Box u)$ and the Cauchy-Schwarz inequality $M\big((\partial_t u)^2\big) \geq \big(M(\partial_t u)\big)^2$ give $(\partial_t^2 - \partial_r^2)\psi \geq r^{-1}(\partial_t\psi)^2$. Integrating this inequality in the characteristic coordinates $t-r$ and $t+r$ leads to a Riccati-type inequality whose solutions become infinite at a finite value of $t+r$ unless the spherical means vanish, and with finite propagation speed this forces $u = 0$. The second statement follows, because a global smooth solution would lie in $C^2([0,\infty)\times\mathbb{R}^3)$. This analysis in characteristic coordinates is not developed in this book. The theorem is due to John [5].

The similar equation of Nirenberg,

$$\Box u = (\partial_t u)^2 - \sum_1^3 (\partial_i u)^2 \qquad t \geq 0,\ x \in \mathbb{R}^3$$

behaves differently: for $f, g \in C_c^\infty(\mathbb{R}^3)$ and small $\epsilon > 0$ it has a global smooth solution with the data

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} = \epsilon g$$

For a smooth function u, the function $v = 1 - e^{-u}$ has $\partial_\mu v = e^{-u}\partial_\mu u$ and

$$\Box v = e^{-u}\Big(\Box u - (\partial_t u)^2 + \sum_1^3 (\partial_i u)^2\Big)$$

so u solves Nirenberg's problem exactly when v solves the *linear* Cauchy problem

$$\Box v = 0 \qquad v|_{t=0} = 1 - e^{-\epsilon f} \qquad \partial_t v|_{t=0} = \epsilon g e^{-\epsilon f} \tag{6.11}$$

and conversely $u = -\log(1-v)$ wherever $v < 1$. By corollary 2.3.5, (6.11) has a global solution v, which is smooth because the spherical means of smooth functions are smooth in (t,x). It therefore suffices that $\|v(t,\cdot)\|_{L^\infty} < 1$ for all $t \geq 0$. Let R be such that the data of v vanish outside the ball B of radius R about the origin. By the expanded form (2.9) of Kirchhoff's formula, $|v(t,x)|$ is at most the average over the sphere $\{y \mid |y-x| = t\}$ of $t|\partial_t v(0,y)| + |v(0,y)| + t|\nabla v(0,y)|$, and that sphere meets B in a set of area at most $16\pi R^2$. Hence

$$\|v(t,\cdot)\|_{L^\infty} \leq \frac{A}{1+t} \qquad t \geq 0$$

where A is a constant, depending on R, times the sum of the L^∞ norms of $v|_{t=0}$, $\nabla v|_{t=0}$ and $\partial_t v|_{t=0}$. By (6.11) these norms are $O(\epsilon)$, so $A < 1$ for ϵ small, and then $u = -\log(1-v)$ is a global smooth solution of Nirenberg's problem.

These examples suggest that in dimension $n = 3$, the question of global existence of smooth solutions of systems of the type $\Box u = B(\partial u, \partial u)$, where each vector component of B is a bilinear form in ∂u, depends strongly on the algebraic structure of B. More generally, for a system of the form $\Box u = F(\partial u)$, where F vanishes to second order at the origin, it is the quadratic part of F that determines the global regularity properties of the equation: for nonlinearities that vanish to third order at the origin, small data give global solutions also when $n = 3$ [10, Week 8, Section 8.1, Remark].

6.5.3 Statement Of Null Condition

Consider a system of N equations

$$\Box u^I = F^I(u, \partial u) \qquad (t, x) \in \mathbb{R}^{1+3}$$

where the unknown u and the given C^∞ function F are $\mathbb{R}^N$-valued:

$$u = (u^1, \dots, u^N) \qquad F = (F^1, \dots, F^N)$$

Definition 6.5.2: Null Vectors

A vector $\xi = (\xi_0, \xi_1, \xi_2, \xi_3) \in \mathbb{R}^4$ is *null* if $\xi \neq 0$ and $\xi_0^2 = \xi_1^2 + \xi_2^2 + \xi_3^2$. In other words, ξ is a nonzero vector on the light cone (or null cone) of Minkowski space $\mathbb{R}^{1+3}$.

Definition 6.5.3: Quadratic Part

Let F^I be as in

$$\Box u^I = F^I(u, \partial u) \qquad (t, x) \in \mathbb{R}^{1+3}$$

Then the *quadratic part* of F^I is

$$F^I_{(2)}(z) = \sum_{|\alpha|=2} \frac{1}{\alpha!} \partial^\alpha F^I(0) z^\alpha$$

where $z \in \mathbb{R}^{N+4N}$ corresponds to $(u, \partial u)$.

The examples of John and Nirenberg above suggest that global smooth solutions for small data need a condition on the quadratic part of F. The relevant condition is the *null condition* of Klainerman and Christodoulou.

Definition 6.5.4: Null Condition

Let F be as in

$$\Box u^I = F^I(u, \partial u) \qquad (t,x) \in \mathbb{R}^{1+3}$$

Then we say that F satisfies the *null condition* if

1. F vanishes to second order at the origin:

$$F(0) = 0 \qquad DF(0) = 0$$

Thus, by Taylor's theorem, $F(z) = F_{(2)}(z) + R(z)$ where R is C^∞ and vanishes to third order at 0.

2. The quadratic part of F is of the form

$$F^I_{(2)}(u, \partial u) = \sum_{J,K=1}^{N} \sum_{\mu,\nu=0}^{3} a^{I_{\mu\nu}}_{JK} \partial_\mu u^J \partial_\nu u^K$$

where the a's are real constants satisfying, for all $I, J, K = 1, \dots, N$,

$$\sum_{\mu,\nu=0}^{3} a^{I_{\mu\nu}}_{JK} \xi_\mu \xi_\nu = 0$$

for all null vectors ξ.

Condition (2) allows $F_{(2)}$ to depend on ∂u only, not on u.

Example 6.5.5: Null Condition

The equation of Nirenberg above,

$$\Box u = (\partial_t u)^2 - \sum_{1}^{3} (\partial_i u)^2 \quad t \geq 0,\ x \in \mathbb{R}^3$$

satisfies the null condition, since its quadratic part has $a^{\mu\nu} = 0$ for $\mu \neq \nu$, $a^{00} = 1$ and $a^{ii} = -1$ for $1 \leq i \leq 3$, so that $\sum_{\mu,\nu} a^{\mu\nu} \xi_\mu \xi_\nu = \xi_0^2 - \xi_1^2 - \xi_2^2 - \xi_3^2$. The equation found by F. John,

$$\Box u = (\partial_t u)^2 \quad t \geq 0,\ x \in \mathbb{R}^3$$

does not, since there $\sum_{\mu,\nu} a^{\mu\nu} \xi_\mu \xi_\nu = \xi_0^2$, which is positive at the null vector $(1, 1, 0, 0)$.

Lemma 6.5.6: Null Form

If B is a real bilinear form on $\mathbb{R}^4 \times \mathbb{R}^4$ such that

$$B(\xi, \xi) = 0$$

for all null vectors ξ, then B is a linear combination, with real coefficients, of the so-called *null forms*:

$$Q_0(\xi, \eta) = \xi_0\eta_0 - \sum_1^3 \xi_i\eta_i$$

$$Q_{\mu\nu}(\xi, \eta) = \xi_\mu\eta_\nu - \xi_\nu\eta_\mu \qquad 0 \le \mu < \nu \le 3$$

***Proof*:**

Write $B(\xi, \eta) = \xi^T A\eta = \sum_{\mu,\nu=0}^3 a^{\mu\nu}\xi_\mu\eta_\nu$, where $A = (a^{\mu\nu})$ is a real 4×4 matrix and ξ, η are column vectors, and split $A = A_s + A_a$ with $A_s = \frac{1}{2}(A + A^T)$ symmetric and $A_a = \frac{1}{2}(A - A^T)$ skew-symmetric. Since $\xi^T A_a\xi = 0$, the hypothesis says that $\xi^T A_s\xi = 0$ for every null vector ξ. Write $A_s = (s_{\mu\nu})$, and let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$.

Step 1: The symmetric part. For $1 \le i \le 3$ the vectors $e_0 \pm e_i$ are null, and

$$0 = (e_0 \pm e_i)^T A_s(e_0 \pm e_i) = s_{00} \pm 2s_{0i} + s_{ii}$$

so $s_{0i} = 0$ and $s_{ii} = -s_{00}$. For $1 \le i < j \le 3$ the vector $\sqrt{2}e_0 + e_i + e_j$ is null, and by what was just shown

$$0 = 2s_{00} + s_{ii} + s_{jj} + 2\sqrt{2}(s_{0i} + s_{0j}) + 2s_{ij} = 2s_{ij}$$

so $s_{ij} = 0$. Hence $A_s = s_{00}\,\mathrm{diag}(1, -1, -1, -1)$, that is, $\xi^T A_s\eta = s_{00}Q_0(\xi, \eta)$, where $s_{00} = a^{00}$.

Step 2: The skew-symmetric part. The entries $\frac{1}{2}(a^{\mu\nu} - a^{\nu\mu})$ of A_a change sign when μ and ν are exchanged, so the diagonal entries vanish and

$$\xi^T A_a\eta = \sum_{0\le\mu<\nu\le3} \frac{1}{2}(a^{\mu\nu} - a^{\nu\mu})(\xi_\mu\eta_\nu - \xi_\nu\eta_\mu) = \sum_{0\le\mu<\nu\le3} \frac{1}{2}(a^{\mu\nu} - a^{\nu\mu})Q_{\mu\nu}(\xi, \eta)$$

Adding the two parts gives

$$B = a^{00}Q_0 + \sum_{0\le\mu<\nu\le3} \frac{1}{2}(a^{\mu\nu} - a^{\nu\mu})Q_{\mu\nu}$$

completing the proof.

Corollary 6.5.7: Further Decomposition

Let F be as in equation

$$\Box u^I = F^I(u, \partial u) \qquad (t,x) \in \mathbb{R}^{1+3}$$

Then F satisfies the null condition if and only if $F^I(u, \partial u)$ is of the form:

$$\sum_{J,K} a^I_{JK} Q_0(\partial u^J, \partial u^K) + \sum_{J,K} \sum_{0 \le \mu < \nu \le 3} b^{I_{\mu\nu}}_{JK} Q_{\mu\nu}(\partial u^J, \partial u^K) + R^I(u, \partial u)$$

where the a's and b's are real constants and R^I is smooth and vanishes to third order at 0

***Proof*:**

Suppose F satisfies the null condition. By definition 6.5.4, $F(0) = 0$ and $DF(0) = 0$, so $F^I = F^I_{(2)} + R^I$ with R^I smooth and vanishing to third order at 0, and

$$F^I_{(2)}(u, \partial u) = \sum_{J,K=1}^{N} B^I_{JK}(\partial u^J, \partial u^K) \qquad B^I_{JK}(\xi, \eta) = \sum_{\mu,\nu=0}^{3} a^{I_{\mu\nu}}_{JK} \xi_\mu \eta_\nu$$

where each real bilinear form B^I_{JK} on $\mathbb{R}^4$ satisfies $B^I_{JK}(\xi, \xi) = 0$ for all null vectors ξ. By lemma 6.5.6, each B^I_{JK} is a real linear combination of Q_0 and the $Q_{\mu\nu}$, and substituting these combinations gives F^I the stated form.

Conversely, suppose each F^I has the stated form. The two sums are homogeneous quadratic polynomials in ∂u, and R^I vanishes to third order, so $F(0) = 0$, $DF(0) = 0$, and the quadratic part $F^I_{(2)}$ is the sum of the two sums. This writes $F^I_{(2)}$ as in definition 6.5.4, with, for each I, J, K, the bilinear form $a^I_{JK} Q_0 + \sum_{\mu<\nu} b^{I_{\mu\nu}}_{JK} Q_{\mu\nu}$. This form vanishes at (ξ, ξ) for every null vector ξ, because $Q_0(\xi, \xi) = \xi_0^2 - \sum_{i=1}^3 \xi_i^2 = 0$ and $Q_{\mu\nu}(\xi, \xi) = \xi_\mu \xi_\nu - \xi_\nu \xi_\mu = 0$. Hence F satisfies the null condition, as we sought to show.

Consider:

$$\Box u^I = F^I(u, \partial u) \qquad (t,x) \in \mathbb{R}^{1+3}$$

with initial data

$$u|_{t=0} = \epsilon f \qquad \partial_t u|_{t=0} = \epsilon g$$

where $f = (f^1, f^2, ..., f^N)$ and $g = (g^1, g^2, ..., g^N)$ belong to $C_c^\infty(\mathbb{R}^3)$ and $\epsilon > 0$. Then:

Theorem 6.5.8: Global Existence Under Right Conditions For Low Dimension

Assume that F in the above equation satisfies the null condition. Then there exists $\epsilon_0 = \epsilon_0(f, g) > 0$ such that the above has a smooth global solution provided $\epsilon < \epsilon_0$

6.5.4 Improved Decay

Null forms satisfy a better pointwise bound than generic bilinear forms. For a smooth v write $|\Gamma v|^2 = \sum_{j=0}^{m} |\Gamma_j v|^2$. For each null form Q there is a constant C with

$$|Q(\partial v, \partial w)(t,x)| \leq \frac{C}{1+|t|+|x|} |\Gamma v(t,x)|\,|\Gamma w(t,x)|$$

for all (t,x) and all smooth v and w, and a bilinear form B on $\mathbb{R}^4$ satisfies the corresponding estimate only if $B(\xi,\xi) = 0$ for every null vector ξ [10, Week 8, Lemma 9 and its corollary]. Applying an invariant vector field to $Q(\partial v, \partial w)$ produces null forms again [10, Week 8, Lemma 10], and combined with this bound that gives the estimate for $\Gamma^\alpha Q(\partial v, \partial w)$ used in the proof of the main theorem below [10, Week 8, Proposition 8].

6.5.5 An energy inequality and Hörmander's estimate

The energy inequality of theorem 3.2.6,

$$\|\partial u(t,\cdot)\|_{L^2} \leq \|\partial u(0,\cdot)\|_{L^2} + \int_0^t \|\Box u(s,\cdot)\|_{L^2} ds \tag{6.12}$$

has a generalization adapted to the invariant vector fields. Recall that the original proof was done by noticing that $u_t \Box u$ is a spacetime divergence:

$$u_t \Box u = \mathrm{div}_{t,x}(e_0, e') \tag{6.13}$$

where $e_0 = \frac{1}{2}|\partial u|^2$ and $e' = -u_t \nabla_x u$. Integrating this identity for fixed t gives:

$$\int u_t \Box u dx = \frac{d}{dt} \int e_0 dx - \int \mathrm{div}_x (u_t \nabla_x u) dx$$

and the last term vanishes by the divergence theorem, if u decays sufficiently fast as $|x| \to \infty$. For the *energy* $E(t) = \int e_0(t,x)dx$ one obtains, after applying the Cauchy-Schwarz inequality to the left hand side of the above identity,

$$E'(t) \leq \sqrt{2E(t)} \|\Box u(t,\cdot)\|_{L^2}$$

and (6.12) follows as in the proof of theorem 3.2.6.

The method generalizes by replacing (6.13) with an identity

$$X(\partial) u \cdot \Box u = \mathrm{div}_{t,x}(e_0, e') \tag{6.14}$$

where $X(\partial)$ is some first order differential operator and (e_0, e') is some spacetime vector involving u, such that the associated energy is non-negative:

$$E(t) = \int e_0(t,x) dx \geq 0$$

Integrating (6.14) then gives a generalized energy inequality. Set

$$X(\partial) = \overline{X} \cdot \partial + 2t \tag{6.15}$$

where as usual ∂ is the spacetime gradient and

$$\overline{X} = (1 + t^2 + |x|^2, 2tx_1, 2tx_2, 2tx_3)$$

Let m be the matrix $\text{diag}(1, -1, -1, -1)$ of the Minkowski metric, and let $\overline{1} = (1, 0, 0, 0)$. Then (6.14) holds with

$$(e_0, e') = X(\partial)u \cdot m(\partial u) - \frac{1}{2}(\partial u)^T m(\partial u)\overline{X} - u^2\overline{1}$$

where ∂u is regarded as a column vector for the purposes of matrix multiplication. Integrating (6.14) then yields the identity

$$\frac{d}{dt}E(t) = \int X(\partial)u \cdot \Box u dx$$

for the energy $E(t) = \int e_0 dx$, and the right hand side is bounded by

$$\|(1 + t + |\cdot|)^{-1}X(\partial)u(t, \cdot)\|_{L^2}\|(1 + t + |\cdot|)\Box u(t, \cdot)\|_{L^2}$$

One then shows that

$$\|(1 + t + |\cdot|)^{-1}X(\partial)u(t, \cdot)\|_{L^2} \leq C\sqrt{E(t)}$$

Putting this all together, we obtain:

$$\sqrt{E(t)} \leq C\sqrt{E(0)} + C\int_0^t \|(1 + s + |\cdot|)\Box u(s, \cdot)\|_{L^2} ds$$

Furthermore, one has

$$E(t) \approx \sum_{|\alpha| \leq 1} \|\Gamma^\alpha u(t, \cdot)\|_{L^2}^2$$

Applying this to $\Gamma^\beta u$ for $|\beta| \leq M$, and commuting $\Box$ with Γ^β by lemma 6.4.2(2), yields the generalized energy inequality.

Proposition 6.5.9: Updated Energy Inequality

For any integer $M \geq 0$, there exists a constant C such that

$$\sum_{|\alpha| \leq M+1} \|\Gamma^\alpha u(t, \cdot)\|_{L^2}$$

$$\leq$$

$$C \sum_{|\alpha| \leq M+1} \|\Gamma^\alpha u(0, \cdot)\|_{L^2} + C \sum_{|\alpha| \leq M} \int_0^t \|(1 + s + |\cdot|)\Gamma^\alpha \Box u(s, \cdot)\|_{L^2} ds$$

for all $t > 0$ and all $u \in C^\infty([0, \infty) \times \mathbb{R}^3)$ with compact support in x for each t.

***Proof Key ideas*:**
The argument is the one sketched above: the identity (6.14) with the stated (e_0, e'), the bound $\|(1+t+|\cdot|)^{-1}X(\partial)u(t, \cdot)\|_{L^2} \leq C\sqrt{E(t)}$ and the comparison of $E(t)$ with $\sum_{|\alpha| \leq 1} \|\Gamma^\alpha u(t, \cdot)\|_{L^2}^2$, applied to each $\Gamma^\beta u$ with $|\beta| \leq M$. These computations with the conformal multiplier $X(\partial)$ are used nowhere else in this book and are not carried out here. The statement is [10, Week 8,

Proposition 9], which refers to Sogge's book [11] for the details.

Theorem 6.5.10: Hörmander's Inequality

There exists a C such that if $F \in C^2([0,\infty)\times\mathbb{R}^3)$ and $\Box u = F$ with vanishing initial data at $t=0$, then

$$(1+t+|x|)|u(t,x)| \leq C \sum_{|\alpha|\leq 2} \int_0^t \int_{\mathbb{R}^3} |\Gamma^\alpha F(s,y)| \frac{dyds}{1+s+|y|}$$

***Proof Key ideas*:**
The proof derives the weighted bound from the explicit form of the fundamental solution of $\Box$ on $\mathbb{R}^{1+3}$, which Kirchhoff's formula (theorem 2.3.4) expresses. The estimate is used only in the outline of the proof of theorem 6.5.8, and its derivation is not carried out here. The statement is [10, Week 8, Theorem 31], which refers to Sogge's book [11] and to Hörmander's lecture notes for the proof.

6.5.6 Proof of the main Theorem

By corollary 6.5.7, the null condition puts the system in the form

$$\Box u^I = \sum_{J,K} a^I_{JK} Q_0(\partial u^J, \partial u^K) + \sum_{J,K} \sum_{0\leq\mu<\nu\leq 3} b^{I_{\mu\nu}}_{JK} Q_{\mu\nu}(\partial u^J, \partial u^K) + R^I(u,\partial u) \tag{6.16}$$

where R^I vanishes to third order at 0, with initial data $u|_{t=0} = \epsilon f$ and $\partial_t u|_{t=0} = \epsilon g$.

***Proof Key ideas*:**
This is the proof of theorem 6.5.8. It uses three tools that this book quotes without proof: the generalized energy inequality (proposition 6.5.9), Hörmander's estimate (theorem 6.5.10), and the improved decay of null forms in the form commuted with the invariant vector fields: for each null form Q and each integer $M \geq 0$,

$$(1+|t|+|x|) \sum_{|\alpha|\leq M} |\Gamma^\alpha Q(\partial v, \partial w)| \leq C_M \big(V_{M+1} W_{M/2+1} + V_{M/2+1} W_{M+1}\big)$$

at every point (t,x), for all smooth v and w, where $V_k = \sum_{1\leq|\alpha|\leq k} |\Gamma^\alpha v(t,x)|$ and W_k is defined in the same way from w. The first tool rests on the conformal multiplier computation sketched before proposition 6.5.9. The second rests on the explicit fundamental solution of $\Box$ on $\mathbb{R}^{1+3}$, which Kirchhoff's formula (theorem 2.3.4) provides, and the missing piece is Hörmander's weighted estimate itself. The third combines the pointwise bound for null forms of section 6.5.4 with the commutation relations between null forms and the invariant vector fields. The estimates that turn these inputs into the three tools are not carried out in this book. The theorem is [10, Week 8, Theorem 30], where the third tool is Proposition 8. The proof there sets $R^I = 0$, as does the outline below. For semilinear systems whose nonlinearity is a quadratic form in ∂u satisfying the null condition plus cubic terms, Klainerman, who introduced the null condition, proved that small data in three space dimensions give global solutions [6].

Step 1: Reduction to an a priori estimate. Let $T^* > 0$ be the lifespan of lemma 6.4.3, in its

version for systems (noted before theorem 5.3.1), and for $0 < T_0 < T^*$ let u be the smooth solution on $S_{T_0} = [0, T_0) \times \mathbb{R}^3$. By lemma 6.4.3, $T^* = \infty$ once ∂u is bounded on $[0, T^*) \times \mathbb{R}^3$, that is, on every S_{T_0} by a bound independent of T_0, and this follows from the estimate

$$\sum_{|\alpha| \leq k} \|\Gamma^\alpha u(t, \cdot)\|_{L^\infty} \leq \frac{A\epsilon}{1+t} \qquad 0 \leq t < T_0 \tag{6.17}$$

for $\epsilon < \epsilon_0$, with $k = 4$ and with A and ϵ_0 independent of T_0.

Step 2: The linear part. Let w_α solve $\Box w_\alpha = 0$ with the data of $\Gamma^\alpha u$ at $t = 0$, and let u_0 solve $\Box u_0 = 0$ with data $(\epsilon f, \epsilon g)$. The vector fields Γ_j commute with $\Box$ up to multiples of $\Box$ (lemma 6.4.2(2)), so $\Box \Gamma^\alpha u_0 = 0$, and the decay estimate $\|v(t, \cdot)\|_{L^\infty} = O(t^{-1})$ for solutions of the homogeneous wave equation on $\mathbb{R}^{1+3}$ with smooth, compactly supported data [10, Week 2, Theorem 6] gives $|\Gamma^\alpha u_0(t, x)| \leq C_\alpha \epsilon (1+t)^{-1}$. The data of $w_\alpha - \Gamma^\alpha u_0$, computed from eq. (6.16), are $O(\epsilon^2)$, so the same estimate gives $|w_\alpha(t, x)| \leq C_\alpha \epsilon (1+t)^{-1}$.

Step 3: The bootstrap. One runs a continuity argument on the set of $T \in [0, T_0)$ for which eq. (6.17) holds on $[0, T]$, under the bootstrap assumption that the left-hand side of eq. (6.17) is at most $2A\epsilon(1+t)^{-1}$ on some $[0, T']$. The generalized energy inequality (proposition 6.5.9), the improved decay of null forms and Gronwall's lemma give

$$\sum_{|\alpha| \leq k+3} \|\Gamma^\alpha u(t, \cdot)\|_{L^2} \leq C_0 (1+t)^{C_1 \epsilon} \sum_{|\alpha| \leq k+3} \|\Gamma^\alpha u(0, \cdot)\|_{L^2}$$

on $[0, T']$. Hörmander's estimate applied to $\Gamma^\alpha u - w_\alpha$, which has vanishing data, together with the improved decay of null forms and this energy bound, then gives

$$(1+t) \sum_{|\alpha| \leq k} \|\Gamma^\alpha u(t, \cdot) - w_\alpha(t, \cdot)\|_{L^\infty} \leq C\epsilon^2$$

for small ϵ. With Step 2, this recovers eq. (6.17) with the constant A once $C\epsilon \leq A/2$, which closes the bootstrap.

7

Future Reading

future reading (following [10]):

1. Strichartz type estimates
2. Application to Maxwell-Klein-Gordon
3. More on well-posedness

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